

硕士学位论文

某些代数簇的超级强逼近性质及其
Brauer-Manin 障碍

**VERY STRONG APPROXIMATION FOR CERTAIN
VARIETIES AND ITS BRAUER-MANIN
OBSTRUCTION**

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二〇二六年六月

国内图书分类号：O187.2

国际图书分类号：511

学校代码：14325

密级：公开

理学硕士学位论文

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学科名称：数学

答辩日期：2026 年 5 月

培养单位：数学系

学位授予单位：南方科技大学

VERY STRONG APPROXIMATION FOR CERTAIN VARIETIES AND ITS BRAUER-MANIN OBSTRUCTION

A dissertation submitted to
Southern University of Science and Technology
in partial fulfillment of the requirement
for the degree of
Master of Science
in
Mathematics

by
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Supervisor: Associate Prof. Hu Yong

May, 2026

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摘 要

本文是关于刘青和徐飞的文章 [57] 中一些重要结果的综述报告。本文回顾了局部-整体原理及其 Brauer-Manin 障碍的理论发展概况, 针对 Harari 和 Voloch 于 2010 年提出的一个猜想, 讨论了代数簇 (特别是双曲型曲线) 上整点的局部-整体原理, 然后着重介绍了文章 [57] 中关于代数簇 (特别是代数环面) 的含 Brauer-Manin 障碍集的超级强逼近性质的一些重要结果。

本文讨论的主定理 1 是关于代数簇 S -adelic 空间的拓扑结构定理, 该定理的证明依赖于一种非常局部的工具——概型的光滑化。得益于主定理 1, 我们可以建立代数簇上有理点的局部逼近性质和其整模型上整点的局部逼近性质之间的桥梁。此外, 借助概型的光滑化, 针对一个 Dedekind 整环 R , 及其分式域上的一个光滑代数簇 X , 我们有下述定理: X 的任意一个 R -整模型均存在一个光滑化, 且该光滑化能够保持整模型的局部点性质。

本文介绍的主定理 2 是上述定理的进一步推广。主定理 2 对任何一个不必光滑的代数簇, 证明了其忠实平坦模型的存在性。同时, 我们还证明了, 代数簇的任意一个整模型均存在一个平坦化, 且该平坦化能够保持整模型的所有整点信息。

基于上述结果, 我们主要研究了某些代数簇的含 Brauer-Manin 障碍集的强逼近和超级强逼近性质。本文主定理 3 证明了函数域上的代数环面, 以及数域上的满足某种秩等式的代数环面, 满足含 Brauer-Manin 障碍集的超级强逼近性质。此外, 我们还证明了满足上述条件的代数环面的子簇也满足含 Brauer-Manin 障碍集的超级强逼近性质。

最后, 我们给出了函数域上 Harari-Voloch 猜想的证明概要, 并由此讨论了刘青和徐飞提出的一个新猜想。

关键词: 代数环面; Brauer-Manin 障碍; Harari-Voloch 猜想; 整模型; 超级强逼近

ABSTRACT

This thesis is a survey report on some important results obtained in Qing Liu and Fei Xu's paper [57], and provides an overview of developments of the local-global principle and its Brauer-Manin obstruction. In particular, we discuss the local-global principle to integral points on hyperbolic curves, which is highly related to a conjecture proposed by Harari and Voloch in 2010. Subsequently, we introduce some crucial results in the paper [57] about very strong approximation properties with Brauer-Manin obstruction for algebraic varieties, especially for algebraic tori.

Thanks to the dilatation and smoothening process of integral models, we obtain the main theorem I, the structure theorem of S -adelic spaces of algebraic varieties, which builds a bridge between local approximation properties of rational points on algebraic varieties and those of integral points on their integral models. Moreover, by means of smoothening, we deduce the theorem that given a smooth algebraic variety X over the fraction field of some Dedekind domain R , every integral R -model of X admits a smoothening that preserves all information of local integral points on X .

Without the hypothesis of smoothness, the main theorem II is a further generalization of the main theorem I, and it guarantees the existence of faithfully flat integral models of algebraic varieties. Moreover, we show that every integral model admits a flattening which preserves all integral points.

Based on the above results, we mainly focus on the strong and very strong approximation properties with Brauer-Manin obstruction for certain algebraic varieties. The main theorem III validates that the very strong approximation with Brauer-Manin obstruction holds for algebraic tori over global function fields, and under a rank equality condition over number fields. Furthermore, we show that the very strong approximation with Brauer-Manin obstruction also holds for subvarieties of those algebraic tori.

Lastly, we give a sketch proof of the Harari-Voloch conjecture over global function fields and discuss a new conjecture of Liu and Xu.

Keywords: algebraic tori; Brauer-Manin obstruction; Harari-Voloch conjecture; integral models; very strong approximation

TABLE OF CONTENTS

摘 要.....	I
ABSTRACT	II
LIST OF SYMBOLS AND ACRONYMS.....	V
CHAPTER 1 INTRODUCTION.....	1
1.1 Research background	1
1.2 Literature review	3
1.2.1 Local approximation of rational points	4
1.2.2 Brauer-Manin obstruction and descent obstruction	7
1.2.3 Insufficiency and effectivity of obstructions	9
1.3 Statements of main problem	9
1.3.1 The Harari-Voloch conjecture	9
1.3.2 Very strong approximation.....	11
1.4 Arrangement of thesis	12
CHAPTER 2 BLOWING-UPS AND THE SMOOTH PROCESS.....	14
2.1 Blowing-ups	14
2.2 Henselization and strict henselization	19
2.3 Integral models over DVRs.....	23
2.4 The Smoothing process.....	26
2.5 Dilatation.....	28
CHAPTER 3 INTEGRAL MODELS OF ALGEBRAIC VARIETIES	33
3.1 Integral models over Dedekind domains.....	33
3.2 Structure theorem of S-adelic spaces	38
3.2.1 Adelic points and adelic spaces	38
3.2.2 Main theorem I	39
3.3 Regular proper models and Néron models.....	42
3.3.1 Smooth models	42
3.3.2 Desingularization and regular proper models	45
3.3.3 Néron models	50

TABLE OF CONTENTS

CHAPTER 4 FAITHFULLY FLAT PROPER MODELS	51
4.1 Flatness and scheme-theoretic image.....	51
4.2 Main theorem II	59
CHAPTER 5 VERY STRONG APPROXIMATION FOR ALGEBRAIC TORI	63
5.1 Basic properties of SAP and VSAP.....	63
5.2 Baby case of VSAP	69
5.3 Main theorem III	72
5.4 The Harari-Voloch conjecture over global function fields.....	80
REFERENCES.....	84
ACKNOWLEDGEMENTS	90
致谢.....	91
RESUME AND ACADEMIC ACHIEVEMENTS.....	94

LIST OF SYMBOLS AND ACRONYMS

k	the global field
k^*	the multiplicative group of k
k^s	the separable closure of k
$\bar{k}$	the algebraic closure of k
Ω_k	the set of places of k
k_v	the completion of k at a place v
$\mathcal{O}_v$	the ring of integers of k_v
$\mathcal{O}_{k,v}$	the associated discrete valuation ring of k with respect to v
$R^\times$	the unit group of a commutative ring R
$\mathbf{A}_k$	the adèle ring of k
$\mathbf{A}_k^S$	prime-to- S adèle ring
$\mathbf{I}_k$	the idèle group of k
$\mathbf{I}_{\bar{k}}$	the idèle group of $\bar{k}$
$X(K)$	the set of K -valued points of an algebraic variety X over a field K
$\mathcal{O}_X$	the structure sheaf of a scheme X
$\text{ht}(\mathfrak{p})$	the height of a prime ideal $\mathfrak{p}$
$\tilde{M}$	the sheaf of modules associated to a module M
SAP-BM	strong approximation property with Brauer-Manin obstruction
VSAP-BM	very strong approximation property with Brauer-Manin obstruction

CHAPTER 1 INTRODUCTION

Let k be a global field, Ω_k denote the set of places of k , and ∞_k be the set of archimedean places of k , where $\infty_k = \emptyset$ if k is a global function field. For any non-empty finite subset S of Ω_k containing ∞_k , we denote by $\mathcal{O}_S$ the ring of S -integers. If k is a number field, $\mathcal{O}_k$ denotes the ring of integers of k . For each finite place v , let $\mathcal{O}_{k,v}$ denote the associated discrete valuation ring of k , and $\mathcal{O}_v$ denote its completion with the fraction field $k_v = \text{Frac}(\mathcal{O}_v)$. For each $v \in \infty_k$, we put $\mathcal{O}_v = \mathcal{O}_{k,v} = k_v$.

Unless otherwise stated, an algebraic variety X over k in this thesis refers to a separated scheme of finite type over k . A k -curve means a variety of dimension 1 over k . A k -curve is called **hyperbolic** if it is the complement of a reduced effective divisor of degree $d \geq 0$ on a projective curve of genus g such that $2g - 2 + d > 0$.

Let $\mathbf{A}_k$ be **the adèle ring of k** . It is defined as the restricted product:

$$\mathbf{A}_k = \prod'_{v \in \Omega_k} (k_v, \mathcal{O}_v) = \left(\prod_{v \in S} k_v \right) \times \mathbf{A}_k^S = \left(\prod_{v \in S} k_v \right) \times \prod'_{v \notin S} (k_v, \mathcal{O}_v),$$

where S is any finite subset of Ω_k containing ∞_k and $\mathbf{A}_k^S = \prod'_{v \notin S} (k_v, \mathcal{O}_v)$ is called **prime-to- S adèle ring**. It is well-known that the adèle ring $\mathbf{A}_k$ is a k -algebra via the diagonal embedding $k \rightarrow \mathbf{A}_k$ and its adelic topology has a topological basis given by $\prod_{v \in S} U_v \times \prod_{v \notin S} \mathcal{O}_v$ for all S and all open sets U_v of k_v .

1.1 Research background

Throughout the history of arithmetic geometry, it has always been a core research topic to discuss the behaviors of rational points on a given algebraic variety. For some conics, such as ellipses and hyperbolas, one can easily find all rational points on them. However, when ones consider cubic curves, especially elliptic curves, or higher dimensional varieties, especially abelian varieties, it is closely related to dozens of most challenging problems in modern mathematics, such as the Birch and Swinnerton-Dyer conjecture^{[4] [90] [93]} and Shafarevich–Tate conjecture^[94], which are both still widely open.

Actually, for an algebraic variety X over a field k , it is difficult to decide whether the set $X(k)$ of k -points on X is empty or not, which is equivalent to the Hilbert's tenth prob-

lem^[76] over k , asking whether there is an algorithm that takes a multivariable polynomial $f(x_1, x_2, \dots, x_n) \in k[x_1, x_2, \dots, x_n]$ as input and outputs YES or NO according to the existence of k -points. On the one hand, for the local field k , there is an algorithm when k is of characteristic 0 while it is still an open question for $\mathbb{F}_q((t))$. On the other hand, for the global field k , no such algorithm exists when k is global function field^{[73] [82]}, and it is unknown whether such an algorithm exists when k is number field.

Although one has obtained the existence of k -points on X , the question whether the set $X(k)$ is finite or not has also been concerned naturally, which can be difficult to answer even for a specific and simple-looking equation^[28] defining a K3 surface. Moreover, it remains unknown whether there exists a K3 surface X over a number field such that $X(k)$ is non-empty and finite. However, when k is a number field and X is a smooth projective geometrically integral curve of genus at least 2, Faltings proved^[29] that the set $X(k)$ is finite, which is a conjecture of Mordell in 1922, and instead, when k is a global function field, there are many examples within infinite k -points in [[77], Section 9.5.2].

In other direction, for a projective variety X over a global field k , if $X(k)$ is infinite, one can try to measure the size and predict the rate of the growth of the set $X(k)$ in some way, for example the height function machine. From a more qualitative perspective, one may expect to detect the Zariski density of rational points. One says that the rational points are **potentially dense** on an algebraic variety X if there exists a finite extension L/k such that $X(L)$ is Zariski dense in X_L . It is conjectured by Campana, [[8], Conjecture 9.20], that for a variety X over a number field k , the potential density is equivalent to a certain geometric condition. Besides, if k is a global field and v is a place of k , one may ask if $X(k)$ is dense in $X(k_v)$ with respect to the v -adic analytic topology, within which there is a conjecture of Mazur^[66].

Notably, for a given algebraic variety X over a global field k , it is a strategy to check first if X has a k_v -point for each place v of k in an attempt to check if X has a k -point.

Definition 1.1: Let k be a global field and X be an algebraic variety over k . If $X(k_v) \neq \emptyset$ for each place v of k implies $X(k) \neq \emptyset$, one says that X satisfies **the local-global principle** or **the Hasse principle**.

The local-global principle of rational points has been a subject of investigation for many years. For one thing, since there are many examples scattered in the literature where the local-global principle fails^[1], even for curves^[98] and surfaces^[47], it is natural to ask how to produce and measure obstructions to the local-global principle.

In 1971, Manin analyzed most of the then known counterexamples^[61] by means of the Brauer group and pointed out that the elements of Brauer group could realize obstructions to the local-global principle. Meanwhile, it was shown that a torsor under an algebraic group G could also give rise to an obstruction, by Chevalley and Weil for the finite G ^[12], by Colliot-Thélène and Sansuc for the commutative G ^{[16] [13] [17]}, and by Harari and Skorobogatov for the general G ^[39]. These cohomological obstructions are always called **Brauer-Manin obstructions** in general.

For another thing, rational points on varieties satisfy the Brauer-Manin obstructions, conversely, whether these conditions determine rational points has also been arguably discussed. Here are some results for the projective curves^[88] and abelian varieties^[78].

Turning attention from the geometric setting to more arithmetic one, mathematicians have studied similar behaviors for integral points on algebraic varieties. More precisely, people do not study the algebraic variety X itself directly, but study nice arithmetic models of X , which are always nice schemes over some general Dedekind domain and can suitably recapture all information that we concern of integral points on X . In the early 21th century, Liu, Lorenzini, and Raynaud systematically studied the properties of arithmetic models of curves^{[54] [55]}. In 2009, Colliot-Thélène and Xu^[21] have introduced the integral Brauer-Manin obstruction to the existence of integral points by means of arithmetic models, where they studied integral points on homogenous spaces of connected linear algebraic groups, in regard to the strong approximation properties, without using the Brauer group of integral models but still that of X .

Based on these and inspired by the conjectures of Scharaschkin^[80] and Skorobogatov^[85], that the Brauer-Manin obstruction is the only obstruction to the Hasse principle for rational points on curves defined over number fields, Harari and Voloch conjecture^[41] that the integral Brauer-Manin obstruction is also the only one for affine hyperbolic curves. After that, Liu and Xu^[57] proposed the very strong approximation property for algebraic varieties, and especially studied it for subvarieties of algebraic tori. As a consequence, they proved the Harari-Voloch conjecture for the rational hyperbolic curves over global function fields. It is this very problem that serves as the research focus of the thesis.

1.2 Literature review

Now, in this section, we will recall some crucial results and examples with respect to the local approximation of rational points.

1.2.1 Local approximation of rational points

One of the earliest examples satisfying the Hasse principle is the smooth quadrics, which is well-known as the Hasse-Minkowski theorem [[31], Theorem 5.7], and is central to the theory of quadratic forms. As a consequence, many mathematicians have devoted themselves to finding out the local-global phenomena in arithmetic and algebraic geometry. Until now, there has already been abundant results with respect to the Hasse principle.

Theorem 1.1: In the following cases, one has the Hasse principle.

- (1) **(Minkowski-Hasse^[75] [72](66:1))** Smooth quadrics over global fields.
- (2) **(Châtelet)**. Severi–Brauer varieties over global fields.
- (3) **(Chernousov, Eichler, Harder, Kneser)**. Principal homogenous spaces of simply connected or adjoint semisimple algebraic groups over number fields.
- (4) **(Hooley)**. Smooth projective cubics in $\mathbb{P}_{\mathbb{Q}}^n$, $n \geq 9$.
- (5) **(Colliot-Thélène^[14], Swinnerton-Dyer^[91])**. Del pezzo (Fano) surfaces of degree at least 5 or degree 1 over global fields.
- (6) **(Selmer^[81])**. The diagonal cubic surfaces $ax^3 + by^3 + cz^3 + dz^3 = 0$ in $\mathbb{P}_{\mathbb{Q}}^3$ with the relation $ab = cd$ or $|abcd| \leq 500$, which is a Fano surface of degree 3.
- (7) **(Colliot-Thélène et al.^[15])**. The smooth affine surfaces over $\mathbb{Q}$ defined by the equation

$$y^2 + z^2 = (c - x^2)(x^2 - c + 1) \neq 0,$$

where $c \equiv 1(\text{mod } 4)$ or $c \equiv 2(\text{mod } 4)$.

- (8) **(Colliot-Thélène et al.^[18])**. Smooth proper models of the intersections of two quadrics over number fields in $\mathbb{P}_k^n$, $n \geq 8$.
- (9) **(Colliot-Thélène et al.^[18])**. Châtelet surfaces over number fields given by the equation $y^2 - az^2 = P(x)$, where $P(x)$ is an irreducible polynomial of degree 4.

Remark 1.1: The example (7) in Theorem (1.1) fails^[15] the local-global principle whenever $c \equiv 3(\text{mod } 4)$, and the case $c = 3$ is given by Iskovskih^[46].

This list is very far from being complete and by several different approach, many results are obtained for complete intersections of two quadrics and fibrations into quadrics.

Indeed, there are many examples violating the local-global principle, even as well for Fano varieties that are generally expected to have many rational points. For example, Birch and Swinnerton-Dyer analyzed a del Pezzo surface of degree 4 violating the local-global principle, using the Brauer-Manin obstruction in [[5], Section 4]. Also, for cubic surfaces, the local-global principle is known to be false due to Cassels et al.^[11] and

Swinnerton-Dyer^[92]. After that, Kresch and Tschinkel^[48] obtained many counterexamples for del pezzo surfaces of degree 2, including the surface over $\mathbb{Q}$ defined by a single weighted homogeneous equation $w^2 = -6x^4 - 3y^4 + 2x^4$. Besides, the failure of Hasse principle on general K3 surfaces was elaborated by Hassett and Várilly-Alvarado^[44]. Most of these counterexamples can be explained by Brauer-Manin obstructions.

Application of Lang-Nishimura lemma in [[77], Thm. 3.6.11] shows that Hasse principle is a k -birational invariant property for smooth proper integral k -varieties.

If the Hasse principle holds, one can ask furthermore a stronger condition.

Definition 1.2: One says that **the weak approximation property** holds for an algebraic variety X over a global field k if the image of the following diagonal map

$$X(k) \rightarrow \prod_{v \in \Omega_k} X(k_v)$$

is dense, where $\prod_{v \in \Omega_k} X(k_v)$ is endowed with the product topology.

In fact, by definition of product topology, an algebraic variety X that is everywhere locally solvable satisfies the weak approximation property if and only if for every finite set S of places and any non-empty open sets $U_v \subseteq X(k_v)$ for $v \in S$, there exists $x \in X(k)$ such that $x \in U_v$ for each $v \in S$. In terms of sequences, weak approximation holds for X if and only if for any given finite subset S and local point $P_v \in X(k_v)$ at $v \in S$, one can find a single sequence of rational points $P_n \in X(k)$ that converges to all of P_v at once.

If an algebraic variety X satisfies the weak approximation property, it must have a k -point by definition. However, one should be aware of that it might be easy to obtain the weak approximation property assuming the existence of k -points, but could be hard to obtain the Hasse principle.

Theorem 1.2: The weak approximation property holds for the following cases.

- (1) Smooth rational varieties over number fields.
- (2) **(Harder, Kneser, Platonov; [[74], Thm 3.1])**. Simply connected, adjoint, or almost simple, semisimple algebraic groups over number fields.
- (3) **(Colliot-Thélène^[14], Swinnerton-Dyer^[91])**. Del pezzo surfaces of degree at least 5 over global fields.
- (4) **(Colliot-Thélène et al.^[18])**. Smooth projective intersections of two quadrics with a k -point in $\mathbb{P}_k^n$, $n \geq 6$.
- (5) **(Colliot-Thélène et al.^[18])**. Châtelet surfaces over number fields given by the equation $y^2 - az^2 = P(x)$, where $P(x)$ is an irreducible polynomial of degree 4.

It is famous due to Kneser that the weak approximation property is a birational invariant of smooth integral varieties over number fields. So, in particular, to explore the weak approximation property for a smooth integral variety over number fields, it is enough to consider a non-empty Zariski open subset.

Certainly, many counterexamples failing weak approximation has been found out and analyzed. Actually, there is a criterion, [[20], Proposition 13.3.12], for the failure of weak approximation. For example, a Châtelet surface X over a number field k , which is a smooth projective model of an affine surface defined by the equation $y^2 - az^2 = P(x)$ with $a \in k^* \setminus (k^*)^2$, may fail the weak approximation property, but it has been shown^{[18] [19]} that $X(k)$ is always Zariski dense in the obstruction set $X(\mathbf{A}_k)^{\text{Br}(X)}$. Moreover, Nakahara and Roven^[71] have recently established a complete description for when X fails weak approximation if $P(x)$ splits into linear factors, and as pointed out in [[71], Example 5.3/5.4], the case that $P(x)$ is a product of two irreducible quadratics remains to be described, which might be complicative because of the huge Brauer group.

On the other hand, the failure of weak approximation for subvarieties of abelian varieties is generally expected^[88] to be explained by a finite descent obstruction (1.6).

There is a much stronger approximation than weak approximation if $\prod_{v \in \Omega_k} X(k_v)$ is replaced by the adelic space $X(\mathbf{A}_k)$. However, the adelic density is too strong to hold for many varieties. For example, if X is affine, $X(k)$ is discrete in $X(\mathbf{A}_k)$ since k is discrete in $\mathbf{A}_k$. Thus, a weaken condition is below.

Definition 1.3: One says that **the strong approximation property off a finite subset** $S \subset \Omega_k$ holds for an algebraic variety X if the image of the following diagonal map

$$X(k) \rightarrow X(\mathbf{A}_k^S)$$

is dense, where $X(\mathbf{A}_k^S)$ is endowed with the restricted adelic topology.

Here, we give a remark comparing the following two sets. One is the set $X(\mathbf{A}_k^S)$ of S -adelic points on X and the other is the set $P_S(X(\mathbf{A}_k))$ of adelic points but prime to S , obtained by the projection map $P_S : X(\mathbf{A}_k) \rightarrow \prod_{v \notin S} X(k_v)$.

Remark 1.2: In general, one always has $P_S(X(\mathbf{A}_k)) \subset X(\mathbf{A}_k^S)$. Namely, there is a natural projecting:

$$\begin{aligned} P_S : X(\mathbf{A}_k) &\rightarrow X(\mathbf{A}_k^S) \\ (x_v)_{v \in \Omega_k} &\mapsto (x_v)_{v \notin S} \end{aligned} \tag{1-1}$$

Conversely, $P_S(X(\mathbf{A}_k)) = X(\mathbf{A}_k^S)$ if and only if the projecting map P_S is surjective. In-

deed, the S -adelic point does not require the existence of local points at each place in S , but when it happens, one can lift an S -adelic point to a full adelic point by choosing arbitrary local points at places in S . An example in which the surjectivity of P_S fails is that a quadric in $\mathbb{P}_{\mathbb{Q}}^4$ defined by the equation $x^2 + y^2 + z^2 + w^2 + t^2$, where we take $S = \{\infty_{\mathbb{Q}}\}$.

Remark 1.3: There is also another definition of strong approximation with a slightly different from the one above (1.3) if we replace the $X(\mathbf{A}_k^S)$ in (1.3) by the projecting image $P_S(X(\mathbf{A}_k))$ in (1-1). The two definitions are equivalent if X has rational points. Without the assumption of the existence of rational points, the later one is useful when studying some problems about arithmetic purity.

Theorem 1.3: Strong approximation off some finite subset of Ω_k holds in the following.

- (1) The affine space $\mathbf{A}_k^n$ for any $n \geq 1$.
- (2) (Cao, Wei, and Xu; [[42], Lemma 1.8]) The complement of a closed subset of codimension at least 2 in $\mathbf{A}_k^n$.
- (3) (Harder, Kneser, Platonov) Simply connected semisimple algebraic groups G with a noncompactness condition that for each almost simple factor G_i of G , there exists a place v in the punctured finite set such that $G_i(k_v)$ is noncompact, [[77], Theorem 5.10.6].
- (4) ([77], Remark 5.11) G is a connected affine algebraic group over number fields with the trivial torus $G_{\text{solvable}}/G_{\text{unipotent}}$, and then its semisimple quotient G/G_{solvable} satisfies the criterion of (3).

As for the failure of strong approximation, a criterion in the case of number fields is posed by Minchev and generalized to global fields by Poonen.

Theorem 1.4 (Minchev^[69], Poonen^[77]): Let k be a global field and S be a finite subset of Ω_k . Let $f : Y \rightarrow X$ be a finite étale morphism of geometrically integral k -varieties. If the S -adelic space $X(\mathbf{A}_k^S)$ is non-empty and f is not an isomorphism, then the image of the inclusion $X(k) \rightarrow X(\mathbf{A}_k^S)$ is not dense. In other words, X will not satisfy the strong approximation property with respect to S .

In particular, if X is a normal geometrically integral k -variety whose étale fundamental group $\pi_{\text{ét}}(X_{k^s}) \neq \{1\}$, then X will not satisfy strong approximation with respect to any finite subset S of places of k by a useful lemma of Harari in [[77], Lemma 3.5.57].

1.2.2 Brauer-Manin obstruction and descent obstruction

Around 1970, Manin presented a way to explain counterexamples to the local-global principle, known at the time, in a unified manner, whose idea is that each nontrivial Brauer class should be responsible for the failure of local-global principle.

Let $\mathrm{Br}(X) = H_{\mathrm{\acute{e}t}}^2(X, \mathbb{G}_m)$ and $\xi \in \mathrm{Br}(X)$. For any $(x_v)_{v \in \Omega_k} \in X(\mathbf{A}_k)$, one has $\mathrm{inv}_v(\xi(x_v)) = 0$ for all but finitely many places $v \in \Omega_k$ by [[77], Proposition 8.2.1], where $\mathrm{inv}_v : \mathrm{Br}(k_v) \rightarrow \mathbb{Q}/\mathbb{Z}$ is the local invariant map from the local class field theory and $\xi(x_v) \in \mathrm{Br}(k_v)$ is the value of $\mathrm{ev}_\xi : X(k_v) \rightarrow \mathrm{Br}(k_v)$ at x_v . Thus ξ defines a map

$$\begin{aligned} \mathrm{ev}_\xi : X(\mathbf{A}_k) &\rightarrow \mathbb{Q}/\mathbb{Z} \\ (x_v)_{v \in \Omega_k} &\mapsto (\xi, (x_v)) := \sum_{v \in \Omega_k} \mathrm{inv}_v(\xi(x_v)). \end{aligned}$$

Definition 1.4: There is a natural pairing between $X(\mathbf{A}_k)$ and the Brauer group $\mathrm{Br}(X)$

$$\begin{aligned} X(\mathbf{A}_k) \times \mathrm{Br}(X) &\rightarrow \mathbb{Q}/\mathbb{Z} \\ ((x_v), \xi) &\rightarrow \sum_{v \in \Omega_k} \mathrm{inv}_v(\xi(x_v)). \end{aligned} \tag{1-2}$$

The above pairing is known as the **Brauer-Manin pairing**. The left kernel of this pairing is called **Brauer-Manin set**, which is defined as

$$X(\mathbf{A}_k)^{\mathrm{Br}} = \left\{ (x_v)_{v \in \Omega_k} \in X(\mathbf{A}_k) \left| \sum_{v \in \Omega_k} \mathrm{inv}_v(\xi(x_v)) = 0 \text{ for all } \xi \in \mathrm{Br}(X) \right. \right\}.$$

It is a quick result of the famous Brauer-Hasse-Noether exact sequence, [[20], Theorem 13.1.8], that $X(k)$ is contained in $X(\mathbf{A}_k)^{\mathrm{Br}}$.

Definition 1.5: If $X(\mathbf{A}_k) \neq \emptyset$ but $X(\mathbf{A}_k)^{\mathrm{Br}} = \emptyset$, then one says that **there is a Brauer-Manin obstruction to the Hasse principle(or the local-global principle) for X** .

Let G be a smooth algebraic group over k and $f : Z \rightarrow X$ be a fppf G_X -torsor over X , where $G_X = G \times_k X$. Then, one can constrain the locations of rational points by torsors.

Definition 1.6: We define **the descent set** of X as follows

$$X(\mathbf{A}_k)^{\mathrm{descent}} = \bigcap_{\text{all smooth affine } G} \left(\bigcap_{\text{all } G\text{-torsor } f : Z \rightarrow X} \left(\bigcup_{\tau \in H^1(k, G)} f^\tau(Z^\tau(\mathbf{A}_k)) \right) \right),$$

where $H^1(k, G)$ denotes the pointed set $\check{H}_{\mathrm{fppf}}^1(X, G)$ and $f^\tau : Z^\tau \rightarrow X$ is the twisted torsor.

We can obtain a possibly better upper bound obstruction set to approximate $X(k)$ further by replacing $Z^\tau(\mathbf{A}_k)$ by the Brauer-Manin set $Z^\tau(\mathbf{A}_k)^{\mathrm{Br}}$.

Definition 1.7 (Skorobogatov): One can define **the étale-Brauer set** as follows

$$X(\mathbf{A}_k)^{\mathrm{\acute{e}t, Br}} = \bigcap_{\substack{\text{finite étale } G \\ \text{all } G\text{-torsor } f : Z \rightarrow X}} \left(\bigcup_{\tau \in H^1(k, G)} f^\tau(Z^\tau(\mathbf{A}_k)^{\mathrm{Br}}) \right).$$

A priori, the **étale-descent set** is defined as

$$X(\mathbf{A}_k)^{\text{ét,descent}} = \bigcap_{\substack{\text{finite étale } G \\ \text{all } G\text{-torsor } f : Z \rightarrow X}} \left(\bigcup_{\tau \in H^1(k, G)} f^\tau(Z^\tau(\mathbf{A}_k)^{\text{descent}}) \right).$$

For the comparison of above refinements, there is a result obtained nearly.

Theorem 1.5 (Cao^[9], Demarche, Harari, Skorobogatov, Stoll): Let k be a number field and X be a smooth projective geometrically integral k -variety X . Then

$$X(\mathbf{A}_k)^{\text{ét,Br}} = X(\mathbf{A}_k)^{\text{ét,descent}} = X(\mathbf{A}_k)^{\text{descent}} = X(\mathbf{A}_k)^{\text{descent,descent}}$$

where the last equality is due to Cao^[9] only assuming that X is quasi-projective over k .

1.2.3 Insufficiency and effectivity of obstructions

On the one hand, in 1999, Skorobogatov^[86] proved that the Brauer-Manin obstruction is insufficient to explain all counterexamples to the local-global principle. Almost one decade ago, Poonen et al.^[78] constructed the first examples for which Skorobogatov's étale Brauer-Manin obstruction does not explain the failure of local-global principle.

Theorem 1.6 (Poonen^[78]): There exists a smooth projective geometrically integral variety over $\mathbb{Q}$ such that $X(\mathbf{A}_k)^{\text{ét,Br}} \neq \emptyset$ but $X(\mathbb{Q}) = \emptyset$.

On the other hand, one can have a procedure attempting to decide whether X has a k -point. Namely, search for k -points while searching for a set that obstructs k -points.

Definition 1.8: For an algebraic variety X over a global field k , one says that **the Brauer-Manin obstruction to the Hasse principle is the only one for X** if the following implication holds

$$X(\mathbf{A}_k)^{\text{Br}} \neq \emptyset \implies X(k) \neq \emptyset.$$

If the Brauer-Manin obstruction is the only one, then above procedure terminates successfully. However, the computation of the obstruction sets appears to be substantially hard even though more reasonable algorithms and a kind of finite description of $X(\mathbf{A}_k)^{\text{Br}}$ have been obtained by Kresch and Tschinkel^[49] under some additional assumptions.

1.3 Statements of main problem

1.3.1 The Harari-Voloch conjecture

One of the purposes of this thesis is to discuss the Brauer-Manin obstruction for integral points on affine curves, with special regard to the following conjecture^[41] proposed

by Harari and Voloch in 2010.

Conjecture 1.1 (Harari-Voloch 2010): Let $X \subset \mathbb{P}_k^1$ be an affine hyperbolic curve, which is an open subscheme of $\mathbb{P}_k^1$ with the complement being a reduced separable divisor of degree at least 3. Let S be a finite subset of Ω_k containing ∞_k . Let $\mathbf{X}$ be a flat model of X over $\mathcal{O}_S$. Then the subset of $\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v)$ consisting of those points which are orthogonal to the group $B_S(X)$ under the following Brauer-Manin-like pairing (1-3) coincides with the image of $\mathbf{X}(\mathcal{O}_S)$ in $\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v)$, where

$$B_S(X) = \ker \left(\text{Br}(X) \rightarrow \prod_{v \in S} \text{Br}(X \times_k k_v) / \text{Br}(k_v) \right) \subset \text{Br}(X),$$

and the Brauer-Manin-like pairing is

$$\begin{aligned} \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \times B_S(X) &\rightarrow \mathbb{Q}/\mathbb{Z} \\ ((x_v), \xi) &\mapsto \sum_{v \in \Omega_k} \text{inv}_v(\xi(x_v)). \end{aligned} \tag{1-3}$$

In other words, the following diagonal map

$$\mathbf{X}(\mathcal{O}_S) \rightarrow \left(\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \right)^{B_S(X)}$$

is bijective.

Remark 1.4: In the conjecture (1.1), the Brauer-Manin-like pairing (1-3) is induced by the integral Brauer-Manin pairing (1-4) as follows

$$\begin{aligned} \left(\prod_{v \in S} X(k_v) \times \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \right) \times B_S(X) &\rightarrow \mathbb{Q}/\mathbb{Z} \\ ((P_v), \alpha) &\mapsto \sum_{v \in \Omega_k} \text{inv}_v(\alpha(P_v)), \end{aligned} \tag{1-4}$$

where for each $v \in S$, $P_v \in X(k_v)$ is chosen arbitrarily. Indeed, denoting the projection map by $\pi_v : X \times_k k_v \rightarrow X$ and its pull-back by $\pi_v^* : \text{Br}(X) \rightarrow \text{Br}(X \times_k k_v)$, since $P_v \in X(k_v) = X_{k_v}(k_v)$ and $\pi_v^*(\alpha) \in \text{Br}(k_v) \subseteq \text{Br}(X_{k_v})$ by definition of $B_S(X)$, one has

$$\alpha(P_v) = \alpha(\pi_v(P_v)) = \pi_v^*(\alpha)(P_v) = \pi_v^*(\alpha) \stackrel{\Delta}{=} \alpha_v \in \text{Br}(k_v) \subseteq \text{Br}(X_{k_v}),$$

which implies that the restriction of $B_S(X)$ to $\text{Br}(X_{k_v})$ is constant^①. Thus, by [[20], Proposition 13.3.1], the pairing (1-4) is well-defined and so is the pairing (1-3).

① Call the Brauer group $\text{Br}(X)$ of a k -variety X be **constant** if the canonical map $\text{Br}(k) \rightarrow \text{Br}(X)$ is surjective.

Let us define $B_S^0(X) = \ker(\text{Br}(X) \rightarrow \prod_{v \in S} \text{Br}(X_{k_v}))$, which is a subgroup of $B_S(X)$. It is claimed that $(P_v)_{v \in \Omega_k}$ is orthogonal to $B_S(X)$ if and only if $(P_v)_{v \notin S}$ is orthogonal to $B_S^0(X)$, namely $\sum_{v \notin S} \text{inv}_v(\alpha(P_v)) = 0$ for each $\alpha \in B_S^0(X)$. More precisely, denoting $C(\alpha) = \sum_{v \in S} \text{inv}_v(\alpha_v)$, it is obvious that the pairing $((P_v)_{v \in \Omega_k}, \alpha)$ can be divided into

$$((P_v)_{v \in \Omega_k}, \alpha) = \sum_{v \in \Omega_k} \text{inv}_v(\alpha(P_v)) = C(\alpha) + \sum_{v \notin S} \text{inv}_v(\alpha(P_v)).$$

If $(P_v)_{v \in \Omega_k}$ is orthogonal to $B_S(X)$, one must have $((P_v)_{v \in \Omega_k}, \alpha) = 0$ for each $\alpha \in B_S^0(X)$. By definition of $B_S^0(X)$, we must have $C(\alpha) = 0$, which implies $\sum_{v \notin S} \text{inv}_v(\alpha(P_v)) = 0$. Conversely, if $(P_v)_{v \notin S}$ is orthogonal to $B_S^0(X)$, by [[20], Remark 13.3.5], one has $B_S(X) = B_S^0(X) \oplus \text{Br}(k)$, and then for each $\alpha \in B_S(X)$, one can write $\alpha = \beta + \gamma$ for some $\beta \in \text{Br}(k)$ and $\gamma \in B_S^0(X)$. Thus, the pair

$$((P_v)_{v \in \Omega_k}, \alpha) = ((P_v), \beta) + ((P_v), \gamma) = \sum_{v \in \Omega_k} \text{inv}_v(\beta(P_v)) = 0,$$

where the last equality follows from the Brauer-Hasse-Noether exact sequence. Thus,

$$\left(\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \right)^{B_S(X)} = \left\{ (P_v) \in \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \mid \sum_{v \notin S} \text{inv}_v(\alpha(P_v)) = 0 \text{ for each } \alpha \in B_S^0(X) \right\}.$$

Moreover, if $g(X) = 0$, the group $B_S(X)$ can be described by the generalized Jacobian T of X in [[41], Theorem 1], and $(P_v)_{v \notin S} \in \overline{\mathcal{T}(\mathcal{O}_S)}$ if and only if $(P_v)_{v \notin S}$ is orthogonal to $B_S(X)$.

It is obvious to see that $\mathbf{X}(\mathcal{O}_S) \subset (\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v))^{B_S(X)}$. When X is an affine curve over a number field k , the set of integral points $\mathbf{X}(\mathcal{O}_S)$ is finite for any integral model $\mathbf{X}$ of X over $\mathcal{O}_S$ by Faltings^[29] and Siegel.

1.3.2 Very strong approximation

The conjecture (1.1) is linked to a notion firstly proposed by Qing Liu and Fei Xu in 2015, and see [[57], Definition 1.1]. The thesis will discuss this notion and related topics.

Definition 1.9: Let X be a k -variety. Let $P_S : X(\mathbf{A}_k) \rightarrow X(\mathbf{A}_k^S)$ be the projection map. One says that X **satisfies the strong approximation property with Brauer-Manin obstruction off S** (abbreviated to SAP-BM off S) if $X(k)$ is dense in $P_S[X(\mathbf{A}_k)^{\text{Br}}]$ under the diagonal map. In other words, each non-empty open subset of $P_S[X(\mathbf{A}_k)^{\text{Br}}]$ must contain some rational point of X . One says that X **satisfies the very strong approximation**

property with Brauer-Manin obstruction off S (abbreviated to VSAP-BM off S) if

$$X(k) = P_S[X(\mathbf{A}_k)^{\text{Br}}].$$

Definition (1.9) generalizes the classical one (1.3) since the two definitions coincide when $\text{Br}(X)$ is constant and P_S is surjective, where $X(\mathbf{A}_k)^{\text{Br}} = X(\mathbf{A}_k)$.

Remark 1.5: As pointed out in the theorem (5.3), the conjecture (1.1) implies VSAP-BM off S for the affine hyperbolic curve X . Indeed, the conjecture (1.1) holds for each flat model $\mathbf{X}$ of X over $\mathcal{O}_S$ if and only if the following diagonal map

$$X(k) \rightarrow X(\mathbf{A}_k^S)^{\text{Bs}(X)}$$

is bijective. It suffices to see

$$X(k) \subset P_S[X(\mathbf{A}_k)^{\text{Br}(X)}] \subset P_S[X(\mathbf{A}_k)^{\text{Bs}(X)}] \subset X(\mathbf{A}_k^S)^{\text{Bs}(X)},$$

where the last step follows from the remark (1.4). If Conjecture (1.1) holds for X , then $X(k) = X(\mathbf{A}_k^S)^{\text{Bs}(X)}$, and hence VSAP-BM off S holds for X .

In Harari and Voloch's paper^[41], they showed that Conjecture (1.1) fails for some elliptic curve over $\mathbb{Q}$ minus one point. After that, Liu and Xu^[57] proved that the conjecture (1.1) holds for the rational affine hyperbolic curves over global function fields. Meanwhile, inspired by the adelic Mordell-Lang conjecture in [Question 3.12, [88]], they proposed a new conjecture, which can imply the conjecture (1.1) over number fields.

Conjecture 1.2 (Liu and Xu): Let k be a global field. Let X be a rational hyperbolic curve over k and $j : X \rightarrow T$ be the closed immersion into a k -torus T from theorem (5.8). If G is a finitely generated subgroup of $T(k)$ and $\bar{G}^S$ is the topological closure of G inside $T(\mathbf{A}_k^S)$, then there is a finite subscheme $Z \subseteq X$ such that $X(\mathbf{A}_k^S) \cap \bar{G}^S \subseteq Z(\mathbf{A}_k^S)$.

1.4 Arrangement of thesis

As an introduction, Chapter 1 provides an overview of the thesis background, domestic and international research status, and the paper's overall structure.

Chapter 2 is about blowing-ups and the smooth process, where we primarily recall the fundamental concepts and key properties of the smoothening process and dilatation of schemes over discrete valuation rings, laying the necessary foundational groundwork for the subsequent discussions in the thesis.

Chapter 3 discusses integral models of algebraic varieties, which serves as versatile tool to make us closer to integral points in geometry. Section 3.2 will verify a useful

property (3.6) to approximate the integral points on algebraic varieties, which plays an important role in the proof of the main theorem I—the structure theorem of S -adelic spaces.

In Chapter 4, we study faithfully flat (proper) models of (proper) algebraic varieties. The main theorem II, that every algebraic variety over the fraction field of some Dedekind domain admits a faithfully flat integral model, will be proved by means of scheme-theoretic image.

Some examples satisfying the very strong approximation properties will be presented in Chapter 5. In particular, the main theorem III, that algebraic tori over global function fields satisfy VSAP-BM and that algebraic tori under a stricter equality condition over number fields satisfy VSAP-BM, will be obtained. After that, the Harari-Voloch conjecture over global function fields will be validated for rational hyperbolic affine curves. In the end, inspired by the adelic Mordell-Lang conjecture, we will discuss a new conjecture (1.2) proposed by Liu and Xu about algebraic tori.

Although most of the research work in this thesis inspired by [41] and [57], the author has also integrated his own thoughts and innovative achievements. More precisely,

- (1) We establish a gluing lemma (3.2) for schemes over some Dedekind domain, by which we get Proposition (3.5) that good reductions induce smooth proper models.
- (2) Inspired by [[57], Remark 2.6], we attain a global version of smoothening process (3.3), and discuss some oversight that might occur in remarks (3.4) and (3.5).
- (3) Besides, with the observations of Harari et al.^[41] and Liu et al.^[57], as well as discussions with my advisor Professor Hu, we propose Proposition (5.1) that VSAP-BM is equivalent to SAP-BM if the set of S -integral points on any integral model is finite.
- (4) Moreover, we obtain an important remark (1.4) on Harari-Voloch conjecture and remarks (1.2) (1.3) on two definitions of strong approximation properties.

CHAPTER 2 BLOWING-UPS AND THE SMOOTH PROCESS

2.1 Blowing-ups

During the study of arithmetic geometry, most of the singularities that appear can be resolved by blowing-up, which is a fundamental notion of birational morphisms. Here, we define the blowing-up as a morphism associated to a graded coherent sheaf.

Definition 2.1: Let X be a locally Noetherian scheme. A **graded $\mathcal{O}_X$ -algebra** $\mathcal{B}$ is a quasi-coherent sheaf of $\mathcal{O}_X$ -algebra with a grading $\mathcal{B} = \bigoplus_{n \geq 0} \mathcal{B}_n$, where $\mathcal{B}_n$ are quasi-coherent sub- $\mathcal{O}_X$ -modules of $\mathcal{B}$. One says that a graded $\mathcal{O}_X$ -algebra $\mathcal{B}$ is **homogenous $\mathcal{O}_X$ -algebra** if $\mathcal{B}_1$ is coherent sheaf and $\mathcal{B}_n$ is locally generated by $\mathcal{B}_1$ as an $\mathcal{O}_X$ -algebra, which means $\mathcal{B}_1^n = \mathcal{B}_n$ for each $n \geq 1$. If $\mathcal{B}$ is homogenous, then every homogenous part $\mathcal{B}_n$ is also coherent.

Example 2.1: Let X be a locally Noetherian scheme. Let $\mathcal{I}$ be a coherent sheaf of ideals of $\mathcal{O}_X$. Then $\mathcal{B} = \bigoplus_{n \geq 0} \mathcal{I}^n$ is a homogenous $\mathcal{O}_X$ -algebra.

Lemma 2.1: Let X be a scheme and $\mathcal{B}$ be a graded $\mathcal{O}_X$ -algebra. Then there exists a unique X -scheme $\pi : \text{Proj}(\mathcal{B}) \rightarrow X$ such that for any open affine subscheme U of X , there is a U -isomorphism $\pi_U : \pi^{-1}(U) \xrightarrow{\sim} \text{Proj}(\mathcal{B}(U))$ that is compatible with the restriction map $\text{Res}_V^U : U \rightarrow V$ for any open affine subscheme $V \subseteq U$.

Proof: The uniqueness comes from the definition directly. For the existence, we can assume that X is an affine scheme firstly and in this case we define $\text{Proj}(\mathcal{B}) = \text{Proj}(\mathcal{B}(X))$. For any affine open subscheme V of X , one has:

$$\text{Proj}(\mathcal{B}(V)) = \text{Proj}(\mathcal{B}(X) \otimes_{\mathcal{O}_X(X)} \mathcal{O}_X(V)) \simeq (\text{Proj}(\mathcal{B}(X))) \times_X V.$$

where the first equality comes from the compatibility of the restriction map $\mathcal{B}(X) \rightarrow \mathcal{B}(V)$ with module structure via the ring homomorphism $\mathcal{O}_X(X) \rightarrow \mathcal{O}_X(V)$ and the second one follows from the universal property of fiber products. In the general case, one can cover X with open affine subschemes X_i and define $\text{Proj}(\mathcal{B}|_{X_i}) = \text{Proj}(\mathcal{B}|_{X_i}(X_i))$. It is $\text{Proj}(\mathcal{B})$ that the glueing process will give along the unique pieces $\text{Proj}(\mathcal{B}|_{X_i \cap X_j})$, which exist actually since as long as $\text{Proj}(\mathcal{B})$ makes sense, then $\text{Proj}(\mathcal{B}|_W)$ exists for any open subscheme W of X with $\text{Proj}(\mathcal{B}|_W) = \text{Proj}(\mathcal{B}) \times_X W$. ■

Now, let me recall some facts about the sheaf of modules and twists, which play

an essential role in the study of quasi-coherent sheaves on projective schemes. Let $B = \bigoplus_{n \geq 0} B_n$ be a graded ring, and $M = \bigoplus_{n \in \mathbb{Z}} M_n$ be a graded B -module, namely $B_n M_m \subseteq M_{n+m}$ for every $n \geq 0$ and $m \in \mathbb{Z}$. For any non-nilpotent $f \in B_+ = \bigoplus_{d > 0} B_d$, we define the abelian group of elements of degree zero in the localized B_f -module M_f :

$$M_{(f)} = \{mf^{-d} \mid m \in M_{d \deg(f)}\}$$

which is a $B_{(f)}$ -module in a natural way. Moreover, if R is a ring and B is a graded R -algebra, for any $n \in \mathbb{Z}$, we define **the n -th twist A -algebra of B** as:

$$B(n) = \bigoplus_{d \geq 0} B_d(n) = \bigoplus_{d \geq 0} B_{d+n}.$$

If $X = \text{Proj}(B)$, we will denote the sheaf of $\mathcal{O}_X$ -modules $\widetilde{B(n)}$ by $\mathcal{O}_X(n)$.

Corollary 2.1: Let X be a scheme, $\mathcal{B}$ be a graded $\mathcal{O}_X$ -algebra. Then there exists a canonical sheaf on $Y = \text{Proj}(\mathcal{B})$, which is denoted by $\mathcal{O}_Y(1)$. If $\mathcal{B}$ is homogenous $\mathcal{O}_X$ -algebra, then $\mathcal{O}_Y(1)$ is an invertible sheaf.

Proof: Let U be an open affine subscheme of X . Then, $Y_U = \text{Proj}(\mathcal{B}(U))$ is endowed with a canonical sheaf $\mathcal{O}(1) = \widetilde{\mathcal{B}(U)}(1)$, where $\mathcal{B}(U)$ is a graded $\mathcal{O}_X(U)$ -module. One can glue these sheaves together when U runs through all open affine subschemes of X . As a consequence, the gluing sheaf is exactly what we think, and denoted by $\mathcal{O}_Y(1)$. When $\mathcal{B}$ is a homogenous $\mathcal{O}_X$ -algebra, for each open affine subscheme U of X , $\mathcal{O}_Y(1)|_{Y_U} = (\widetilde{\mathcal{B}(U)}(1))$ is an invertible sheaf and so is $\mathcal{O}_Y(1)$. This is because for any homogenous element $f \in \mathcal{B}(U)$ of degree 1, when localized at f , one always has the equality $\mathcal{B}(U)_{(f)}(n) = f^n \mathcal{B}(U)_{(f)}$ and then on the affine open subset $D_+(f) = \text{Spec}(\mathcal{B}(U)_{(f)})$, one has $\mathcal{O}_Y(n)|_{D_+(f)} = f^n \mathcal{O}_Y|_{D_+(f)}$, which is free of rank 1 over $D_+(f)$, while the homogeneity of $\mathcal{B}$ means that $\mathcal{B}(U)$ can be covered by $D_+(f)$ with only f of degree 1. ■

Definition 2.2: Let X be a locally Noetherian scheme and $\mathcal{I}$ be a coherent sheaf of ideals. The **blowing-up of X with center (or along) $V(\mathcal{I}) = \{x \in X \mid \mathcal{I}_x \neq \mathcal{O}_{X,x}\}$** refers to a scheme over X with a morphism $\text{Proj}(\bigoplus_{n \geq 0} \mathcal{I}^n) \rightarrow X$. For the convenience, the scheme $\text{Proj}(\bigoplus_{n \geq 0} \mathcal{I}^n)$ is always denoted by $\widetilde{X}$.

Lemma 2.2: If one replaces the sheaf of ideals $\mathcal{I}$ by its arbitrary power $\tilde{\mathcal{I}} = \mathcal{I}^k$, where $k \in \mathbb{N}$, then the blowing up will not change up to isomorphism with $V(\mathcal{I}) = V(\tilde{\mathcal{I}})$.

Proof: It suffices to consider the affine open local pieces. Take an affine open subscheme U of X , and then the blowing-up $\widetilde{X}$ of X restricted to U is $\text{Proj}((\bigoplus_{n \geq 0} \mathcal{I}^n)(U))$. It remains to discuss by commutative algebra. Let $B = (\bigoplus_{n \geq 0} \mathcal{I}^n)(U)$, which is a graded ring and can be written as $B = \bigoplus_{d \geq 0} B_d$, with $B_d = \mathcal{I}^d(U)$. Let $k \geq 1$ be an integer. Consider $C =$

$\bigoplus_{\geq 0} C_d$, where $C_d = B_{kd}$, which denotes the k -th power of B . We claim that $\text{Proj}(B) \simeq \text{Proj}(C)$. Namely, for any homogenous element $f \in B_+$, by [[53], proposition 2.3.38], one has $D_+^B(f) = \{p \in \text{Proj}(B) \mid f \notin p\} \simeq \text{Spec}(B_{(f)})$. However, $\text{Spec}(B_{(f)})$ is isomorphic to $\text{Spec}(C_{(f^k)})$ by the k -th power map from B to C . Thus, $D_+^B(f) \simeq D_+^C(f^k)$. ■

Example 2.2: Let $X = \text{Spec}(k[x, y])$, where k is a field. Let $I_1 = (x, y)$ and $I_2 = (x^2, y^2, xy)$. The blowing-ups of X along $V(I_2)$ is the same as that of $V(I_1)$.

Now, we list some of crucial properties of blowing-ups but without given proof.

Proposition 2.1: Let X be a locally Noetherian scheme and $\mathcal{I}$ be a quasi-coherent sheaf of ideals on X . Let $\pi : \tilde{X} \rightarrow X$ be the blowing-up of X with center $V(\mathcal{I})$. Then:

- (1) π is an isomorphism if and only if $\mathcal{I}$ is an invertible sheaf on X .
- (2) π is a proper morphism. (acutally projective, and see [[34], Proposition 13.96])
- (3) π induces an isomorphism $\pi^{-1}(X \setminus V(\mathcal{I})) \rightarrow X \setminus V(\mathcal{I})$.
- (4) If X is integral and $\mathcal{I} \neq 0$, then $\tilde{X}$ is integral, and π is birational and surjective.
- (5) The image of $\pi^*\mathcal{I} \rightarrow \pi^*\mathcal{O}_X = \mathcal{O}_{\tilde{X}}$ in $\mathcal{O}_{\tilde{X}}$, which is denoted by $\mathcal{JO}_{\tilde{X}}$ or $(\pi^{-1}\mathcal{I})\mathcal{O}_{\tilde{X}}$, is equal to $\mathcal{O}_{\tilde{X}}(1)$. In particular, $\mathcal{JO}_{\tilde{X}}$ is an invertible sheaf on $\tilde{X}$.
- (6) If X is affine, then $\mathcal{JO}_{\tilde{X}} = \mathcal{O}_{\tilde{X}}(1)$ is a very ample line bundle relative to π .
- (7) In general, $\mathcal{JO}_{\tilde{X}} = \mathcal{O}_{\tilde{X}}(1)$ is a very ample line bundle relative to π .
- (8) Blowing-ups commute with the flat base extension. Let $Z \rightarrow X$ be a flat morphism with Z locally Noetherian and $\tilde{\pi} : \tilde{Z} \rightarrow Z$ be the blowing-up of Z along $V(\mathcal{JO}_Z)$. Then one has $\tilde{Z} \simeq \tilde{X} \times_X Z$.

Proof: see [[53], Proposition 8.1.12], [[34], Proposition 13.96 and Corollary 13.97]. ■

Corollary 2.2: Let X be a locally Noetherian scheme, $\mathcal{I}$ be a coherent sheaf of ideals. Then for any invertible sheaf $\mathcal{L} \subseteq \mathcal{O}_X$, the coherent sheaf $\mathcal{L} \otimes \mathcal{I} = \mathcal{LI}$ defines the same blowing-up as $\mathcal{I}$.

Proof: It suffices to consider the open affine local pieces and apply the proposition (2.1). ■

Blowing-ups have a universal property, which is useful when a scheme is compared before and after its blowing-up.

Proposition 2.2 (Universal property of blowing-ups): Let $f : W \rightarrow X$ be a morphism of locally Noetherian schemes and $\mathcal{I}$ be a quasi-coherent sheaf of ideals on X . Let $\mathcal{J} = (f^{-1}\mathcal{I})\mathcal{O}_W$ be the image of $f^*\mathcal{I} \rightarrow f^*\mathcal{O}_X = \mathcal{O}_W$ in $\mathcal{O}_W$, which is a quasi-coherent sheaf of ideals on W . Let $\pi : \tilde{X} \rightarrow X$ and $\rho : \tilde{W} \rightarrow W$ be the blowing-ups of X and W along the closed subschemes $V(\mathcal{I})$ and $V(\mathcal{J})$ respectively. Then, there exists a unique

morphism $\tilde{f} : \widetilde{W} \rightarrow \widetilde{X}$ such that the following commutative diagram

$$\begin{array}{ccc} \widetilde{W} & \xrightarrow{\tilde{f}} & \widetilde{X} \\ \downarrow \rho & & \downarrow \pi \\ W & \xrightarrow{f} & X \end{array}$$

holds.

In particular, if $W \rightarrow X$ is a closed immersion and the image of W is not contained in $V(\mathcal{I})$, then the morphism $\widetilde{W} \rightarrow \widetilde{X}$ is also a closed immersion. One calls the closed subscheme $\widetilde{W}$ **the strict transform of W** , which is the Zariski closure of $\pi^{-1}(W \setminus V(\mathcal{I}))$ in $\widetilde{X}$ from a set-theoretical point of view.

Proof: See [[53], Proposition 8.1.15 and Corollary 8.1.17]. ■

Here is a widely useful version of Proposition (2.2).

Corollary 2.3: Let X be a locally Noetherian scheme and $\mathcal{I}$ be a quasi-coherent sheaf of ideals on X . Let $\pi : \widetilde{X} \rightarrow X$ be the blowing-up of X along $V(\mathcal{I})$. For any morphism $f : W \rightarrow X$ between locally Noetherian schemes, if $(f^{-1}\mathcal{I})\mathcal{O}_W$ is an invertible sheaf on W , then there exists a unique X -morphism $\tilde{f} : W \rightarrow \widetilde{X}$ such that the following commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{\tilde{f}} & \widetilde{X} \\ & \searrow f & \downarrow \pi \\ & & X \end{array}$$

holds.

Proof: This follows from Proposition (2.2) quickly. Indeed, by Proposition (2.1), the blowing-up of W along the center $V((f^{-1}\mathcal{I})\mathcal{O}_W)$ is isomorphic to W if and only if $(f^{-1}\mathcal{I})\mathcal{O}_W$ is an invertible sheaf. ■

By lemma (4.5), for a given sheaf of ideals $\mathcal{I}$, there is a natural closed subscheme structure on the closed subset $V(\mathcal{I})$. The blowing up of X depends on the closed subscheme structure of $V(\mathcal{I})$ instead of only the closed subset $V(\mathcal{I})$. More precisely, blowing up along the different sheaf of ideals with the same underlying space might give the different results.

Lemma 2.3: Let $X = \text{Spec}(A)$ be a Noetherian integral affine scheme, and $\mathcal{I}$ be an ideal of A . If the ideal $\mathcal{I}$ is generated by finite elements $f_1, f_2, \dots, f_n$, then the blowing-ups of

$\tilde{X} \rightarrow X$ along $V(\tilde{\mathcal{I}})$ is the union of the affine open subschemes $\text{Spec}(A_i)$, $1 \leq i \leq n$, where A_i is the sub- A -algebra $A[\frac{f_1}{f_i}, \dots, \frac{f_n}{f_i}]$ of $\text{Frac}(A)$ generated by $f_j f_i^{-1}$, $1 \leq j \leq n$.

Proof: It follows from the construction of blowing-ups and see [[53], Lemma 8.1.4]. ■

Example 2.3: Let k be a field with $\text{Char}(k) \neq 2$ and $X = \text{Spec}(k[x, y])$. Consider the ideals $\mathcal{I} = (x, y)$ and $\tilde{\mathcal{I}} = (x^2, y)$. Then the blowing-ups $\pi_1 : \tilde{X}_1 \rightarrow X$, $\pi_2 : \tilde{X}_2 \rightarrow X$ along $V(\mathcal{I})$, $V(\tilde{\mathcal{I}})$ respectively are not isomorphic. Moreover, there does not even exist any morphism of X -schemes from one to the other even though they are birational.

Proof: By the local explicit formula (2.3) of the blowing-ups, one can write $\tilde{X}_1$ as the union of two open affine pieces as follows:

$$\begin{aligned}\tilde{X}_1(z) &= \text{Spec}\left(k[x, y, \frac{y}{x}]\right) = \text{Spec}\left(\frac{k[x, y, z]}{(y - xz)}\right) \\ \tilde{X}_1(w) &= \text{Spec}\left(k[x, y, \frac{x}{y}]\right) = \text{Spec}\left(\frac{k[x, y, w]}{(x - yw)}\right)\end{aligned}$$

Similarly, one can also write $\tilde{X}_2$ as the union of two open affine pieces as follows:

$$\begin{aligned}\tilde{X}_2(z) &= \text{Spec}\left(k[x, y, \frac{y}{x^2}]\right) = \text{Spec}\left(\frac{k[x, y, z]}{(y - x^2 z)}\right) \\ \tilde{X}_2(w) &= \text{Spec}\left(k[x, y, \frac{x^2}{y}]\right) = \text{Spec}\left(\frac{k[x, y, w]}{(x^2 - yw)}\right)\end{aligned}$$

By the Jacobian criterion, it is easy to see that $\tilde{X}_1$ is smooth over k while $\tilde{X}_2$ has a singularity at the piece $\tilde{X}_2(w)$. Hence, $\tilde{X}_1$ is not isomorphic to $\tilde{X}_2$ even though they are birational to each other. Indeed, there are no X -morphism from $\tilde{X}_1$ to $\tilde{X}_2$, and vice versa. To show this, by the universal property of blowing-up (2.3), it suffices to show the ideal sheaf $\tilde{\mathcal{I}}$ is not invertible in $\tilde{X}_2$, and the ideal sheaf $\mathcal{I}$ is not invertible in $\tilde{X}_1$. For $\mathcal{I} = (x, y)$, in the piece $\tilde{X}_2(z)$, it can be written as $\mathcal{I} = (x, y) = (x, x^2 z) = x(1, xz) = (x)$, which is invertible on $\tilde{X}_2(z)$; in the piece $\tilde{X}_2(w)$, it can be written as $\mathcal{I} = (x, y)$ which is not generated by one element at the singular point corresponding to the ideal $\mathfrak{m}_1 = (x, y, w)/(x^2 - yw)$. Namely, if $(x, y)_{\mathfrak{m}_1}$ is a principal ideal and generated by an element f , then $f|x$ and $f|y$, which is impossible because x and y are both irreducible and $k[x, y, w]/(x^2 - yw)$ is a normal ring. For $\tilde{\mathcal{I}} = (x^2, y)$, in the piece $\tilde{X}_1(w)$, it can be written as $\tilde{\mathcal{I}} = (x^2, y) = ((yw)^2, y) = (y)(yw^2, 1) = (y)$, which is invertible; in the piece $\tilde{X}_1(z)$, it can be written as $\tilde{\mathcal{I}} = (x^2, y) = (x^2, xz) = (x)(x, z)$, which is not invertible at the point corresponding to the ideal $\mathfrak{m}_2 = (x, y, z)/(y - xz)$ by the same argument as before. Hence, one has $\text{Hom}_X(\tilde{X}_1, \tilde{X}_2) = \emptyset = \text{Hom}_X(\tilde{X}_2, \tilde{X}_1)$. ■

2.2 Henselization and strict henselization

In order to make a scheme as closer to smoothness as possible without altering its functor of points, it is necessary to take the étale topology and study the local structure of étale morphism, in which henselian local rings link these two things intimately together.

Definition 2.3: Let $f : X \rightarrow Y$ be a morphism of finite type between locally Noetherian schemes. One says that f is **unramified at** $x \in X$ if the finite extension of residue fields $k(f(x))/k(x)$ is separable, and if the local homomorphism $\mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$ verifies $\mathfrak{m}_{f(x)}\mathcal{O}_{X,x} = \mathfrak{m}_x$, which is equivalent to $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_{f(x)}\mathcal{O}_{X,x}$. One says that f is **étale at** x if f is flat and unramified at x . One says f is unramified (resp. étale) if it is unramified (resp. étale) at every point of X . One says that a homomorphism of Noetherian local rings $A \rightarrow B$ is **étale** if it is unramified, flat, and B is a localization of a finitely generated A -algebra.

Example 2.4: Let L/K be a finite field extension. Then $\text{Spec}(L) \rightarrow \text{Spec}(K)$ is unramified, also étale if and only if L/K is separable.

Example 2.5: Let L/K be a field extension of number fields, and $\mathcal{O}_L, \mathcal{O}_K$ be the ring of integers respectively. Then $\text{Spec}(\mathcal{O}_L) \rightarrow \text{Spec}(\mathcal{O}_K)$ is flat by lemma (4.2). For any prime ideal $\mathfrak{q} \in \text{Spec}(\mathcal{O}_L)$, setting $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}_K$, the residue field extension $k(\mathfrak{q})/k(\mathfrak{p})$ is finite separable extension because of finite fields. Then $\text{Spec}(\mathcal{O}_L) \rightarrow \text{Spec}(\mathcal{O}_K)$ is unramified, hence étale, at a prime ideal $\mathfrak{q} \in \text{Spec}(\mathcal{O}_L)$ if and only if the maximal ideal $\mathfrak{q}(\mathcal{O}_L)_{\mathfrak{q}}$ of the local ring $(\mathcal{O}_L)_{\mathfrak{q}}$ is generated by the ideal $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}_K$ in $(\mathcal{O}_L)_{\mathfrak{q}}$, which coincides with the usual definition of unramifiedness from algebraic number theory.

Example 2.6: Let k be a field, $P(x) \in k[x]$ be a monic polynomial. Let us consider $X = \text{Spec}(k[x]/P(x))$. Then for any point $x \in X$, there is an associated irreducible factor $Q(x)$ corresponding to x . Then X is étale at x if and only if $Q(x)$ is a simple factor of $P(x)$, and is also a separable polynomial, i.e without repeated roots in $\bar{k}$.

The unramified property is essentially a fibration property, and so is étale.

Lemma 2.4: Let $f : X \rightarrow Y$ be a morphism of finite type between locally Noetherian schemes. Then f is unramified if and only if for every $y \in Y$, the fiber X_y is a finite, reduced scheme of finite type over $k(y)$, and $k(x)$ is separable over $k(y)$ for any $x \in X_y$.

Proof: If f is unramified, then for any $y \in Y$, the fiber X_y is of zero dimension and reduced since for each $x \in X_y$, the maximal ideal $\mathfrak{m}_x\mathcal{O}_{X_y,x}$ of the fiber X_y at x is nothing but $(\mathfrak{m}_y\mathcal{O}_{X,x})\mathcal{O}_{X_y,x} = \mathfrak{m}_y\mathcal{O}_{X_y,x} = 0$ by definition of the $k(y)$ -algebra $\mathcal{O}_{X_y,x}$. Conversely, by an argument of commutative algebra, the quotient $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ remains stable when

passing from X to the fiber X_y . Hence, X_y is finite reduced over $k(y)$. ■

By [[53], Corollary 6.2.11], a smooth morphism is locally an étale morphism into a relative affine space followed by a projection, where the local structure of étale morphism provides local coordinates and can be used to reduce questions about smooth schemes to the case of étale schemes.

Proposition 2.3: Let $X \rightarrow Y$ be a smooth morphism of schemes. If Y is reduced (resp. normal, resp. regular), so is X .

Proof: See [[77], Proposition 3.5.51]. ■

From a geometric point of view, the schemes associated to the henselian and strictly henselian local rings satisfy certain inverse function theorem.

Definition 2.4: Let R be a local ring with the unique maximal ideal $\mathfrak{m}$ and the residue field k . The local scheme $S = \text{Spec}(R)$ is **henselian** if each étale morphism $X \rightarrow S$ is a local isomorphism at all points x of X over s with trivial residue field extension $k(x) = k(s)$. More precisely, for each point $x \in X$ over s with $k(x) = k(s)$, the local rings satisfy $\mathcal{O}_{X,x} \simeq \mathcal{O}_{S,s}$. In addition, if the residue field $k(s)$ is separably closed for every $s \in S$, then S is called **strictly henselian**.

Remark 2.1: In algebraic number theory, one always defines the henselian rings in terms of Hensel's lemma, which is certainly equivalent to our definition (2.4). Besides, in (2.4), for any étale morphism $X \rightarrow S$ with a trivial extension $k(s) = k(x)$ at point x over s , it is automatically valid that, by [[53], Proposition 4.3.26], the two formal completion $\widehat{\mathcal{O}_{X,x}}$, $\widehat{\mathcal{O}_{S,s}}$ for the topology defined by the respectively maximal ideals are isomorphic. The isomorphism $\widehat{\mathcal{O}_{X,x}} \simeq \widehat{\mathcal{O}_{S,s}}$ is often viewed as a local isomorphism in the analytic sense, which is not a Zariski local isomorphism in general. By Artin approximation or other arguments, for example passing to henselization, this formal isomorphism might be algebraized to an isomorphism of actual Zariski-local rings $\mathcal{O}_{X,x} \simeq \mathcal{O}_{S,s}$.

As we all expected, the notion of henselian local rings and henselian local schemes are equivalent.

Proposition 2.4: Let R be a local ring with its residue field k and $S = \text{Spec}(R)$. Then the followings are equivalent:

- (1) R is a henselian ring.
- (2) S is a henselian scheme.
- (3) Each finite R -algebra A is a finite product of local rings.
- (4) For each finite R -algebra A , the canonical map between the sets of idempotent

elements

$$\text{Idempotent}(A) \rightarrow \text{Idempotent}(A \otimes_R k)$$

is bijective.

(5) For each quasi-finite morphism $X \rightarrow S$, and for each point x over the closed point $s \in S$, there exists an open neighborhood U of x such that $U \rightarrow S$ is finite morphism.

Proof: See [[6], Proposition 2.4]. ■

There is also a Hensel's lemma in the context of algebraic geometry.

Proposition 2.5: Let R be a henselian local ring with its residue field k and X be a smooth R -scheme. Then the canonical map $X(R) \rightarrow X(k)$ from the set of R -valued points of X to the set of k -valued points of X is surjective. In other words, every k -rational point of X can be uniquely lift to a section of $X \rightarrow \text{Spec}(R)$. Moreover, if R is strictly henselian, then the set of k -rational points of $X_k = X \otimes_R k$ which can be lift to R -valued points of X is Zariski dense in X_k .

Proof: See [[53], Corollary 6.2.13], [[77], Proposition 3.5.70] and [[6], Proposition 2.5]. ■

Examples of henselian rings occur in the algebraic number theory, for example the ring of p -adic algebraic integers $\mathbb{Z}_p \cap \overline{\mathbb{Q}}$, and occur in the analytic geometry such as rings of germs of holomorphic functions.

Example 2.7: Let $(A, \mathfrak{m})$ be a Noetherian local ring. If A is complete for the $\mathfrak{m}$ -adic topology, then A is henselian. Thus, every Noetherian local ring is dominated by a henselian local ring. Indeed, there is always a “smallest” henselian local ring A^h that dominates A , which is called the henselization of A as follows.

Definition 2.5: Let R be a commutative local ring. A **henselization** of R is a henselian local ring R^h together with a local inclusion homomorphism $i : R \rightarrow R^h$ satisfying a universal property: every local homomorphism from R to a henselian local ring A factors through R^h uniquely.

Theorem 2.1: For each local ring, there exists a unique henselization up to isomorphism.

Proof: The uniqueness comes from the universal property. For the existence, in view of Definition (2.4), the henselization R^h of R must be the union of all local rings $\mathcal{O}_{X,x}$ of étale R -schemes X at points x above the closed point $s \in S = \text{Spec}(R)$, whose residue fields coincides with the residue field k of R . For the construction, one considers the family of étale local R -algebra $\{R_i\}_{i \in I}$ consists of all isomorphism classes of R -algebras appearing as local rings of étale R -schemes at points over the closed point $s \in \text{Spec}(R)$ with the

same residue field as R . One can easily check $\{R_i\}_{i \in I}$ is a direct system and the direct limit is exactly the henselization of R , i.e. $R^h = \varinjlim_{i \in I} R_i$ ■

Example 2.8: If R is a discrete valuation ring with its fraction field $K = \text{Frac}(R)$, then R^h is equal to $\widehat{R} \cap \overline{K}$, where $\widehat{R}$ is the completion of R , and $\overline{K}$ is an algebraic closure of K .

Definition 2.6: Let R be a local ring with its residue field k . A **strict henselization** of R is a strictly henselian local ring R^{sh} , whose residue field coincides with the separable closure k^s of k , together with a local inclusion morphism $i : R \rightarrow R^{sh}$ satisfying a universal property: for any local morphism $R \rightarrow A$ with A a strictly henselian local ring, and for any k -morphism $\alpha : k^s \rightarrow k^A$ with the residue field k^A of A , there exists a unique local morphism $u^{sh} : R^{sh} \rightarrow A$ such that $u^{sh} \circ i = u$, where u^{sh} induces the map α .

Theorem 2.2: For each local ring, the strictly henselization exists uniquely.

Proof: The uniqueness comes from the universal property also. For the construction, one can consider the family of all isomorphism classes of R -algebras $\{R_i\}_{i \in I}$ which occur as local rings of étale R -schemes at points over the closed point $s \in \text{Spec}(R)$. Define:

$$I^{sh} = \{(R_i, \alpha_{ij}) \mid i \in I, \alpha_{ij} : R_i \rightarrow k^s\}$$

where α_{ij} runs over all R -morphisms into a fixed separable closure k^s of k . One can verify that I^{sh} is a direct system, whose direct limit $R^{sh} = \varinjlim_{(i,j) \in I^{sh}} (R_i, \alpha_{ij})$. ■

The henselization of rings does not alter many properties of the base ring.

Proposition 2.6: If R is a reduced (resp. normal, resp. regular, resp. Noetherian) local ring, so is the (strictly) henselization of R . In particular, if R is a discrete valuation ring with uniformizer π , then the (strictly) henselization of R is also a discrete valuation ring, and π gives rise to a uniformizer.

Proof: The regularness, reducedness and normality follow from the directed limit by proposition (2.3). For the discrete valuation ring $(R, \mathfrak{m})$, the maximal ideal $\mathfrak{m}$ generates the maximal ideal $\mathfrak{m}_i$ of each R_i in the construction of R^h (resp. R^{sh}) since $R_i/\mathfrak{m}_i = R/\mathfrak{m}$. Thus, it is clear that $\mathfrak{m}$ also generates the maximal ideal of R^h (resp. R^{sh}). ■

In the end of this section, we compare the local ring R with its henselization R^h, R^{sh} . By construction, one has the ring extensions:

$$R \hookrightarrow R^h \hookrightarrow R^{sh}$$

where R^{sh} can be also interpreted as the strict henselization of R^h , and then the extension $R^h \rightarrow R^{sh}$ is an integral extension. Namely, the morphism $\text{Spec}(R^{sh}) \rightarrow \text{Spec}(R^h)$ is a finite morphism. If R is normal, then R^h and R^{sh} are normal by Proposition (2.6), hence

integral domains. In this case, one can take the fraction fields:

$$K \hookrightarrow K^h \hookrightarrow K^{sh}$$

which are separable and algebraic over K . Moreover, K^{sh}/K^h is a Galois extension and the associated Galois group $\text{Gal}(K^{sh}/K^h)$ acts on R^{sh} , which is isomorphic to the Galois group $\text{Gal}(K^s/K)$. Furthermore, the fixed subring of R^{sh} is precisely R^h .

Proposition 2.7: Let R be a local ring. The ring extensions $R \hookrightarrow R^h \hookrightarrow R^{sh}$ are both faithfully flat.

Proof: Since étale morphisms are flat, by construction (2.1), (2.2), R^h and R^{sh} are direct limits of flat ring extensions and, hence they are flat ring extensions of R . ■

For a closer study of these ring extensions, one can see [[6], Proposition 2.11] for more details.

2.3 Integral models over DVRs

Let R be a DVR, with its fraction field K , residue field k and uniformizer π . For example, one could have $R = \mathbb{Z}_p, K = \mathbb{Q}_p, \pi = p, k = \mathbb{F}_p$. Let X be a proper K -variety, one can make sense of the reduction of X modulo π , which is a k -variety. Let's consider $\text{Spec}(R)$, which consists of two points: the generic point $\eta = \text{Spec}(K)$ corresponding to zero ideal (0) , and the closed point $s = \text{Spec}(k)$ corresponding to the maximal ideal (π) .

Definition 2.7: Let X be a R -scheme, the **generic fiber** of X is the K -scheme $X_K = X \times_{\text{Spec}(R)} \text{Spec}(K)$, and the **special fiber** of X is the k -scheme $X_k = X \times_{\text{Spec}(R)} \text{Spec}(k)$.

Schemes of finite type over some DVR or general Dedekind domain, for example the ring of S -integers of number fields, are often called **arithmetic schemes** if they satisfy some more attached technical conditions depending on different authors. When the relative dimension $\dim_x(f)$ of $f : X \rightarrow \text{Spec}(R)$ equals to one for each $x \in X$, they are called **arithmetic surfaces** due to the 1-dimensional base ring R . More precisely, if X is flat over R , then $\dim_x(f) = \dim f^{-1}(f(x)) = \dim(X) - \dim(R)$ for each $x \in X$. For example, $X = \text{Spec}(\mathbb{Z}[x])$ is an arithmetic surface, whose fiber over each prime number p is a Zariski-closed subset and isomorphic to the affine line $\mathbb{A}_{\mathbb{F}_p}^1 = \text{Spec}(\mathbb{F}_p[x])$.

Definition 2.8: Let X be a K -scheme. An R -**model** of X is an R -scheme X_R equipped with an isomorphism $X_R \times_R K \xrightarrow{\sim} X$ as K -schemes, which means X can be embedded into X_R as its generic fiber. In this thesis, we require an R -model to be separated and of finite type, making the special fiber an algebraic variety.

Given a K -variety X , the R -model of X is not unique up to isomorphism in general.

Example 2.9: Let $X = \text{Proj}(\mathbb{Q}_p[x, y, z]/(xy - pz^2))$ be a projective $\mathbb{Q}_p$ -curve, where p is a prime number. Then the schemes $X_1 = \text{Proj}(\mathbb{Z}_p[x, y, z]/(xy - z^2))$ and $X_2 = \text{Proj}(\mathbb{Z}_p[x, y, z]/(xy - pz^2))$ are both $\mathbb{Z}_p$ -model of X . Indeed, by the construction of fiber products, one could have X_1 and X_2 have non-isomorphic special fiber,

$$X_1 \times_{\text{Spec}(\mathbb{Z}_p)} \text{Spec}(\mathbb{F}_p) = \text{Proj}(\mathbb{Z}_p[x, y, z]/(xy - z^2) \otimes \mathbb{F}_p) = \text{Proj}(\mathbb{F}_p[x, y, z]/(xy - z^2))$$

$$X_2 \times_{\text{Spec}(\mathbb{Z}_p)} \text{Spec}(\mathbb{F}_p) = \text{Proj}(\mathbb{Z}_p[x, y, z]/(xy - pz^2) \otimes \mathbb{F}_p) = \text{Proj}(\mathbb{F}_p[x, y, z]/(xy)).$$

The first one is a conic in $\mathbb{P}_{\mathbb{F}_p}^2$ while the second one is degenerate, a reduced union of two projective lines intersecting at a single point. Hence, X_1 and X_2 are non-isomorphic. But one could show that they have the same generic fiber, which is X . Namely, on the one hand, since p is invertible in $\mathbb{Q}_p$, one may consider the change of variables $y' = py$ for $X_2 \times_{\text{Spec}(\mathbb{Z}_p)} \text{Spec}(\mathbb{Q}_p) = \text{Proj}(\mathbb{Q}_p[x, y, z]/(xy - z^2))$. So, the generic fiber of X_2 is isomorphic to X . On the other hand, it is obvious that the generic fiber of X_1 is X .

Now, a reduction modulo π of a variety X over K , can be described as the special fiber of an R -model of X . Thus, the R -model $\mathbf{X}$ of X has captured two of the most important fibrations. One is the variety X , described by the generic fiber of $\mathbf{X}$ and the other is the reduction modulo π of X , described by the special fiber of $\mathbf{X}$.

Lemma 2.5: Let R be a discrete valuation ring with the residue field k and the fraction field K . If X is an integral scheme of finite type over R with non-empty special fiber X_k and generic fiber X_K , then X is flat over R , hence faithfully flat over R . In particular, every non-empty open subscheme of X must intersect with the generic fiber X_K while it is not true for the special fiber X_k .

Proof: The flatness of X follows from Lemma (4.2) directly. Let $f : X \rightarrow \text{Spec}(R)$ be the structure morphism of X . For every non-empty open subscheme U of X , since f is flat, $f(U)$ is open subset of $\text{Spec}(R)$. Thus, the generic point (0) of $\text{Spec}(R)$ is contained in $f(U)$ and the generic fiber $X_K \cap U = \emptyset$. ■

As pointed out in Example(2.9), there are many non-isomorphic models, so it is necessary to choose a good R -model of X in some sense.

Definition 2.9: Let R be a DVR, $K = \text{Frac}(R)$ and X be a smooth proper K -variety. One says that X has **good reduction** if there exists a smooth proper R -model of X . In this case, the reduction modulo π of X is a smooth proper variety over the residue field.

In the Example(2.9), by the Jacobian criterion of smoothness, it is easy to see that

X_1 is a good reduction but X_2 is not.

Proposition 2.8: Let R be a DVR, and $K = \text{Frac}(R)$. Let $\widehat{R}$ be the completion of R and $\widehat{K} = \text{Frac}(\widehat{R})$ be the completion of K . Let X_K be a smooth proper K -variety. If X_K has good reduction, so does $X_{\widehat{K}} = X_K \times_K \widehat{K}$. Moreover, if X_R is a smooth proper R -model of X_K , then $X_{\widehat{R}} = X_R \times_R \widehat{R}$ is a smooth proper $\widehat{R}$ -model of $X_{\widehat{K}}$.

Proof: Since X_K has good reduction, there is a smooth proper R -model X_R , such that $X_R \times_R K \simeq X_K$. It is claimed that $X_{\widehat{R}} = X_R \times_R \widehat{R}$ is actually a $\widehat{R}$ -model of $X_{\widehat{K}}$. Namely,

$$X_{\widehat{R}} \times_{\widehat{R}} \widehat{K} \simeq (X_R \times_R \widehat{R}) \times_{\widehat{R}} \widehat{K} \simeq X_R \times_R \widehat{K} \simeq (X_R \times_R K) \times_K \widehat{K} \simeq X_K \times_K \widehat{K} = X_{\widehat{K}}$$

The smoothness and properness of $X_{\widehat{R}}$ follows from X_R . ■

Remark 2.2: The converse of Proposition(2.8) is not true in general, and see [[65], Example 5.3] for a counterexample, which is a model of some K3-surface. Arguably, the notion of good reduction would have better properties if it were defined in terms of algebraic spaces. It can happen that a K -variety X does not extend to a smooth proper R -scheme but does extend to a smooth proper R -algebraic space.

In the very local case, the existence of a good reduction of a K -variety X could be completely determined by the smoothness of its reduction to the residue field k . More precisely, under the properness and flatness, the smoothness of the special fiber implies smoothness of the total family.

Proposition 2.9: Let $S = \text{Spec}(R)$ be a Noetherian local scheme, where R is a Noetherian local ring, $\mathfrak{m}$ is the unique closed point of S , and $k = R/\mathfrak{m}$ is the residue field. Let $f : X \rightarrow S$ be a proper flat morphism. If the special fiber X_k is smooth over k , then X is smooth over S .

Proof: By [[53], Corollary 6.2.12], it is clear that the smooth locus X_{smooth} of f is an open subset of X . Let $Z = X \setminus X_{\text{smooth}}$, which is a closed subset of X . It suffices to show that Z is empty. Let us consider the image $f(Z)$, which is also a closed subset of the local scheme S because of the properness of f . Since f is flat, and the special fiber X_k is smooth over k , by [[53], Definition 4.3.35], the fiber-by-fiber criterion of smoothness shows that f is smooth at X_k . Thus, $X_k \cap Z = \emptyset$ and then $\mathfrak{m} \notin f(Z)$. However, S is a local scheme and each non-empty closed subset of S must contain the unique closed point $\mathfrak{m}$ because $\mathfrak{m}$ is the unique maximal point of S under specialization, which implies $f(Z) = \emptyset$. Hence, $Z = \emptyset$, $X_{\text{smooth}} = X$, and X is smooth over S . ■

Remark 2.3: In Proposition (2.9), it is necessary to require the flatness to make sure that the smooth proper special fiber X_k admits a smooth proper R -model. Without the

flatness, the fiber criterion of the smoothness might fail and the smoothness of the fiber does not imply the smoothness of the morphism. A counterexample is the disjoint union of two smooth proper curve over S and over k respectively, which is proper over S with smooth special fiber but not smooth over S , even not flat over S . For similar results of the properness, one can see [[53], Remark 3.3.28] and [[35], IV.15.7.10].

2.4 The Smoothing process

In this section, we will focus on the very local cases in arithmetic geometry. Let R be a discrete valuation ring with its fraction field $K = \text{Frac}(R)$, the unique maximal ideal $\mathfrak{m}$, a uniformizer π , and its residue field $k = R/\mathfrak{m}$. We denote R^h a henselization of R and R^{sh} a strict henselization of R , both of which are discrete valuation rings with uniformizing element π by Proposition (2.6), and the residue field of R^{sh} is the separable closure k^s of k by construction.

Now, we will introduce one of the most important applications of blowing-ups, making a scheme closer to smoothness without breaking the structure of its functor of points.

Definition 2.10: Let X be an R -scheme of finite type whose generic fiber $X_K = X \times_R K$ is smooth over K . A **smoothing of X is an R -morphism $f : X' \rightarrow X$ which satisfies the following conditions:**

(1) f is proper and is also an isomorphism on generic fibers, namely $f_K : X'_K \rightarrow X_K$ is an isomorphism.

(2) For each étale R -algebra R' , each R' -valued point of X lifts uniquely to an R' -valued point of X' which factors through the smooth locus X'_{smooth} of X' . Namely, the canonical map $X'_{\text{smooth}}(R') \rightarrow X(R')$ is a bijection.

In fact, to check a proper R -morphism whether a smoothing morphism is essentially a local problem.

Lemma 2.6: Let R be a Noetherian ring. If R is semilocal, then for any finite étale R -algebra R' , R' is also a Noetherian semilocal ring. If R is reduced(resp. normal, resp. regular), so is R' . In particular, if R is a Dedekind domain, then any étale R -algebra is a regular Noetherian semilocal ring.

Proof: For any finite étale R -algebra R' , one has a finite étale morphism $\varphi : \text{Spec}(R') \rightarrow \text{Spec}(R)$, which is in particular unramified, of finite type and smooth. So by lemma (2.4), for each prime ideal $\mathfrak{p} \in \text{Spec}(R)$, $\varphi^{-1}(\mathfrak{p}) = \{\beta \in \text{Spec}(R') | \beta \cap R = \mathfrak{p}\} = \text{Spec}(k(\mathfrak{p}) \otimes_R R')$ is a finite étale subscheme over $k(\mathfrak{p})$, where $k(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} = \text{Frac}(R/\mathfrak{p})$. Since

R' is finite over R , i.e the ring extension R'/R is integral, by [[3], Theorem 5.10], φ has going-up property. More precisely, for any maximal ideal $\beta \in \text{Spec}(R')$, the image $\varphi(\beta) = \beta \cap R$ is also a maximal ideal of R . Otherwise, one can choose a maximal ideal $\mathfrak{m}'$ of R containing $\beta \cap R$ and by going-up property, there exists a prime ideal $\beta \subseteq \mathfrak{Q}$ such that $\mathfrak{Q} \cap R = \mathfrak{m}'$, which contradicts with the choice of β . Thus, the maximal ideals of R' lie over the maximal ideals of R , and so R' is semilocal by unramifiedness of φ . Since φ is smooth, by proposition (2.3), if R' is reduced(resp.normal, resp. regular), so is R . If R is a Dedekind domain and semilocal, then $\text{Spec}(R)$ is a finite set. ■

Proposition 2.10: Let X be an R -scheme of finite type whose generic fiber X_K is smooth over K . Let $f : X' \rightarrow X$ be a proper R -morphism which is an isomorphism at the generic point, i.e $f_K : X'_K \xrightarrow{\cong} X_K$ is an isomorphism. Then f is a smoothening of X if and only if the canonical map $X'_{\text{smooth}}(R^{sh}) \rightarrow X(R^{sh})$ is bijective.

Proof: It suffices to check the second condition in the definition(2.10) above. By lemma (2.6), each étale R -algebra R' is semilocal. By means of spreading out and the glueing lemma, in order to check the second condition, it suffices to consider the local étale extension of R , which is a Noetherian regular local ring with exactly one non-zero prime ideal from the proof of lemma (2.6), hence a DVR and flat over R . Without loss of the generality, one may assume that R' is a local étale R -algebra, which is a DVR and flat over R , with the fraction field $L = \text{Frac}(R')$. Let $s : \text{Spec}(R') \rightarrow X$ be a R' -valued point of X . Consider the following commutative diagram:

$$\begin{array}{ccc}
 \text{Spec}(L) & \xrightarrow{\tilde{s}} & X' \\
 \downarrow \tau & \searrow j & \uparrow i \\
 & X_K \cong X'_K & \\
 & \searrow \pi & \\
 \text{Spec}(R') & \xrightarrow{s} & X
 \end{array}$$

where thanks to the flatness of R' over R , the L -valued point $s \circ \tau : \text{Spec}(L) \rightarrow X$ of X factors through X_K and induces the morphism $j : \text{Spec}(L) \rightarrow X_K$. Define $\tilde{s} = i \circ j : \text{Spec}(L) \rightarrow X'$ and one establishes the commutative diagram. By the valuative criterion of properness, there exists a unique morphism $s' : \text{Spec}(R') \rightarrow X'$. Thus, the canonical map $X'(R') \rightarrow X(R')$ deduced from f is bijective for any local étale R -algebra R' . Suppose that f is a smoothening of X , and the second condition (2.10) says that, for each local étale extension R' of R , the canonical map $X'(R') \rightarrow X(R')$ factors through the smooth locus X'_{smooth} of X' . Since the strict henselization (2.2) R^{sh} is the direct limit of all local étale

extensions of R , if f is a smoothening, then one can take limit and obtain the bijection:

$$X'_{\text{smooth}}(R^{sh}) = X'_{\text{smooth}}(\varinjlim R') = \varinjlim X'_{\text{smooth}}(R') \xrightarrow{\cong} \varinjlim X(R') = X(\varinjlim R') = X(R^{sh})$$

where due to locally of finite presentation of X , by [[77], Remark 3.1.11], the functor of points commutes with the direct limits. Conversely, if $X'_{\text{smooth}}(R^{sh}) \rightarrow X(R^{sh})$ is bijective, then for any local étale R -algebra R' and $s : \text{Spec}(R') \rightarrow X$, there exists a local étale R -algebra $\tilde{R}$ such that $s \in X'_{\text{smooth}}(\tilde{R}) \cap X'(R') \subseteq X'_{\text{smooth}}(R')$. So $X'_{\text{smooth}}(R') \rightarrow X(R')$ is bijective. ■

Smoothening can be obtained by a finite sequence of blowing-ups.

Theorem 2.3 (Smoothening Process): Let R be a DVR with its fraction field $K = \text{Frac } R$ and residue field k . Let X be an R -scheme of finite type whose generic fiber X_K is smooth over K . Then X admits a smoothening $f : X' \rightarrow X$. Furthermore, one can obtain f by a finite sequence of blowing-ups along closed subschemes of the special fiber X_k of X . In particular, if X is quasi-projective over R , and so is X' .

Proof: See [[6], Theorem 2.3.3]. ■

Corollary 2.4: Let R be a DVR with its fraction field $K = \text{Frac } R$ and X be a scheme of finite type over R . If X_K is a smooth K -scheme, X_K admits a smooth R -model. Namely, there exists an R -morphism $f : \tilde{X} \rightarrow X$ from a smooth R -scheme $\tilde{X}$ of finite type to X such that

- (1) $\tilde{X}$ is birational to X if X is integral.
- (2) f is an isomorphism on generic fibers $f_K : \tilde{X}_K \xrightarrow{\cong} X_K$.
- (3) the canonical map $\tilde{X}(R^{sh}) \rightarrow X(R^{sh})$ is bijective.
- (4) for each étale R -algebra R' , the canonical map $\tilde{X}(R') \rightarrow X(R')$ is bijective.

Proof: It follows from the smoothening process (2.3) that X admits a smoothening $h : X' \rightarrow X$. Take $\tilde{X}$ as the smooth locus X'_{smooth} of X' , and everything follows from proposition (2.10). ■

2.5 Dilatation

It is claimed that a smoothening process can be obtained by a finite sequence of blowing-ups along closed subschemes of special fibers in theorem (2.3). Besides, we also concern what happens to the set $X(R)$ consisting of R -valued points on the scheme X when such a blowing up is applied to X . Here is a basic example.

Example 2.10: Let R be a discrete valuation ring with a uniformizer π and the residue

field $k = R/(\pi)$. It is known that there exists a unique discrete valuation on R , denoted by v , which can be extended to K and induces an absolute value $|\cdot|_K$ on K . Thus, $R = \{x \in K \mid |x|_K \leq 1\}$. Let $X = \text{Spec}(R[x_1, x_2, \dots, x_n])$, the ideal $I = (\pi, x_1, x_2, \dots, x_n)$ with the associated sheaf $\mathcal{I}$ and $Y_k = \text{Spec}(R[x_1, x_2, \dots, x_n]/(\pi, x_1, x_2, \dots, x_n)) = \text{Spec}(R[x_1, x_2, \dots, x_n]/I)$, where Y_k is a reduced closed subscheme of X consisting of the origin of the special fiber X_k of X .

There are bijections between the set of R -valued points on X and the set of rational points on the affine n -space $\mathbb{A}_K^n$ as follows.

$$\begin{aligned} X(R) &= \text{Hom}_{\text{Spec}(R)}(\text{Spec}(R), X) = \text{Hom}_{R\text{-alg}}(R[x_1, x_2, \dots, x_n], R) \\ &\leftrightarrow \{a = (a_1, \dots, a_n) \mid a_i \in R\} \\ &= \{a_K = (a_1, \dots, a_n) \in \mathbb{A}_K^n \mid |x_i(a_K)|_K = |a_i|_K \leq 1\} \end{aligned}$$

where the one to one correspondence above follows from the evaluation maps at variables x_i and coordinates respectively. More generally, by the same argument, it is confirmed that such a bijection between the set of R -valued points on a R -scheme X of finite type and the rational points on its generic fiber X_K bounded by the absolute could be demonstrated.

Furthermore, the R -valued points of X which specialize into Y_k correspond bijectively to the rational points $a_K \in \mathbb{A}_K^n$ with $|x_i(a_K)| \leq |\pi|$. More precisely,

$$\{q \in X(R) \mid q \in Y_k(k) \bmod \pi\} \leftrightarrow \{a_K = (a_1, \dots, a_n) \in \mathbb{A}_K^n \mid |x_i(a_K)|_K = |a_i|_K \leq |\pi|_K\}$$

Now, let $f : \tilde{X} \rightarrow X$ be the blowing-up of X along the closed subscheme Y_k . Then by lemma (2.3), $\tilde{X}$ is covered by the affine open subschemes of the form:

$$\begin{aligned} \tilde{X}_\pi &= \text{Spec}\left(R[x_1, \dots, x_n]\left[\frac{x_1}{\pi}, \dots, \frac{x_n}{\pi}\right]\right) = \text{Spec}\left(R\left[\frac{x_1}{\pi}, \dots, \frac{x_n}{\pi}\right]\right) \\ &= \text{Spec}\left(\frac{R[x_1, \dots, x_n][y_1, \dots, y_n]}{\pi y_i - x_i}\right) (1 \leq i \leq n) \\ \tilde{X}_{x_i} &= \text{Spec}\left(R[x_1, \dots, x_n]\left[\frac{\pi}{x_i}, \dots, \frac{x_n}{x_i}\right]\right) (1 \leq i \leq n). \end{aligned}$$

Note that when restricted to the local pieces $\tilde{X}_{x_i}$ (resp. $\tilde{X}_\pi$), the inverse image of ideal sheaf $\mathcal{I}\mathcal{O}_{\tilde{X}}$ is exactly the sheaf associated to the ideal $I \cdot \Gamma(\tilde{X}_{x_i}, \mathcal{O}_{\tilde{X}_{x_i}})$ (resp. $I \cdot \Gamma(\tilde{X}_\pi, \mathcal{O}_{\tilde{X}_\pi})$), which is generated by x_i (resp. π) because of $x_j = \frac{x_j}{x_i} x_i$ for each j . Thus, in particular, $\tilde{X}_\pi$ is the set of points of $\tilde{X}$ at which $\mathcal{I}\mathcal{O}_{\tilde{X}}$ is generated by π and there is an R -morphism:

$$\tilde{X}_\pi = \text{Spec}\left(R\left[\frac{x_1}{\pi}, \dots, \frac{x_n}{\pi}\right]\right) \rightarrow X = \text{Spec}(R[x_1, \dots, x_n])$$

which is induced by sending x_i to x_i for each i .

It is seen that the set of R -valued points $\tilde{X}_\pi(R)$ of $\tilde{X}_\pi$ is mapped bijectively onto the

set of those R -valued points $Y_k(R)$ which specialized into Y_k .

$$\tilde{X}_\pi(R) = \text{Hom}_{R\text{-alg}}\left(R\left[\frac{x_1}{\pi}, \dots, \frac{x_n}{\pi}\right], R\right) \leftrightarrow \{a_K \in \mathbb{A}_K^n \mid |x_i(a_K)|_K \leq |\pi|_K\}$$

where the first bijection is obtained by evaluating $a_i = f(x_i) \in R$ and $f(\frac{x_i}{\pi}) \in R$ for any R -morphism f . Furthermore, two points $a, b \in \tilde{X}_\pi(R)$ have the same specialization over k if and only if $|x_i(a_K) - x_i(b_K)|_K \leq |\pi^2|_K$ for all i . Call $\tilde{X}_\pi$ **the dilatation of Y_k in X** .

In order to illustrate dilatations of more general type, one should consider an arbitrary R -scheme X of finite type and a closed subscheme Y_k of the special fiber X_k .

Theorem 2.4: Let R be a discrete valuation ring with a uniformizer π , the fraction field $K = \text{Frac}(R)$ and the residue field $k = R/(\pi)$. Let X be an R -scheme of finite type and Y_k be a closed subscheme of its special fiber X_k , whose associated sheaf of ideals is denoted by $\mathcal{I} \subseteq \mathcal{O}_X$. Let $f : \tilde{X} \rightarrow X$ be the blowing-up of X along Y_k and $u : \tilde{X}_\pi \rightarrow X$ denote the restriction of f to the open subscheme of $\tilde{X}$ where the inverse image ideal sheaf $\mathcal{I}\mathcal{O}_{\tilde{X}}$ is generated by π . Then:

- (1) $\tilde{X}_\pi$ is a flat R -scheme even if X is not, and the induced morphism on the special fibers $u_k : (\tilde{X}_\pi)_k \rightarrow X_k$ factors through Y_k .
- (2) For any flat R -scheme Z and R -morphism $v : Z \rightarrow X$ such that $v_k : Z_k \rightarrow X_k$ factors through Y_k , then v must factor through $\tilde{X}_\pi$ uniquely. More precisely, there exists a unique R -morphism $\tilde{v} : Z \rightarrow \tilde{X}_\pi$ such that $v = u \circ \tilde{v}$.
- (3) If X is flat over R (2.5), then $\tilde{X}$ is flat over R .

One calls $(\tilde{X}_\pi, u)$ **the dilatation of Y_k on X** .

Proof: Because of the universal property of property (2), the pair $(\tilde{X}_\pi, u)$ is unique up to isomorphism in the class of all pairs (Z, v) satisfying the property (1).

Without loss of generality, one may assume that X is an integral scheme, and to obtain the general results in [[6], Section 3.2], the same approach is employed, but it requires a more cumbersome discussion because the local description of blowing-ups becomes more complicative. As we have shown, the blowing up $\tilde{X}$ of X along $Y_k = V(\mathcal{I})$ is defined as the homogenous projective scheme (2.2) $\tilde{X} = \text{Proj}(\bigoplus_{n \geq 0} \mathcal{I}^n)$, which is defined locally and glued together by lemma (2.1). Over any affine open subscheme $U = \text{Spec}(A)$ of X which is not contained in the generic fiber X_K , where A is an integral domain, finitely generated algebra over R , the sheaf of ideals $\mathcal{I}$ is attached to an ideal $I = \mathcal{I}(U)$ of A . Since $Y_k = V(\mathcal{I}) \subset X_k$, one must have $\pi \in I$. Then, the ideal I could be generated by finitely many elements $t_0 = \pi, t_1, \dots, t_n$ of A .

Over the affine open subscheme U , by Lemma (2.1), the blowing-up $\tilde{X}$ of X is the

projective scheme $\tilde{X}(U) = \text{Proj}(\oplus_{d \geq 0} I^d)$, which is isomorphic to the closed subscheme of $\mathbb{P}_A^n$ given by the homogenous ideal:

$$\tilde{I} = \ker(A[T_0, T_1, \dots, T_n] \rightarrow \oplus_{d \geq 0} I^d),$$

where the graded homomorphism is surjective and given by sending the variables T_i to $t_i \in I^1 = I$ because of the Noetherian property of A by [[3], Proposition 10.7]. Namely, one has $\tilde{X}(U) \simeq \text{Proj}(A[T_0, T_1, \dots, T_n]/\tilde{I})$ and $J = (t_i T_j - t_j T_i)_{1 \leq i, j \leq n} \subseteq \tilde{I}$, with the equality if $\{t_0, \dots, t_n\}$ forms a minimal system of generators of I by [[53], Lemma 8.1.2]. Let $V_i = D^+(T_i)$ be the affine open subscheme of $\mathbb{P}_A^n$ where T_i does not vanish. Then $\tilde{X}(U) \cap V_i$ is affine and defined as the spectrum of its global section in the lemma (2.3):

$$\tilde{X}(U) \cap V_i = \text{Spec}\left(A\left[\frac{t_0}{t_i}, \dots, \frac{t_n}{t_i}\right]\right) = \text{Spec}\left(\frac{A[T_0, \dots, T_{i-1}, T_{i+1}, \dots, T_n]}{(t_j - t_i T_j)_{j \neq i}}\right).$$

Then, $\tilde{X}(U)$ is flat over U , and for the general case that X is not integral, it suffices to divide by the t_i -torsion part of $\tilde{X}(U)$. Thus, if X is flat over R , then $\tilde{X}$ is flat over R .

Returning to the case of global R -scheme X , let us consider the set:

$$\tilde{X}_\pi = \{x \in \tilde{X} \mid \text{the inverse image ideal sheaf } \mathcal{IO}_{\tilde{X}} \text{ is generated by } \pi \text{ at } x\}$$

which is an open subscheme of $\tilde{X}$. More precisely, over an affine open subscheme $U = \text{Spec}(A)$ of X as above, it is clear that $(\mathcal{IO}_{\tilde{X}})|_{\tilde{X}(U)}$ is equal to $IA[\frac{t_0}{t_i}, \dots, \frac{t_n}{t_i}]$ and is generated by t_i when restricted to the affine piece $D^+(t_i) = \tilde{X}(U) \cap V_i$ for every $0 \leq i \leq n$ because $t_j = \frac{t_j}{t_i} t_i$. Therefore, over $U = \text{Spec}(A)$, $\tilde{X}_\pi$ consists of nothing but the affine A -scheme $D^+(\pi) = \text{Spec}(\tilde{A}_\pi)$, where

$$\tilde{A}_\pi = A\left[\frac{t_1}{\pi}, \dots, \frac{t_n}{\pi}\right] = \frac{A[T_1, \dots, T_n]}{(t_j - \pi T_j)_{1 \leq j \leq n}}.$$

Hence, $\tilde{X}_\pi \cap \tilde{X}(U)$ is flat over R by lemma (4.1) for any U and then $\tilde{X}_\pi$ is flat over R even if X is not. Let $u : \tilde{X}_\pi \rightarrow X$ be the canonical morphism obtained by the open immersion followed by the blowing-up. It is obvious that $u_k : (\tilde{X}_\pi)_k \rightarrow X_k$ factors through Y_k since

$$\tilde{A}_\pi \otimes_A k \simeq \frac{(A/\pi)[T_1, \dots, T_n]}{(t_1, \dots, t_n)} = \frac{(A/\pi)[T_1, \dots, T_n]}{\bar{I}}.$$

For the universal property of the pair $(\tilde{X}_\pi, u)$, suppose that Z is a flat R -scheme with an R -morphism $v : Z \rightarrow X$ satisfying the property (2), it is essentially a local problem on X and Z . One may assume that both schemes are affine, namely $X = \text{Spec}(A)$ and $Z = \text{Spec}(B)$, where A and B are both integral domain and B is flat over R . Let $v^* : A \rightarrow B$ be the induced ring homomorphism. Under notations as before, the fact that $v_k : Z_k = \text{Spec}(B/\pi B) \rightarrow Y_k = \text{Spec}(A/\pi)$ factors through $Y_k = \text{Spec}(A/I)$ implies that

the ideal IB is contained in πB . Then, there exists elements $b_i \in B$ with $v^*(t_i) = \pi b_i$, and the elements b_i are unique because of the flatness (no π -torsion part) of B over R . Thus, one can define an A -morphism $A[T_0, \dots, T_n] \rightarrow B$ sending T_i to b_i , which yields a morphism $\tilde{v}^* : \tilde{A}_\pi \rightarrow B$, hence a morphism $w : Z \rightarrow \tilde{X}_\pi$ such that $v = u \circ \tilde{v}$. ■

There are some elementary properties of dilatations.

Proposition 2.11: Let R be a discrete valuation ring with a uniformizer π and the residue field $k = R/(\pi)$.

(1) Let X be a closed subscheme of an R -scheme W , and Y_k be a closed subscheme of X_k . Then the dilatation $\tilde{X}_\pi$ of Y_k on X can be realized as a closed subscheme of the dilatation $\tilde{W}_\pi$ of Y_k on W , where Y_k is viewed as a closed subscheme of W_k .

(2) Dilatations commute with flat base change $R \rightarrow R'$, where R' is a discrete valuation ring with a uniformizer π also. Namely, one has the canonical isomorphism $\tilde{X}_\pi \otimes_R R' = (\tilde{X} \otimes_R R')_\pi$ for any R -scheme X of finite type.

(3) All dilatations factor through the largest flat R -subscheme of X , which is the support of the sheaf of π -torsion in $\mathcal{O}_X$, i.e the Zariski closure of the generic fiber X_K .

(4) Dilatations commute with fiber products. Let X_i be R -schemes of finite type, and Y_k^i be closed subschemes of $(X_i)_k$ for $i = 1, 2$. Then the dilatation of $Y_k^1 \times_k Y_k^2$ on $X_1 \times_R X_2$ is the product $\tilde{X}_{1\pi} \times_R \tilde{X}_{2\pi}$ of the dilatations of Y_k^i on X_i . In particular, if X is an R -group scheme, and Y_k is a subgroup scheme of X_k , the dilatation $\tilde{X}_\pi$ of Y_k on X is an R -group scheme and the canonical map $u : \tilde{X}_\pi \rightarrow X$ is a group homomorphism.

Proof: For (1), by the universal property of dilatations, the closed immersion $f : X \rightarrow W$ induces a morphism on the dilatations $f_\pi : \tilde{X}_\pi \rightarrow \tilde{W}_\pi$, which is a closed immersion from the construction (2.4). For (2), it follows from the properties of blowing-ups (2.1) and the construction of dilatations essentially. For (3), it follows from the universal property of scheme-theoretic image and the proposition (4.2) because of the generic fibers $(\tilde{X}_\pi)_K \hookrightarrow (\tilde{X})_K \simeq X_K$, where actually $(\tilde{X}_\pi)_K \simeq X_K$ if X is an integral scheme. For (4), it follows from the universal property of fiber products. ■

Finally, we present that dilatations preserve the smoothness.

Proposition 2.12: Let R be a discrete valuation ring with a uniformizer π and the residue field $k = R/(\pi)$. Let X be a smooth R -scheme, and Y_k be a closed subscheme of X_k , which is smooth over k . Then the dilatation $\tilde{X}_\pi$ of Y_k on X is also smooth over R .

Proof: See [[6], Proposition 3.2.3] and essentially from [[53], Theorem 8.1.19]. ■

CHAPTER 3 INTEGRAL MODELS OF ALGEBRAIC VARIETIES

3.1 Integral models over Dedekind domains

The definition (2.7) makes sense for any integral domain R with only one generic fiber but many special fibers. A common situation in arithmetic geometry, generalizing the case of DVR, is where R is a Dedekind domain, that is, an integrally closed Noetherian integral domain of dimension at most 1.

Definition 3.1: A **Dedekind scheme** is a normal locally Noetherian scheme of dimension at most 1.

Proposition 3.1: Let X be a Noetherian integral scheme. Then X is a Dedekind scheme if and only if $\mathcal{O}_X(U)$ is a Dedekind domain for every open subset U of X .

Proof: It follows from the local property of the normality essentially. For any open subset U of X , let $\alpha \in \text{Frac}(\mathcal{O}_X(U))$ which is integral over $\mathcal{O}_X(U)$. It suffices to show $\alpha \in \mathcal{O}_X(U)$. For any affine open subset V of U , since X is integral, the restriction map $\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_X(V)$ is injective, then α is integral over $A = \mathcal{O}_X(V)$ and $\alpha \in \text{Frac}(A)$. Let us consider the ideal $I = \text{ann}((\alpha A + A)/A)$. If $I \neq A$, then I is contained in some prime ideal $\mathfrak{p}$ of A . As X is normal, $A_{\mathfrak{p}}$ is integrally closed, then $\alpha \in A_{\mathfrak{p}}$ because of $\text{Frac}(A_{\mathfrak{p}}) = \text{Frac}(A)$. Thus, there exists an $s \in A \setminus \mathfrak{p}$ such that $s\alpha \in A$, then $s \in I$, whence a contradiction with $I \subseteq \mathfrak{p}$. Hence, $1 \in I$ and $\alpha \in A = \mathcal{O}_X(V)$. By choice of V , one has $\alpha \in \mathcal{O}_X(U)$. Conversely, let $x \in X$ and V be an affine open neighborhood of x . Then $\mathcal{O}_X(V)$ is integrally closed by assumption, by [[3], Proposition 5.13], so is $\mathcal{O}_{X,x}$. ■

The following examples of Dedekind schemes which we are interested in arithmetic.

Example 3.1: Let R be a Dedekind domain, then $X = \text{Spec}(R)$ is a Dedekind scheme.

Example 3.2: Let k be a global field and S be a finite subset of places of k containing archimedean places ∞_k . Then, the scheme $U = \text{Spec}(\mathcal{O}_S)$ associated to the ring of S -integers $\mathcal{O}_S$ is a Dedekind scheme. Indeed, if k is a number field, $U = \text{Spec}(\mathcal{O}_S)$ is an affine open subscheme of $X = \text{Spec}(\mathcal{O}_k)$, where $\mathcal{O}_k$ is the ring of integers of k . If k is the function field of a smooth projective integral curve X over some finite field, $U = \text{Spec}(\mathcal{O}_S)$ is an affine open subscheme of X .

Example 3.3: Regular curves over a field are Dedekind schemes.

Example 3.4: Let $X = \text{Spec}(R)$ be a normal Noetherian local scheme of dimension 2 with closed point x , then $U = X \setminus \{x\}$ is a non-affine Dedekind scheme. More precisely, by the proof in the proposition (3.1), U is a normal Noetherian scheme. It is obvious that $\dim(U) = 1$ due to the absence of the maximal ideal s . Thus, U is a Dedekind scheme. It is claimed that U is not affine scheme. Otherwise, by proposition (3.1), U can be written as $U = \text{Spec}(A)$ for some Dedekind domain A and since $\text{codim}(s, X) = 2$, by an analogous of Hartogs' lemma [[53], Theorem 4.1.14], one has $R = \mathcal{O}_X(X) \simeq \mathcal{O}_X(U) = A$, which contradicts. In particular, $U = \text{Spec}(k[x, y]_{\mathfrak{m}}) \setminus \{\mathfrak{m}k[x, y]_{\mathfrak{m}}\}$ is a Dedekind scheme, where k is a field and $\mathfrak{m}$ is a prime ideal of $k[x, y]$.

Here are some basic properties of Dedekind schemes.

Proposition 3.2: Let X be a Dedekind scheme. Then:

- (1) X is a finite disjoint union of integral Dedekind schemes.
- (2) X is covered by finitely many affine open Dedekind scheme.
- (3) For any closed point $x \in X$, the local ring $\mathcal{O}_{X,x}$ is a DVR.
- (4) Every open subscheme of X is affine.

Proof: For (1), the irreducible components passing through a given point x are in bijection with the the minimal primes of $\mathcal{O}_{X,x}$ of which there is just the zero ideal. Thus, each x lies in exactly one irreducible component. For (2), it follows from the Noetherian property and (1) directly. For (3), it is obvious that $\mathcal{O}_{X,x}$ is the local ring of some affine open subscheme containing x , whose global section is a Dedekind domain by proposition (3.1). For (4), it is highly nontrivial and see for [[53], Exercise 4.1.9]. ■

For proper schemes over a Dedekind domain, their integral points are determined by that of rational points.

Proposition 3.3: Let R be an integral domain, and $K = \text{Frac}(R)$. Let X be an R -scheme. Then:

- (1) If X is separated over R , then $X(R) \rightarrow X(K)$ is injective.
- (2) If X is proper over R , and R is a Dedekind domain, then $X(R) \rightarrow X(K)$ is bijective.

Proof: (1) It follows from [[43], II.Theorem 4.7] or [[53], Proposition 3.3.11].

(2) Let $f \in X(K)$. It remains to extend $f : \text{Spec}(K) \rightarrow X$ to an R -morphism $\text{Spec}(R) \rightarrow X$. By spreading out of schemes of finite presentation over integral schemes [[77], Theorem 3.2.1], there exists an open dense subscheme $U \subset \text{Spec}(R)$ such that f extends to a U -morphism $f_U : U \rightarrow X_U$. It is claimed that $\text{Spec}(R) - U$ is a finite union of

non-zero prime ideals of R . Indeed, $\text{Spec}(R) - U$ is of the form $V(I)$ for some ideal I of R , whence, it is a classical result that I can be factored uniquely into the product of prime ideals $I = P_1^{e_1} P_2^{e_2} \cdots P_n^{e_n}$. Thus, one gets $\text{Spec}(R) - U = V(I) = \{P_1, P_2, \dots, P_n\}$. Now, it suffices to extend f_U to $U \cup \mathfrak{p}$ for one prime $\mathfrak{p}$ and we can repeat the same process.

For each closed point $\mathfrak{p}$, $R_{\mathfrak{p}}$ is a DVR and, by the valuative criterion of properness [[43], II.Theorem 4.7], f extends to a morphism $\text{Spec}(R_{\mathfrak{p}}) \rightarrow X$. Again, by spreading out to an open neighborhood of a point [[77], Remark 3.2.2], there exists an open dense subscheme $V \subset \text{Spec}(R)$ such that the morphism $\text{Spec}(R_{\mathfrak{p}}) \rightarrow X$ is spread out to an V -morphism $f_V : V \rightarrow X_V$. The restrictions of f_U and f_V to $U \cap V$ must agree by separatedness and $U \cap V \neq \emptyset$. Hence, one can glue them to obtain an extension to $U \cup V$, which contains both U and $\mathfrak{p}$. ■

Remark 3.1: Thanks to the author's advisor, Professor Yong Hu, who told me the essential technique of the "Spreading out", and see for [[53], Exercise 3.2.4, Exercise 3.2.5]. More precisely, if R is a Noetherian ring, X is a R -scheme of finite type, and $\mathfrak{p} \in \text{Spec}(R)$, then for every R -morphism $\alpha : \text{Spec}(R_{\mathfrak{p}}) \rightarrow X$, there exists $f \in R$ such that $f \notin \mathfrak{p}$ and an R -morphism $\tilde{\alpha} : \text{Spec}(R_f) \rightarrow X$, which extends α . It follows from the bijection between the set $X(R_{\mathfrak{p}})$ of $R_{\mathfrak{p}}$ -valued points of X and the set of pairs $(x, \mathcal{O}_{X,x} \rightarrow R_{\mathfrak{p}})$.

Furthermore generally, we have the following result analogous to proposition (3.3) if the base scheme S is a Dedekind scheme.

Proposition 3.4: Let S be a locally Noetherian scheme. Let T be an integral Dedekind S -scheme with the function field $K = \mathbf{k}(T)$. If X is a separated (resp. proper) S -scheme, then the canonical map $f : X(T) \rightarrow X(K)$ is an injection (resp. bijection).

Proof: It is clear that $X_T(T) = X(T)$ and $X_T(K) = X(K)$. Since the properness and separatedness are stable under base extension, without loss of generality, we may assume that X is a separated (resp. proper) scheme over T . By proposition (3.2), there are finitely many affine open covering $\{U_i = \text{Spec}(R_i)\}$ of T , where each R_i is a Dedekind domain with the same function field K . For any given two T -valued points of X : $\varphi_k : T \rightarrow X$ with $k = 1, 2$, satisfying $f(\varphi_1) = f(\varphi_2)$, it is obvious that $f(\varphi_k)$ factors through each U_i , then by proposition (3.3), one has $\varphi_1|_{U_i} = \varphi_2|_{U_i}$, thus $\varphi_1 = \varphi_2$ by separatedness and f is an injection. Conversely, if X is proper, let $\psi : \text{Spec}(K) \rightarrow X$ be a K -point of X . For each U_i , by proposition (3.3), ψ can extend to $\psi_i : U_i \rightarrow X$ and $\psi_i|_{U_i \cap U_j} = \psi_j|_{U_i \cap U_j}$ for any i, j , since $U_i \cap U_j$ is also affine due to the separatedness. Hence, one can glue ψ_i . ■

Corollary 3.1: Let S be a locally Noetherian scheme and T be an integral Dedekind S -scheme with its function field K . If X is proper over S , then every S -morphism $f : U \rightarrow X$ defined on a non-empty open subset U of X can extend uniquely to a morphism $\tilde{f} : T \rightarrow X$. In particular, every rational map from a regular curve to a proper variety is a morphism.

Proof: It follows from Proposition (3.4) fastly. ■

One of the most important subjects for studying integral models is to find out a suitable models for a given algebraic variety that admits a good reduction to deal with.

Definition 3.2: Let R be a Dedekind domain, and $K = \text{Frac}(R)$. Let $\mathfrak{p}$ be a nonzero prime ideal of R . One says that a smooth proper K -variety has **good reduction at $\mathfrak{p}$** if X has a smooth proper $R_{\mathfrak{p}}$ -model. One says that X has **good reduction** if it has good reduction at every nonzero prime ideal $\mathfrak{p}$.

The following lemmas are very useful when we pass from the local case to global.

Lemma 3.1: Let R be a local ring with its maximal ideal $\mathfrak{m}$, X be a R -scheme. If $\{X_i\}$ is an open covering of X , then $\bigcup X_i(R) = X(R)$.

Proof: It suffices to show that for a given $f : \text{Spec}(R) \rightarrow X$, f factors through some X_i uniquely. Since R is a local ring, $\text{Spec}(R)$ has a unique closed point $\mathfrak{m}$ and possibly other generic points. Let $x = f(\mathfrak{m})$, and $x \in X_i$ for some open subset X_i . Then $\mathfrak{m} \in f^{-1}(X_i)$, which is an open subset. But the only open neighbourhood of $\mathfrak{m}$ is the whole space $\text{Spec}(R)$. So $\text{Spec}(R) = f^{-1}(X_i)$. Thus, f factors through X_i uniquely. ■

Remark 3.2: The lemma 3.1 is not true for arbitrary rings.

Lemma 3.2: Let R be a Dedekind domain with $K = \text{Frac}(R)$. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n \in \text{Spec}(R)$ be nonzero prime ideals of R and $U = \text{Spec}(R) \setminus \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$. Suppose that we are given separated schemes of finite type $\mathbf{Y}$ over U , and $\mathbf{Y}_i$ over $R_{\mathfrak{p}_i}$. If $\mathbf{Y}$ and $\mathbf{Y}_i$ have the isomorphic generic fibers, namely given isomorphisms $f_i : \mathbf{Y}_K \rightarrow (\mathbf{Y}_i)_K$ between generic fibers, then, up to isomorphism, there exists a unique separated scheme of finite type $\mathbf{X}$ over R such that $\mathbf{X}_U = \mathbf{X} \times_R U \simeq \mathbf{Y}$ and $\mathbf{X}_{R_{\mathfrak{p}_i}} = \mathbf{X} \times_R R_{\mathfrak{p}_i} \simeq \mathbf{Y}_i$. Moreover, if $\mathbf{Y}, \mathbf{Y}_1, \dots, \mathbf{Y}_n$ are group schemes and f_i are isomorphisms of group schemes, then $\mathbf{X}$ is a group scheme over R .

Proof: Let $f_{ij} = f_j \circ f_i^{-1} : (\mathbf{Y}_i)_K \rightarrow (\mathbf{Y}_j)_K$, which is also an isomorphism by assumption. Then, by definition, one has $f_{jk} \circ f_{ij} = (f_k \circ f_j^{-1}) \circ (f_j \circ f_i^{-1}) = f_k \circ f_i^{-1} = f_{ik}$. By spreading out [[77], Theorem 3.2.1] or remark (3.1), there are open neighborhoods U_i of $\mathfrak{p}_i$ such that f_i extend to U_i , denoted by $\tilde{f}_i : \mathbf{Y} \times_R U_i \rightarrow \mathbf{Y}_i \times_R U_i$, which is also an isomorphism actually. On the affine pieces $U_i \cap U_j$, let $\tilde{f}_{ij} = \tilde{f}_j \circ \tilde{f}_i^{-1} : \mathbf{Y}_i \times_R (U_i \cap U_j) \rightarrow \mathbf{Y}_j \times_R (U_i \cap U_j)$.

Thus, the identity $\widetilde{f}_{ik} = \widetilde{f}_{jk} \circ \widetilde{f}_{ij}$ holds on the non-empty open subsets $U_i \cap U_j \cap U_k$. It follows from the usual gluing lemma [[53], Lemma 2.3.33] to construct the R -scheme $\mathbf{X}$ with open immersions $g_i : \mathbf{Y}_i \times_R U_i \hookrightarrow \mathbf{X}$. Namely, $\mathbf{X} = \cup_i g_i(\mathbf{Y}_i \times_R U_i)$, which is separated and finite type over R since $\mathbf{Y} \rightarrow U$ and $\mathbf{Y}_i \rightarrow R_{\mathfrak{p}_i}$ are. Moreover, through $\widetilde{f}_i$, one has $\mathbf{X}_U \simeq \mathbf{X} \times_R U \simeq \cup(\mathbf{Y}_i \times_R U_i) \times_R U \simeq \cup(\mathbf{Y} \times_R U_i) \times_R U = \mathbf{Y} \times_R U = \mathbf{Y}$. Similarly, one has $\mathbf{X}_{R_{\mathfrak{p}_i}} \simeq \mathbf{X} \times_R R_{\mathfrak{p}_i} \simeq \mathbf{Y}_i$. The assertion on the structure of group scheme is straightforward. ■

Proposition 3.5: Let R be a Dedekind domain and $K = \text{Frac}(R)$. Let X be a smooth proper K -variety. If X has good reductions, then there is a smooth proper R -model of X .

Proof: Since X admits a good reduction, there are smooth proper schemes $\mathbf{X}_i$ over $R_{\mathfrak{p}_i}$ for every non-zero prime ideal $\mathfrak{p}_i$ respectively. By the gluing process as the lemma (3.2), one can glue $\{\mathbf{X}_i\}$ to obtain an R -scheme $\mathbf{X}$, which is smooth over R since each special fiber of $\mathbf{X}$ is smooth. It is validated that properness is local on the base and $\mathbf{X}$ is proper over R from [[43], Corollary 4.8]. More precisely, $\mathbf{X}_i$ is proper over R since $\text{Spec}(R_{\mathfrak{p}_i}) \rightarrow \text{Spec}(R)$ is separated, and then by construction, $\mathbf{X}_i \times_R U_i \rightarrow U_i$ is a proper morphism for some open subscheme $\mathfrak{p}_i \in U_i$. By local property of properness on the base, $\mathbf{X}$ is proper over R . ■

In general, there does not always exist a good reduction of algebraic varieties, even for nice curves, and many examples are discussed by Liu, Poonen, and Silverman in [64] and [62]. It is well-known, named after Shafarevich Conjecture, that the number of isomorphic classes of elliptic curves over a given number field with good reductions outside a finite set of primes is finite. Also, for abelian varieties, the similar results are shown in [[29], Theorem 6], based on which Faltings proved the Mordell Conjecture.

Besides, the existence of good reduction can be detected by some unramified cohomology groups, usually the first and the second cohomology group. For example, for an abelian variety A over a fraction field K of some discrete valuation ring R whose residue field is k , by the Neron-Ogg-Shafarevich criterion [[6], Theorem 7.4.5], A has good reductions if and only if for each prime $l \neq \text{Char}(k)$, the first cohomology group $H^1(A_{\overline{K}}, \mathbb{Q}_l)$ is unramified, equivalently, the Tate module $T_l(A) = \varprojlim_{\nu} {}_l A(K^\nu)$ is unramified over R , where ${}_l \nu$ denotes the kernel of the l^ν -multiplication $l_A^\nu : A \rightarrow A$. Similar results has been demonstrated for $K3$ surfaces by Matsumoto^[65].

3.2 Structure theorem of S-adelic spaces

Thanks to integral models, one could describe the adelic points more precisely. In this section, let k be a global field and Ω_k denote the set of all places of k . Let S be a non-empty finite subsets of places of k containing ∞_k .

3.2.1 Adelic points and adelic spaces

Definition 3.3: Let

$$\mathbf{A}_{k,S} = \left(\prod_{v \in S} k_v \right) \times \left(\prod_{v \notin S} \mathcal{O}_v \right)$$

Call $\mathbf{A}_{k,S}$ **the ring of S-adèles**, which is a subring of $\mathbf{A}_k$. Obviously, $\mathbf{A}_k = \varinjlim \mathbf{A}_{k,S}$. (1)

Let X be an algebraic variety over k and $\mathbf{X}$ be an integral model of X over $\mathcal{O}_S$.

Definition 3.4: The set of adelic points on X is defined as:

$$X(\mathbf{A}_k) = \text{Hom}_k(\text{Spec}(\mathbf{A}_k), X) \subset \prod_{v \in \Omega_k} X(k_v)$$

which can be canonically equipped with a locally compact Hausdorff topology [[23], section 4]. One may also refer to $X(\mathbf{A}_k)$ as **the adelic space of X** .

When X is affine, the adelic topology of $X(\mathbf{A}_k)$ is nothing but the analytic subspace topology of affine spaces. When X is proper, $X(\mathbf{A}_k)$ is compact, and by the valuative criterion for properness [[43], Theorem II.4.7], one has $X(\mathbf{A}_k) = \prod_{v \in \Omega_k} X(k_v)$.

Let T be another finite subset of Ω_k containing S . It is clear that an $\mathbf{A}_{k,T}$ -point of $\mathbf{X}$ gives an $\mathbf{A}_k$ -point of $\mathbf{X} \times_{\mathcal{O}_S} \mathbf{A}_k$. Indeed, $\mathbf{A}_k$ -points of $\mathbf{X} \times_{\mathcal{O}_S} \mathbf{A}_k$, the adelic points of integral model $\mathbf{X}$, can be identified with $\mathbf{A}_k$ -points of X because one has $\mathbf{X} \times_{\mathcal{O}_S} \mathbf{A}_k = (X \times_{\mathcal{O}_S} k) \times_k \mathbf{A}_k = X \times_k \mathbf{A}_k$ thanks to $\mathcal{O}_S \subset k \subset \mathbf{A}_k$. Hence, to study the adelic points of X , it suffices to consider the adelic points of the integral model $\mathbf{X}$. Namely, it gives rise to a bijective map^[23]:

$$\varinjlim \mathbf{X}(\mathbf{A}_{k,T}) \rightarrow \mathbf{X}(\mathbf{A}_k) = X(\mathbf{A}_k) = X(\varinjlim \mathbf{A}_{k,T})$$

where the direct limit runs over finite subset T containing S and the construction of the adelic space $X(\mathbf{A}_k)$ is independent on S .

Similarly, one also has $\mathbf{X}(\mathbf{A}_k^S) = X(\mathbf{A}_k^S)$ and $\mathbf{X}(k) = X(k)$.

More precisely, the following natural map is bijective, see [[23], Thm 3.6]:

$$\mathbf{X}(\mathbf{A}_{k,T}) \xrightarrow{\sim} \left(\prod_{v \in T} X(k_v) \right) \times \left(\prod_{v \notin T} \mathbf{X}(\mathcal{O}_v) \right)$$

which gives T -adelic points $\mathbf{X}(\mathbf{A}_{k,T})$ a locally compact Hausdorff topology. If $T \subset T'$, then $\mathbf{X}(\mathbf{A}_{k,T}) \rightarrow \mathbf{X}(\mathbf{A}_{k,T'})$ is an open embedding. In this way, one can see that a subset of $X(\mathbf{A}_k)$ is open if its intersection with each $\mathbf{X}(\mathbf{A}_{k,T})$ is open. Thus, $\mathbf{X}(\mathbf{A}_{k,T})$ form an open covering of $X(\mathbf{A}_k)$ when T runs over finite subset of Ω_k containing S .

3.2.2 Main theorem I

Now, we relate the analytic open subsets to integral points of integral models.

Proposition 3.6: Let R be a discrete valuation ring with a uniformizer π , its residue field k and fraction field $K = \text{Frac}(R)$. Let $\widehat{R}$ be its completion and $\widehat{K} = \text{Frac}(\widehat{R})$. Let X be an algebraic variety over K . For each $\widehat{K}$ -point $q \in X(\widehat{K})$ and an open neighborhood U of q with respect to the analytic topology induced by $\widehat{K}$, there exists an integral model $\mathbf{X}$ of X over R such that $q \in \mathbf{X}(\widehat{R}) \subseteq U$.

Proof: Firstly, we construct an integral model $\mathbf{X}_0$ of X over R such that $q \in \mathbf{X}_0(\widehat{R})$. Let $W \subseteq X$ be an affine Zariski open neighborhood of $q \in X(\widehat{K})$. Since W has a finite description involving only finitely many polynomials over K , one can write each coefficient of those as a fraction of elements of R , and let R' be the localization of R obtained by adjoining the inverses of all the denominators that appear. Then W extends to an affine model $\mathbf{W}$ over some open dense subscheme $V = \text{Spec}(R')$ of $\text{Spec}(R)$. As shown in the example (2.10), up to scaling a system of coordinates on W by multiplying a power of the uniformizer, we can assume that $q \in \mathbf{W}(\widehat{R})$. Thus, we can glue X and $\mathbf{W}$ along W to obtain an integral model $\mathbf{X}_0$ of X over R with $q \in \mathbf{X}_0(\widehat{R})$.

It remains to refine and shrink $\mathbf{X}_0$. It is claimed that there exists a finite sequence of dilatations (2.5):

$$\mathbf{X}_d \rightarrow \mathbf{X}_{d-1} \rightarrow \cdots \rightarrow \mathbf{X}_0 \quad (3-1)$$

such that $\mathbf{X}_d(\widehat{R}) \subseteq U$.

Let $x_0 \in (\mathbf{X}_0)_k(k)$ be the specialization of q at the special fiber $(\mathbf{X}_0)_k$ of $\mathbf{X}_0$, namely the image of the closed point of $\text{Spec}(\widehat{R})$ by $q : \text{Spec}(\widehat{R}) \rightarrow \mathbf{X}_0$, the singleton consisting of which can be endowed with a reduced closed subscheme structure of $(\mathbf{X}_0)_k$, denoted by $Y_k = \text{Spec}(\mathbf{k}(x_0))$. Let $u_1 : \mathbf{X}_1 \rightarrow \mathbf{X}_0$ be the dilatation of Y_k on $\mathbf{X}_0$. It is obvious that $\widehat{R}$ is torsion-free over R , hence flat over R by lemma (4.1). By the universal property of the dilatation (2.4), any $p \in \mathbf{X}_0(\widehat{R})$, whose specialization in $(\mathbf{X}_0)_k(k)$ is equal to x_0 , can be lift to $\mathbf{X}_1$ uniquely. Conversely, for any $p \in \mathbf{X}_1(\widehat{R})$, by theorem (2.4), the image of p under the canonical map $\mathbf{X}_1(\widehat{R}) \rightarrow \mathbf{X}_0(\widehat{R})$ must specialize into x_0 at $(\mathbf{X}_0)_k(k)$,

which is the image of $(u_1)_k : (\mathbf{X}_1)_k \rightarrow (\mathbf{X}_0)_k$. Hence, the canonical map $u_1(\widehat{R}) : \mathbf{X}_1(\widehat{R}) \rightarrow \mathbf{X}_0(\widehat{R}) \subseteq X(\widehat{R})$ is injective, whose image precisely consists of the points of $X(\widehat{R})$ with specializations x_0 in $(\mathbf{X}_0)_k(k)$. In particular, $q \in \mathbf{X}_1(\widehat{R})$. Thus, we can define inductively a sequence $u_{i+1} : \mathbf{X}_{i+1} \rightarrow \mathbf{X}_i$ of dilatation of $Y_i = \text{Spec}(\mathbf{k}(x_i))$ on $\mathbf{X}_i$, where $x_i \in (\mathbf{X}_i)_k(k)$ is the specialization of q .

Let us show that $\mathbf{X}_d(\widehat{R}) \subseteq U$ when d is large enough. As the dilatation commutes with flat base change (2.11), we can assume that R is a complete DVR in what follows. By construction (2.4), the dilatation $u_{i+1} : \mathbf{X}_{i+1} \rightarrow \mathbf{X}_i$ is obtained by gluing along affine bases, and for any closed subscheme Z_i of $(\mathbf{X}_i)_k$ and any open subscheme $\mathbf{W}$ of $\mathbf{X}_i$, the dilatation of $Z_i \cap \mathbf{W}$ on $\mathbf{W}$ is the restriction to $\mathbf{W}$ of the dilatation $u_{i+1} : \mathbf{X}_{i+1} \rightarrow \mathbf{X}_i$. So, it suffices to make the base affine and do it locally. Without loss of generality, we can assume that $\mathbf{X}_0$ is affine, so is X . Then, there is a closed immersion $\tau_0 : \mathbf{X}_0 \hookrightarrow \mathbf{A}_0 = \text{Spec}(R[t_1, \dots, t_n])$ such that $\tau_0(q)$ is the original point of $\mathbf{A}_0(R) = R^n$. For any integer $i \geq 0$, let $B(0, i) = \pi^i R^n \subseteq \mathbf{A}_0(R) = R^n$, which form a basis of open neighborhoods of $\tau_0(q)$ under analytic topology of $\mathbf{A}_0(K) = K^n$. Since the analytic topology on $\mathbf{X}_0(R)$ coincides with the topology induced from $X(K)$, both of which are induced from K^n essentially, there exists $d \geq 0$ such that $\tau_0^{-1}(B(0, d)) = B(0, d) \cap \mathbf{X}_0(R) \subseteq U$, where $\mathbf{X}_0(R) \subseteq X(K)$, because $\mathbf{X}_0$ is separated over R .

By construction of the dilatation (2.4) and example (2.10), $\mathbf{X}_d$ is affine over R and τ induces a closed immersion:

$$\tau_d : \mathbf{X}_d \hookrightarrow \mathbf{A}_d := \text{Spec}\left(R\left[\frac{x_1}{\pi^d}, \dots, \frac{x_n}{\pi^d}\right]\right) \quad (3-2)$$

Therefore, we have

$$q \in \mathbf{X}_d(R) \subseteq \tau_d^{-1}(\mathbf{A}_d(R)) \cap X(K) = \tau_0^{-1}(B(0, d)) \cap X(K) \subseteq U.$$

where the equality follows from (2.10) that $\mathbf{A}_0(R) = B(0, d)$. ■

Remark 3.3: Let X be an R -scheme of finite type and Y_k be a closed subscheme of X_k . Let $\mathbf{X}$ be the dilatation of Y_k on X . From the univocal property of dilatation (2.4) and the same argument in the proof (3.6), it is validated that for each extension R'/R of ramification index 1 with residue fields k'/k as defined in [[6], Definition 3.6.1], the canonical map $\mathbf{X}(R') \rightarrow X(R)$ is always injective, whose image consists of the points of $X(R')$ specializing into $Y_k(k')$. Namely, $\mathbf{X}(R')$ corresponds to $\{x \in X(R') \mid x \in Y_k(k')(\text{mod } \pi)\}$ bijectively.

Applying the very local proposition (3.6) above, one can deal with the S -adelic space

locally by choosing a suitable integral model. In particular, the integral models provide a topological basis of the S -adelic space $X(\mathbf{A}_k^S)$ of algebraic variety X .

Theorem 3.1: Let X be an algebraic variety over a global field k and S be a finite subset of places of k containing ∞_k . For any prime-to- S adelic point $z \in X(\mathbf{A}_k^S)$ and an open neighborhood $U \subseteq X(\mathbf{A}_k^S)$ of z , there exists an integral model $\mathbf{X}$ of X over $\mathcal{O}_S$ such that

$$z \in \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \subseteq U.$$

In other words, the subsets $\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \subseteq X(\mathbf{A}_k^S)$, where $\mathbf{X}$ runs through all integral models of X over $\mathcal{O}_S$, form a basis of the topological space $X(\mathbf{A}_k^S)$.

Moreover, if X is a (smooth) algebraic group over k , then the subsets of the form $\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v)$, where $\mathbf{X}$ runs through all (smooth) group schemes model of X over $\mathcal{O}_S$, form a fundamental system of neighborhood of 1.

Proof: Since U is an open neighborhood of $z \in X(\mathbf{A}_k^S)$, by definition of the adelic topology that the open subsets of $X(\mathbf{A}_k^S)$ come from that of adelic space $X(\mathbf{A}_k)$ by the projection $P_S : X(\mathbf{A}_k) \rightarrow X(\mathbf{A}_k^S)$, there exists an integral model $\mathbf{Y}$ of X over $\mathcal{O}_T$ for some finite subset of Ω_k containing S such that

$$P_S \left(\prod_{v \in T} U_v \times \prod_{v \notin T} \mathbf{Y}(\mathcal{O}_v) \right) = \prod_{v \in T \setminus S} U_v \times \prod_{v \notin T} \mathbf{Y}(\mathcal{O}_v) \subseteq U$$

where U_v is non-empty open subset of $X(k_v)$ for each $v \in T \setminus S$.

One can write the prime-to- S adelic point $z \in X(\mathbf{A}_k^S)$ as $z = (z_v)_{v \notin S}$. For any $v \in T \setminus S$, by proposition (3.6), there exists an integral model $\mathbf{X}_v$ of X over the discrete valuation ring $\mathcal{O}_{k,v}$ such that $z_v \in \mathbf{X}_v(\mathcal{O}_v) \subseteq U_v$. It follows from the lemma (3.2), where $U = \text{Spec}(\mathcal{O}_S) \setminus (T \setminus S) = \text{Spec}(\mathcal{O}_T)$, that we obtain an integral model $\mathbf{X}$ of X over $\mathcal{O}_S$ by gluing the schemes $\mathbf{X}_v, v \in T \setminus S$ with $\mathbf{Y}$ along the generic fiber X .

The case of algebraic groups is proved in the similar way but using dilatation along the unit section. ■

Furthermore, if we define the adelic space with respect to an algebraic variety over some function field of an integral Dedekind scheme, the structure theorem (2.4) holds.

Definition 3.5: Let T be an integral Dedekind scheme with the generic point ξ and the function field $K = \mathbf{k}(\xi)$. Let X be an algebraic variety over K . Let S be a finite subset of

T containing ξ . Define **the prime-to- ξ adelic space of X** as the restriction product:

$$X(\mathbf{A}_K^\xi) = \varinjlim_S \left(\prod_{t \in S \setminus \{\xi\}} X(K_t) \right) \times \left(\prod_{t \notin S} \mathbf{X}(\widehat{\mathcal{O}_{T,t}}) \right)$$

where $K_t = \text{Frac}(\widehat{\mathcal{O}_{T,t}})$ for each point $t \in T$ and $\mathbf{X}$ is a separated scheme of finite type over some open dense subset $T \setminus (S \setminus \{\xi\})$ of T .

Example 3.5 (Varieties over d -local function fields): A 0-local field is a finite field. For an integer $d > 0$, a **d -local field** is a complete discrete valuation field with its residue field a $(d - 1)$ -local field. Thus 1-local fields are just local fields in the usual sense. Throughout, let X be a smooth projective geometrically integral curve over some d -local field. Let $K = k(X)$ be the function field of X . For any closed point v of X , let K_v be the completion of K with respect to v , which is a $(d + 1)$ -local field.

Theorem 3.2: The theorem also holds in the setting of (3.5).

Proof: See Theorem (3.1). ■

At last of this section, we indicate that thanks to the structure theorem (3.1), it is established some basic properties of SAP-BM (5.1) (resp. VSAP-BM (5.2)) which serve as a bridge between VSAP-BM (1.9) and the Harari-Voloch conjecture (1.1).

3.3 Regular proper models and Néron models

Now, to study the integral Hasse principle and the Harari-Voloch conjecture, it is necessary to discuss whether the choice of different integral models will make the obstruction set different and how to select a suitable integral model. In general, there are usually two types of integral models considerable, one of which is the smooth model, and the other is the proper model. However, it is difficult to make an integral smooth and proper simultaneously as pointed out before in the page 37. In the following, we will carry out a discussion on it, in particular, with regard to the integral Hasse principle and the Harari-Voloch conjecture in this section and chapter 4.

3.3.1 Smooth models

Theorem 3.3: Let R be a Dedekind domain with $K = \text{Frac}(R)$. Let X be a smooth algebraic variety (resp. smooth algebraic group) over K . For any integral R -model $\mathbf{X}$ of X , there exists a birational morphism $f : \tilde{\mathbf{X}} \rightarrow \mathbf{X}$ such that $\tilde{\mathbf{X}}$ is a smooth (group) integral model of X over R . Moreover, for any non-zero prime ideal $\mathfrak{p}$ and local ring extension

R'_p/R_p of ramification index 1, the canonical maps $\tilde{\mathbf{X}}(R'_p) \rightarrow \mathbf{X}(R'_p)$ and $\tilde{\mathbf{X}}(R) \rightarrow \mathbf{X}(R)$ are bijective. In particular, the canonical map $\tilde{\mathbf{X}}(\widehat{R_p}) \rightarrow \mathbf{X}(\widehat{R_p})$ is bijective.

In addition, if $\mathbf{X}(R_p) \neq \emptyset$ for each prime ideal p of R , then $\tilde{\mathbf{X}}$ is faithfully flat over R . If X is a smooth proper algebraic variety over K , f is an open immersion followed by a proper morphism.

Proof: By spreading out in [[77], Theorem 3.2.1(ii)], there exist a dense open subscheme $U \subseteq \text{Spec}(R)$ such that $\mathbf{X}_U = \mathbf{X} \times_R U$ is smooth over U . Since R is a Dedekind domain, there are only finitely many (non-zero) prime ideals not contained in U , denoted by $p_1, \dots, p_n$, and for each such p_i , $\mathbf{X}_{p_i} = \mathbf{X} \times_R R_{p_i}$ is an integral model of X over R_{p_i} . It is reduced to the local case. For each discrete valuation ring R_{p_i} , it follows from the smoothening process (2.3), that there exists a proper birational morphism $f_i : \mathbf{X}'_{p_i} \rightarrow \mathbf{X}_{p_i}$. Thus, one can take the smooth locus of $\mathbf{X}'_{p_i}$, denoted by $\widetilde{\mathbf{X}}_{p_i}$, and by the gluing process (3.2) and (3.5), one can glue $\widetilde{\mathbf{X}}_{p_i}$ and $\mathbf{X}_U$ along the generic fiber X to obtain a smooth R -scheme $\tilde{\mathbf{X}}$ with the generic fiber $\tilde{\mathbf{X}}_K \simeq X$ and a birational morphism $f : \tilde{\mathbf{X}} \rightarrow \mathbf{X}$. Moreover, for any non-zero prime ideal p and ring extension R'_p/R_p of ramification index 1, if $p \in U$, then $\tilde{\mathbf{X}}(R'_p) = \mathbf{X}_U(R'_p) = \mathbf{X}(R'_p)$. If $p \notin U$, by [[6], Corollary 3.6.6], the smoothening process commutes with base change of ramification index 1, thus we have

$$\tilde{\mathbf{X}}(R'_p) = \widetilde{\mathbf{X}}_p(R'_p) = (\widetilde{\mathbf{X}}_p)_{R'_p}(R'_p) = (\widetilde{\mathbf{X}}_p)_{R'_p}(R'_p) = (\mathbf{X}_p)_{R'_p}(R'_p) = \mathbf{X}_p(R'_p) = \mathbf{X}(R'_p)$$

where the third equality follows from the base change theorem above and the fourth one follows from the univocal property of smoothening process (2.3). Note that $R = \bigcap_p R_p$, it is obvious that the canonical map $\tilde{\mathbf{X}}(R) \rightarrow \mathbf{X}(R)$ is bijective.

Moreover, if $\mathbf{X}(R_p) \neq \emptyset$ for each prime ideal p of R , then so is $\tilde{\mathbf{X}}(R_p) \neq \emptyset$, which means that the structure morphism $\tilde{\mathbf{X}} \rightarrow \text{Spec}(R)$ is surjective because for each p , the structure morphism $\text{Spec}(R_p) \rightarrow \text{Spec}(R)$ factors through $\tilde{\mathbf{X}}$. By [[34], Theorem 14.24], since the smoothness implies the flatness, $\tilde{\mathbf{X}}$ is also flat over R . Hence, in this case, $\tilde{\mathbf{X}}$ is faithfully flat over R . If X is proper over K , one may assume $\mathbf{X}_U$ is smooth proper over U . It is easy to see that the smooth R -scheme $\tilde{\mathbf{X}}$ is precisely equal to the smooth locus of the R -scheme $\mathbf{X}'$, which is obtained by gluing $\mathbf{X}_U$ and $\mathbf{X}'_{p_i}$ along the generic fiber X . Since the properness is local on the base, f is an open immersion $\mathbf{X}'_{\text{smooth}} \rightarrow \mathbf{X}'$ followed by a proper morphism $f' : \mathbf{X}' \rightarrow \mathbf{X}$.

If X is furthermore a smooth algebraic group, and $\mathbf{X}$ is a (separated) group scheme, then $\tilde{\mathbf{X}}$ can be chosen to be a smooth group scheme, which depends on the existence of group smoothening process in [[6], 7.1.5]. The existence over a Dedekind base scheme

follows from the local case also by the gluing process (3.2). It is also true that $\tilde{\mathbf{X}}(R'_p) = \mathbf{X}(R'_p)$ for each non-zero prime ideal p and local extension R'_p/R_p of ramification index 1 because the group smoothening is constructed by a finite sequence of dilatations, which can commute with the passage to base change of the ramification index 1. ■

Remark 3.4: For one thing, the fact that the structure morphism $\mathbf{X} \rightarrow \operatorname{Spec}(R)$ is surjective can not imply $\mathbf{X}(R_p) \neq \emptyset$ for each prime ideal p in general, whose counterexample is the surjective morphism $\mathbf{X} = \operatorname{Spec}(\mathbb{Z}[\sqrt{-1}]) \rightarrow \operatorname{Spec}(\mathbb{Z})$ but with $\mathbf{X}(\mathbb{Z}_{(2)}) = \emptyset$. For another thing, the existence of faithfully flat integral models of any algebraic variety X (without the hypothesis of smoothness) over the fraction field K of some Dedekind domain R follows from the Main theorem II and its corollary. Contrary to our expectations, even if X is a smooth proper algebraic variety over K and then we can start with a smooth proper model over some open dense subscheme U of $\operatorname{Spec}(R)$, the very local smoothening process will produce a non-proper smooth model in general, however, the compactification process gives rise to singularities and violates the smoothness, which obstructs the existence of good reduction of X .

Remark 3.5: Theorem (3.3) takes inspiration from [[57], Remark 2.6]. With valuable discussions with my advisor, Professor Yong Hu, we suspect that there might be a minor oversight in their argument — specifically, they claimed that $f : \tilde{\mathbf{X}} \rightarrow \mathbf{X}$ is a quasi-projective morphism. However, as pointed out in [[34], Remark 13.70], the property of being quasi-projective (resp. projective) of morphism of schemes is not local on the base scheme. Indeed, Hironaka gave an example of a birational proper non-projective morphism $f : X \rightarrow S$ of proper schemes over $\mathbb{C}$, where X and S are smooth over $\mathbb{C}$ and irreducible of dimension 3, where S is additionally projective over $\mathbb{C}$ and where $S = U_1 \cup U_2$, with U_i is affine open such that the restriction $f^{-1}(U_i) \rightarrow U_i$ of f is projective. (see [[43], Appendix B, Example 3.4.1]) Besides, there exist proper morphisms which are not quasi-projective with the earliest counterexamples due to Nagata and a simple class of more recent counterexamples given by toric varieties associated to non-polyhedral fans. Thus, in the process (3.3), although the smoothening process provides quasi-projective morphisms f_i on the local, there is insufficient information available to account for f being of quasi-projective.

Example 3.6 (Related to the integral Hasse principle): Let X be an algebraic variety over a global field k with a finite subset S of places containing ∞_k . When X has local points at every $v \notin S$, it is natural to study the integral points of X over $\mathcal{O}_S$. If X is

smooth over K , it is reasonable to restrict ourselves to smooth models in the sense that for any given integral model $\mathbf{X}$ of X over $\mathcal{O}_S$ with $\mathbf{X}(\mathcal{O}_v) \neq \emptyset$ for any $v \notin S$, by theorem (3.3), there exists a birational morphism $f : \tilde{\mathbf{X}} \rightarrow \mathbf{X}$ of R -models of X such that $\tilde{\mathbf{X}}$ is smooth over R and the canonical maps $\tilde{\mathbf{X}}(\mathcal{O}_S) \rightarrow \mathbf{X}(\mathcal{O}_S)$ and $\tilde{\mathbf{X}}(\mathcal{O}_v) \rightarrow \mathbf{X}(\mathcal{O}_v)$ are bijective. In particular, $\tilde{\mathbf{X}}(\mathcal{O}_v) \neq \emptyset$ for any $v \notin S$. Furthermore, $\tilde{\mathbf{X}}$ is also faithfully flat over $\mathcal{O}_S$. If X is proper over K , by theorem (3.3), f is an open immersion followed by a proper morphism.

3.3.2 Desingularization and regular proper models

For a smooth proper variety, it is possible to find no good reduction since a smooth proper model does not always exist, but we can find models with weaker properties.

Lemma 3.3: Let X be a scheme and Z a closed subset of X . Then for all $x \in X$, we have

$$\text{codim}(\overline{\{x\}}, X) = \dim(\mathcal{O}_{X,x}) = \text{ht}(\mathfrak{m}_x) = \sup \{n | \mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_n \subseteq \mathfrak{m}_x; \mathfrak{p}_i \in \text{Spec}(\mathcal{O}_{X,x})\}$$

$$\text{codim}(Z, X) = \min_{z \in Z} \dim(\mathcal{O}_{X,z})$$

Proof: Let $U = \text{Spec}(A)$ be an affine open subset of X containing x . Since $\overline{\{x\}}$ is an irreducible closed subset of X , the codimension $\text{codim}(\overline{\{x\}}, X)$ is defined as the supremum of lengths of strictly ascending chains consisting of irreducible closed subsets between $\overline{\{x\}}$ and X , between which and chains of irreducible closed subsets of U containing x , there exists a bijection given by intersecting with U for one direction and taking Zariski closures in X for the other. Namely, suppose that there is a chain of irreducible closed subsets:

$$\overline{\{x\}} \subsetneq U_1 \subsetneq U_2 \subsetneq \cdots \subsetneq U_n \subsetneq X. \quad (3-3)$$

One can intersect (3-3) with U , then get a chain of irreducible closed subsets of U containing x . Conversely, if there is a such chain in U , one can take the Zariski closure in X without changing the length of chains since there is no irreducible closed subset V containing U , otherwise $V = (V \setminus U) \cup \overline{U}$. So, it suffices to study in U locally. Let x be a prime ideal in A . By Hilbert's Nullstellensatz, the irreducible closed subsets of U containing x correspond bijectively to the prime ideal of A containing x . Hence, $\text{codim}(\overline{\{x\}}, X) = \text{ht}(x) = \dim(A_x) = \dim(\mathcal{O}_{X,x})$. For every closed subset Z of X , by the same argument, one has $\text{codim}(Z, X) = \min_{z \in Z} \text{codim}(\overline{\{z\}}, X)$. ■

Proposition 3.7: Let X be a normal scheme. Then its singular locus $\text{Sing}(X)$ has codimension at least two. In other words, one has $\text{codim}(\text{Sing } X, X) \geq 2$.

Proof: For any point $x \in X$ of codimension 1, $\mathcal{O}_{X,x}$ is a normal local ring of dimension

1 since one has $\text{codim}(\overline{\{x\}}, X) = \dim(\mathcal{O}_{X,x})$ by Lemma (3.3). It follows that $\mathcal{O}_{X,x}$ is a discrete valuation ring by [[3], Proposition 9.2], hence a regular local ring. Thus, X is regular at each point of codimension 1 and $\text{codim}(\text{Sing}(X), X) \geq 2$. ■

Theorem 3.4: Let R be a complete discrete valuation ring, $K = \text{Frac}(R)$. If X is a smooth projective integral curve over K , there admits a regular proper R -model of X .

Proof: Since X is a projective curve over K , one can choose a closed immersion $X \hookrightarrow \mathbb{P}_K^n$ for some n . It is obvious that $\mathbb{P}_K^n$ is the generic fiber of $\mathbb{P}_R^n$, then one has the embeddings:

$$X \hookrightarrow \mathbb{P}_K^n \hookrightarrow \mathbb{P}_R^n$$

Take X'_R as the Zariski closure of X in $\mathbb{P}_R^n$. In other words, X'_R is the scheme-theoretic image of X in $\mathbb{P}_R^n$ by Proposition (4.1). Thus, X'_R is a proper R -model of X . Take the normalization X''_R of X'_R , whose normalization morphism $\pi : X''_R \rightarrow X'_R$ is finite because X'_R is an integral excellent scheme. Indeed, by [[53], Theorem 8.2.39], a complete discrete valuation ring R is an excellent ring and any locally of finite type scheme over an excellent ring is also excellent. Thus, X''_R is proper over R . However, the normalization morphism $\pi : X''_R \rightarrow X'_R$ is actually birational by [[53], Proposition 4.1.22], thus by Lemma (2.5), one has $X''_R \times_R K \simeq X'_R \times_R K \simeq X$. So, the arithmetic surface X''_R is a normal proper R -model of X but regular except at isolated closed points, because the $\text{codim}(\text{Sing}(X''_R), X''_R) \geq 2$ by Proposition (3.7). The existence of the regular proper R -model of X is produced from the resolution of singularities for arithmetic surfaces^[50], alternately blowing up singularities and normalizing. Consider the following sequence of proper birational morphisms:

$$\cdots \rightarrow X_{n+1} \rightarrow X_n \rightarrow \cdots \rightarrow X_1 \rightarrow X'_R, \quad (3-4)$$

where $X_1 = X''_R$, $X_1 \rightarrow X'_R$ is the normalization of X'_R , and for every $i \geq 1$, $X_{i+1} \rightarrow X_i$ is the blowing-up $X'_i \rightarrow X_i$ along the singular locus $\text{Sing}(X_i) = X_i \setminus \text{Reg}(X_i)$ followed by the normalization $X_{i+1} \rightarrow X'_i$. By Lipman's result^[50], remark (3.6), the sequence (3-4) stops if n is sufficiently large, where X_n is regular and morphisms are proper and birational. ■

Definition 3.6: Let X be a reduced Noetherian scheme. A proper birational morphism $\pi : Z \rightarrow X$ with regular scheme Z is called a **desingularization of X or a resolution of singularities of X** , where the birational morphism means that π is of finite type and if $\xi_1, \xi_2, \dots, \xi_n$ are generic points of X , then Z admits exactly n generic points $\xi'_1, \xi'_2, \dots, \xi'_n$ such that $\pi^{-1}(\xi_i) = \xi'_i$ as well as $\mathcal{O}_{X,\xi_i} \rightarrow \mathcal{O}_{Z,\xi'_i}$ is an isomorphism for each i . If π is an isomorphism above $\text{Reg}(X)$, one says that it is a **desingularization in the strong form**.

Remark 3.6 (Lipman,[50] [51]): Let X be an excellent, reduced, Noetherian scheme

of dimension 2. Consider the following sequence of proper birational morphisms

$$\cdots \rightarrow X_{n+1} \rightarrow X_n \rightarrow \cdots \rightarrow X_1 \rightarrow X, \quad (3-5)$$

where $X_1 \rightarrow X$ is the normalization of X and X_{i+1} is the normalization of the blowing-ups of X_i along $\text{Sing}(X_i)$ for every $i \geq 1$. Then the main results of Lipman is that the sequence (3-5) stops for sufficiently large n with regular schemes X_n . In particular, X **admits a desingularization in the strong form**. Indeed, as the paper's reviewer Wallace says, Lipman's object is the following theorem: If Y is a surface with the normalization $\bar{Y}$, there exists a desingularization if and only if the following conditions hold:

- the normalization morphism $\bar{Y} \rightarrow Y$ is finite;
- the normalization $\bar{Y}$ has at most finitely many singular points;
- for every $y \in \bar{Y}$, the completion $\widehat{\mathcal{O}_{\bar{Y},y}}$ of the local ring $\mathcal{O}_{\bar{Y},y}$ is normal.

Since proofs of the desingularization theorem were already known at that time, the essential point of Lipman's paper was the method of proof. The main step is to show that, among all normal proper surfaces which are birationally equivalent to Y , there is one surface Z whose arithmetic genus is minimal. It is then shown that this minimal property of Z is equivalent to the property that all singularities of Z are **pseudo-rational**, which could be removed by a sequence of blowing-ups. More precisely, if Y is a surface with only pseudo-rational singular points, finite in number, such that, for each singular point y , the completion of the local ring $\mathcal{O}_{\bar{Y},y}$ is normal, there exists an iterated blowing-up of Y which is regular. For surfaces over a field, it was proven independently by Zariski and Abhyankar^[52]. For the main result of Lipman, see the presentation by M. Artin^[2].

Here, we provided a standard process (3-4) in the proof of Theorem (3.4) by the methods of blowing-ups and normalizations as mentioned in Remark (3.6).

Lemma 3.4: Let $f : X \rightarrow Y$ be an affine morphism of noetherian separated schemes. Then, for any quasi-coherent sheaf $\mathcal{F}$ on X , there are natural isomorphisms for all $i \geq 0$,

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

In particular, if f is a closed immersion or a finite morphism, one has the above results.

Proof: Take an affine covering $\mathcal{U} = \{U_i\}_{i=1}^n$ of Y . Since f is affine morphism, $\mathcal{U}' = \{f^{-1}(U_i)\}$ is also an affine covering of X . Also, since affine morphism is quasi-compact and separated, $f_*\mathcal{F}$ is also a quasi-coherent sheaf on Y by [[43], Proposition II.5.8]. By [[43], Theorem III.4.5], it is clear that the Čech cohomology coincide with the derived sheaf cohomology because X and Y are both separated, and $\mathcal{F}$ and $f_*(\mathcal{F})$ are both quasi-

coherent. Hence, we can apply the Čech complexes

$$H^i(X, \mathcal{F}) \cong \check{H}(\mathcal{U}', \mathcal{F}) = h^i(\mathcal{C}^\cdot(\mathcal{U}', \mathcal{F})) = h^i(\mathcal{C}^\cdot(\mathcal{U}, f_*\mathcal{F})) \cong \check{H}(\mathcal{U}, f_*\mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

■

The blowing-ups can obtain the desingularization of curves without normalization.

Theorem 3.5: Let X be a projective integral curve over a field k . Then there exists a sequence consisting of all finite birational morphisms as follows

$$\cdots \rightarrow X_{n+1} \rightarrow X_n \rightarrow \cdots \rightarrow X_1 \rightarrow X_0 = X. \quad (3-6)$$

The sequence (3-6) stops for some n with regular curves X_n . In other words, X is desingularized by a finite number of consecutive blowing-ups with regular centers.

Proof: If X is regular, take $n = 0$. Otherwise, let $X_1 \rightarrow X_0 = X$ be the blowing-up of X_0 along the singular locus endowed with the reduced induced closed subscheme structure of X_0 . If X_1 is still singular, we apply a blowing-up $X_2 \rightarrow X_1$ in the same manner, and do this process if necessary, we can get a sequence as above (3-6). Now, let $\pi_n : X_n \rightarrow X_{n-1}$ be the blowing-up of X_{n-1} along $\text{Sing}(X_{n-1})$, which is a proper birational morphism between integral curves, actually a finite birational morphism by Zariski's main theorem [[53], Corollary 4.4.7], whose quasi-finiteness follows from the fact that the closed subsets of integral curves are of dimension zero. Then, there is an exact sequence of sheaves

$$0 \rightarrow \mathcal{O}_{X_{n-1}} \rightarrow (\pi_n)_*\mathcal{O}_{X_n} \rightarrow \mathcal{F}_n \rightarrow 0, \quad (3-7)$$

where $\mathcal{F}_n$ is the skyscraper sheaf with the support contained in $\text{Sing}(X_{n-1})$. Indeed, the injectivity of (3-7) follows from the fact that X_{n-1} is reduced and π is birational, hence scheme-theoretically dominant, and the surjectivity follows from that π_n is an isomorphism over the regular locus $\text{Reg}(X_{n-1})$, thus the cokernel $\mathcal{F}_n$ is supported on the singular locus $\text{Sing}(X_{n-1})$, which is a finite subset. Deduced from (3-7), we have

$$\begin{aligned} 0 \rightarrow H^0(X_{n-1}, \mathcal{O}_{X_{n-1}}) &\rightarrow H^0(X_{n-1}, (\pi_n)_*\mathcal{O}_{X_n}) \rightarrow H^0(X_{n-1}, \mathcal{F}_n) \\ &\rightarrow H^1(X_{n-1}, \mathcal{O}_{X_{n-1}}) \rightarrow H^1(X_{n-1}, (\pi_n)_*\mathcal{O}_{X_n}) \rightarrow H^1(X_{n-1}, \mathcal{F}_n) \rightarrow 0, \end{aligned} \quad (3-8)$$

where $H^i(X_{n-1}, \mathcal{F}_n) = 0$ ($i \geq 1$) and $H^i(X_{n-1}, (\pi_n)_*\mathcal{O}_{X_n}) \cong H^i(X_n, \mathcal{O}_{X_n})$ ($i \geq 0$) by lemma (3.4) since π is finite. Thus, we derive a relation between the arithmetic genera

$$\begin{aligned} p_a(X_{n-1}) &= 1 - \chi(\mathcal{O}_{X_{n-1}}) = 1 - \chi(\mathcal{O}_{X_n}) + \dim_k H^0(X_{n-1}, \mathcal{F}_n) \\ &= p_a(X_n) + \dim_k H^0(X_{n-1}, \mathcal{F}_n). \end{aligned}$$

Let $d = [K(X) \cap \bar{k} : k]$, where $K(X)$ denotes the function field of X . By [[77], Proposition

2.2.22], it is clear that $H^0(X_n, \mathcal{O}_{X_n}) \subset K(X) \cap \bar{k}$. Thus, we have

$$\begin{aligned} p_a(X_n) &= p_a(X_{n-1}) - \dim_k H^0(X_{n-1}, \mathcal{F}_n) \\ &\geq p_a(X_{n-1}) = 1 - \dim_k H^0(X_{n-1}, \mathcal{O}_{X_{n-1}}) + \dim_k H^1(X_{n-1}, \mathcal{O}_{X_{n-1}}) \\ &\geq 1 - \dim_k H^0(X_{n-1}, \mathcal{O}_{X_{n-1}}) \geq 1 - d. \end{aligned}$$

So, $\{p_a(X_n)\}_n^\infty$ is a descending sequence bounded below by $1 - d$, consequently, which stops for sufficiently large n . Let n be such that $p_a(X_n) = p_a(X_{n-1})$. It is claimed that X_{n-1} is regular, then π_n is a blowing-up with regular center, which implies X_n is regular also by [[53], Theorem 8.1.19]. Namely, in this case, we have $H^0(X_{n-1}, \mathcal{F}_n) = 0$, which implies $\mathcal{F}_n = 0$ because $\mathcal{F}_n = 0$ is a skyscraper sheaf. Thus, $\mathcal{O}_{X_{n-1}} \rightarrow (\pi_n)_* \mathcal{O}_{X_n}$ is an isomorphism. As π_n is finite, hence affine, it is an isomorphism. Indeed, if we cover X_{n-1} by affine open subsets $\{V_i\}$, then $\{\pi_n^{-1}(V_i)\}$ is an affine open cover of X_n , and $\pi_n|_{\pi_n^{-1}(V_i)} : \pi_n^{-1}(V_i) \rightarrow V_i$ is an isomorphism because $\mathcal{O}_{X_{n-1}}(V_i) \rightarrow \mathcal{O}_{X_n}(\pi_n^{-1}(V_i))$ is an isomorphism. It follows from Proposition (2.1) that $\text{Sing}(X_{n-1})$ is defined by an invertible sheaf, which contradicts with $\text{codim}(\text{Sing}(X_{n-1}), X_{n-1}) \geq 2$. ■

Remark 3.7: On the one hand, if X is regular, then take $n = 0$. Otherwise, one can just take the normalization of X . However, it is not easy to determine the integral closure for some integral domain A in its fraction field $K = \text{Frac}(A)$. But when A is a finitely generated k -algebra of Krull dimension 1, Theorem (3.5) and Lemma (2.3) make it possible to calculate the integral closure explicitly. On the other hand, by Zariski's main theorem, it is well-known that if $f : X \rightarrow Y$ is a proper birational morphism of Noetherian integral schemes with Y normal, then $\mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$ is an isomorphism automatically and for every $y \in Y$, the fiber $f^{-1}(y)$ is connected. Furthermore, the arithmetic genus is a birational invariant among the smooth projective curves over algebraically closed fields.

If moreover R is a complete DVR, and the smooth projective curve X over $K = \text{Frac}(R)$ has the genus at least one, then among all regular proper R -models of X , there is a unique one satisfying a minimality property such that every birational R -morphism from a regular proper R -model to it is actually a morphism, see [[77], Section 9.3.1.6]. This result, essentially related to the topics of fibered surfaces, is analogous to the theory of minimal models for surfaces over fields, discussed in [[43], pp.418-419].

Conjecture 3.1: Let R be a complete DVR. It is conjectured that a smooth projective geometrically integral K -variety X of any dimension admits a regular proper R -model. This conjecture could be called as **the resolution of singularities for arithmetic schemes**.

3.3.3 Néron models

Let R be a DVR and $K = \text{Frac}(R)$. Let X be a smooth projective geometrically integral K -variety. If X has a good reduction, i.e X has a smooth proper R -model $\mathbf{X}$, then $X(K) \simeq \mathbf{X}(R)$ by the valuative criterion of the properness. This is very useful when studying the integral points of algebraic varieties. However, even if X has bad reduction, it might also have a smooth not-necessarily-proper R -model $\mathcal{N}$ satisfying $\mathcal{N}(R) \simeq X(K)$. To determine the R -model $\mathcal{N}$, it is not sufficient to specify $\mathcal{N}(R)$, but to specify the whole functor of points. By Yoneda's lemma [[77], Lemma 2.3.4], it suffices to specify the functor of points of $\mathcal{N}$ over smooth R -schemes. Thus, we have the following definition.

Definition 3.7: Let R be a Dedekind domain. Let $K = \text{Frac}(R)$. Let X be a smooth K -variety. A **Néron model** of X is a smooth R -scheme $\mathcal{N}$ with an isomorphism $\mathcal{N}_K = \mathcal{N} \times_R K \cong X$ such that for every smooth R -scheme T , the induced map

$$\mathcal{N}(T) = \mathcal{N}_K(T_K) \xrightarrow{\sim} X(T_K)$$

is a bijection.

Remark 3.8: Some authors also call **Néron lft-model** for the above definition (3.7), where “lft” refers to “locally of finite type”, and call **Néron model** only if $\mathcal{N}$ is of finite type^[56]. Actually, Definition (3.7) can be generalized to integral Dedekind schemes, and certainly if X is an algebraic group and $\mathcal{N}$ exists, then $\mathcal{N}$ is a smooth group scheme.

Néron has his name for the concept because he proved that this model exists when X is an abelian variety over the fraction field of a discrete valuation ring, and see [[77], Section 5.7.5]. Here is a more recent result from a different view point of Liu and Tong.

Theorem 3.6 (Liu-Tong, [56]): Let S be an integral Dedekind scheme with its function field $K = \mathbf{k}(S)$. Let X be a smooth projective geometrically integral curve over K of genus at least 1. If $\mathbf{X}$ is the minimal regular proper model of X over S , which exists always. Then the smooth locus $\mathbf{X}^{\text{Smooth}}$ of $\mathbf{X} \rightarrow S$ is a Néron model of X .

For more discussion of Néron models of algebraic varieties, we refer to [6].

CHAPTER 4 FAITHFULLY FLAT PROPER MODELS

In this chapter, we will discuss why one might replace a non-flat model by the flat one when studying the integral points of algebraic varieties. The existence of faithfully flat models of algebraic varieties will be shown precisely, which is a result of Liu et al.^[57].

4.1 Flatness and scheme-theoretic image

The following lemmas are useful when studying the flatness over Dedekind schemes.

Lemma 4.1 ([79], Corollary C-5.104): Let R be a Dedekind domain. Then an R -module M is flat if and only if it is torsion-free.

Remark 4.1: It is known that the flat modules over an integral domain are torsion-free. Conversely, torsion-free modules over Prüfer domain (in particular, Dedekind domain) are flat. There are some simple examples of torsion-free non-flat modules^[63].

Lemma 4.2: Let Y be an integral Dedekind scheme, and $f : X \rightarrow Y$ be a non-constant morphism (i.e. $f(X)$ is not a singleton) with an integral scheme X . Then f is flat.

Proof: Since f is not constant and X is integral, $f(X)$ is irreducible, then its Zariski closure $\overline{f(X)}$ is also irreducible. But Y is of dimension one and irreducible, so one has $\overline{f(X)} = Y$, hence X dominates Y through f . By [[53], Proposition 3.9], f is flat. ■

Lemma 4.3: Let S be an integral Dedekind scheme with its fraction field $K = \mathbf{k}(S)$. Let X and Y be two integral S -schemes. If X and Y are birational to each other, then they have the isomorphic generic fibers. In other words, we have $X \times_S K \simeq Y \times_S K$.

Proof: Since X and Y are birational, there exist non-empty open subsets $U \subset X$ and $V \subset Y$ respectively such that $U \simeq V$. Let ξ_X, ξ_Y and ξ_S denote the unique generic point of X, Y and S respectively. Let $X_K = X \times_S K$ and $Y_K = Y \times_S K$ be the generic fiber of X and Y respectively also. Consider the following diagram:

$$\begin{array}{ccccccc}
 U_K & \hookrightarrow & U & \xrightarrow{\simeq} & V & \hookleftarrow & V_K \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 X_K & \hookrightarrow & X & \xrightarrow{\text{bir}} & Y & \hookleftarrow & Y_K \\
 \downarrow & & \downarrow f & \swarrow g & & & \\
 \text{Spec}(K) & \hookrightarrow & S & & & &
 \end{array}$$

where $U_K = U \times_S K$ and $V_K = V \times_S K$. By Lemma (4.2), f and g are both dominant flat

morphisms, which implies $f(\xi_X) = \xi_S$ and $f(\xi_Y) = \xi_S$. Thus, $\xi_X \in X_K$ and $\xi_Y \in Y_K$. Since X and Y are integral, we also have $\xi_X \in U$ and $\xi_Y \in V$. So, U_K and V_K are non-empty. By [[77], Proposition 3.4.2], f and g are both open mappings, which implies that $f(U)$ and $g(V)$ are open in S and hence contain ξ_S . Thus, we have $X_K \subset U$ and $Y_K \subset V$. Consequently, $U_K = X_K$ and $V_K = Y_K$. By the fact that the isomorphism is stable under the base extension, we have $X_K = U_K \simeq V_K = Y_K$. ■

Corollary 4.1: Let S be an integral Dedekind scheme. Let X be an integral S -scheme. Then for every non-empty open subset U of X , the generic fiber of X is contained in U .

Proof: See the proof of Lemma (4.3). ■

In fact, we have applied Lemma (4.3) before in the proof of Theorem (3.4).

Lemma 4.4: Let X be an affine algebraic variety over a field k . Then there exists a projective variety $\bar{X}$ over k such that X is isomorphic to an open dense subscheme of $\bar{X}$.

Proof: One can write $X = \text{Spec}(k[x_1, \dots, x_n]/I)$ for some ideal I , where I is finitely generated and one can suppose $I = (f_1, \dots, f_m)$ for some $f_i \in k[x_1, \dots, x_n]$. Consider the Zariski closure $\bar{X}$ of X in $\mathbb{P}_k^n$ under the embeddings $X \hookrightarrow \mathbb{A}_k^n \hookrightarrow \mathbb{P}_k^n$

$$\bar{X} = \text{Proj}(k[x_0, x_1, \dots, x_n]/J),$$

where $J = (g_1, \dots, g_m)$ and g_i is the homogenization of f_i obtained by multiplying each monomial of f_i with a lower degree by a suitable power of x_0 in the polynomial ring $k[x_0, x_1, \dots, x_n]$, hence g_i is homogenous, $x_0 \nmid g_i$ and $g_i|_{x_0=1} = f_i$. It is claimed that $\bar{X}$ is a projective variety over k and exactly what we want. More precisely, one has

$$D_+(x_0) \cap \bar{X} = \text{Spec}((k[x_0, x_1, \dots, x_n]/J)_{x_0}) \cong \text{Spec}(k[x_1, \dots, x_n]/I) = X,$$

where the second isomorphism follows from the following isomorphism

$$\begin{aligned} F : k[x_1, \dots, x_n] &\rightarrow (k[x_0, x_1, \dots, x_n]/J)_{x_0} \\ x_i &\mapsto \frac{x_i}{x_0}. \end{aligned}$$

It is clear that F factors through $k[x_0, \dots, x_n]/I$ since $F(f_i) = g_i/x_0^{\deg(g_i)}$ and $I \subseteq \ker(F)$. Let $\tilde{F}$ be the map factoring F whose inverse $\tilde{F}^{-1}$ is defined by evaluating elements of $(k[x_0, x_1, \dots, x_n]/J)_{x_0}$ at $x_0 = 1$, which is well-defined as $(g_i/x_0^{\deg(g_i)})_{x_0=1} = f_i$. Hence, $X \cong D_+(x_0) \cap \bar{X}$, which is an open dense subscheme of $\bar{X}$. ■

Remark 4.2: Indeed, after choosing the ambient space, there are many projective varieties containing the given affine variety X as an open dense subscheme, among which there exists a smallest one, the projective closure of X in the ambient space. However, it

is not easy to compute the projective closure of a given affine variety X because merely homogenizing the set of generators of the defining ideals I is not sufficient, for example [[30], Example 6.35]. Instead, it suffices to homogenize a Gröbner basis of the defining ideal I of X , which makes the problem finding $\overline{X}$ computationally feasible since the Gröbner basis of I consists of finitely many polynomials and see for [[25], Corollary 2.5.6].

In addition, if X is a separated scheme of finite type over a Noetherian scheme Y , a theorem of Nagata^[24] says that there exists a proper scheme $\overline{X}$ over Y such that X is isomorphic to an open dense subscheme of $\overline{X}$.

Corollary 4.2: A morphism of finite type between affine schemes is quasi-projective. In other words, let $X = \text{Spec}(A)$ and $Y = \text{Spec}(B)$, where A is a finitely generated B -algebra by the ring homomorphism $\varphi : B \rightarrow A$, then $\varphi^* : X \rightarrow Y$ is quasi-projective.

Proof: It suffices to show that X is an open subscheme of a closed subscheme of $\mathbb{P}_Y$. By the same argument as Lemma (4.4), we can find an open subscheme of a projective scheme over Y such that it is isomorphic to X . ■

Untill now, we have seen that when studying integral models, it is a common method to take the Zariski closure, which is often called as the scheme-theoretic closure.

Definition 4.1: Let X be a ringed topological space. Let $\mathcal{J}$ be a sheaf of ideals of $\mathcal{O}_X$, namely for any open subset U of X , $\mathcal{J}(U)$ is an ideal of $\mathcal{O}_X(U)$. Let us define

$$V(\mathcal{J}) := \{x \in X \mid \mathcal{J}_x \neq \mathcal{O}_{X,x}\} = \{x \in X \mid (\mathcal{O}_X/\mathcal{J})_x = \mathcal{O}_{X,x}/\mathcal{J}_x \neq 0\} = \text{Supp}(\mathcal{O}_X/\mathcal{J})$$

If X is a scheme, by the operator V , there is a bijection between the set of closed subschemes of X and the set of quasi-coherent ideal sheafs on X by [[43], Proposition II.5.9].

Here is a useful lemma when we use the operator V .

Lemma 4.5: Let X be a ringed topological space and $\mathcal{J}$ a sheaf of ideals of $\mathcal{O}_X$. Let $j : V(\mathcal{J}) \rightarrow X$ denote the inclusion. Then, $V(\mathcal{J})$ is a closed subset of X , $(V(\mathcal{J}), j^{-1}(\mathcal{O}_X/\mathcal{J}))$ is also a ringed space, from which we have a closed immersion $(j, j^\#)$ into $(X, \mathcal{O}_X)$, where $j^\#$ is the canonical surjection $j^\# : \mathcal{O}_X \rightarrow j_*(j^{-1}(\mathcal{O}_X/\mathcal{J})) = \mathcal{O}_X/\mathcal{J}$.

Proof: If $x \in X \setminus V(\mathcal{J})$, one has $\mathcal{O}_{X,x} = \mathcal{J}_x$. Thus, there exists an open neighborhood U of x and $f \in \mathcal{J}(U)$ such that $f_x = 1$. By the definition of sections at stalks, there is an open neighborhood $V \subset U$ of x , such that $f|_V = 1$. Thus, $V \subset X \setminus V(\mathcal{J})$ and $V(\mathcal{J})$ is a closed subset of X . Moreover, for each $x \in V(\mathcal{J})$, one has

$$(j_*(j^{-1}(\mathcal{O}_X/\mathcal{J})))_x = j^{-1}(\mathcal{O}_X/\mathcal{J})_x = (\mathcal{O}_X/\mathcal{J})_{j(x)} = (\mathcal{O}_X/\mathcal{J})_x = \mathcal{O}_{X,x}/\mathcal{J}_x.$$

■

Remark 4.3: Let $f : X \rightarrow Y$ be a continuous map of topological spaces and $\mathcal{F}$ (resp. $\mathcal{G}$) be a sheaf on X (resp. Y). There exist canonical morphisms of sheaves:

$$\mathcal{G} \rightarrow f_* f^{-1} \mathcal{G}, \quad f^{-1} f_* \mathcal{F} \rightarrow \mathcal{F} \quad (4-1)$$

Moreover, if f is a closed immersion, $\mathcal{G} \rightarrow f_* f^{-1} \mathcal{G}$ is surjective; while if f is an open immersion, $f^{-1} f_* \mathcal{F} \rightarrow \mathcal{F}$ is an isomorphism. Indeed, (f^{-1}, f_*) is an adjoint pair from the category of sheaves of abelian groups on X to the category of sheaves of abelian groups on Y . More precisely, we have a canonical bijection $\text{Hom}_X(f^{-1} \mathcal{G}, \mathcal{F}) \cong \text{Hom}_Y(\mathcal{G}, f_* \mathcal{F})$, where f_* is the push-forward and f^{-1} is the inverse image.

Furthermore, if f is a morphism between schemes and $\mathcal{G}$ is a $\mathcal{O}_Y$ -module, then $f^{-1} \mathcal{G}$ is not a $\mathcal{O}_X$ -module in general but a $f^{-1} \mathcal{O}_Y$ -module. To make it a $\mathcal{O}_X$ -module, we consider the pull-back defined as $f^* \mathcal{G} := f^{-1} \mathcal{G} \otimes_{f^{-1} \mathcal{O}_Y} \mathcal{O}_X$, where by the adjoint pair (f^{-1}, f_*) and the bijection (4-1), we have $f^{-1} \mathcal{O}_Y \rightarrow \mathcal{O}_X$. In this case, (f^*, f_*) is an adjoint pair from the category of sheaves of $\mathcal{O}_X$ -module to the category of sheaves of $\mathcal{O}_Y$ -module. To be precise, for any $\mathcal{O}_X$ -module $\mathcal{F}$ and $\mathcal{O}_Y$ -module $\mathcal{G}$, there is a natural isomorphism of groups

$$\text{Hom}_{\mathcal{O}_X}(f^* \mathcal{G}, \mathcal{F}) \cong \text{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_* \mathcal{F}).$$

In general, the support of $\mathcal{F}$ could not be a closed subset of X . Lemma (4.5) tells us the cokernel sheaf of the sheaf of ideals has a good support.

Definition 4.2: Let $f : X \rightarrow Y$ be a morphism of schemes. One says that the morphism f is **quasi-compact** if the inverse image of any open affine subset is quasi-compact. Suppose that f is quasi-compact. Let $\mathcal{J} = \ker(f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X)$. By [[27], Chapter V.1], $V(\mathcal{J})$ is a scheme and $f(X)$ is dense in $V(\mathcal{J})$. We call $V(\mathcal{J}) \subseteq Y$ be **the scheme-theoretic image of X under the morphism f** . Indeed, the scheme-theoretic image $V(\mathcal{J})$ can be computed locally. Let $\{U_i\}$ be an affine open cover of Y , then

$$V(\mathcal{J}) = \{y \in Y \mid \mathcal{J}_y \neq \mathcal{O}_{Y,y}\} = \bigcup_i \{y \in U_i \mid (\mathcal{J}|_{U_i})_y \neq \mathcal{O}_{U_i,y}\} = \bigcup_i V(\mathcal{J}|_{U_i}).$$

It suffices to show that $V(\mathcal{J}|_{U_i})$ is a scheme and so is $V(\mathcal{J})$ by the gluing lemma. Without loss of the generality, one can assume that Y is affine with $Y = \text{Spec}(A)$. By [[70], Theorem 2.1], there is a bijection between $\text{Hom}_{\text{Sch}}(X, \text{Spec}(A))$ and $\text{Hom}_{\text{Ring}}(A, \mathcal{O}_X(X))$, thus f is determined by $\tilde{f} : A \rightarrow \mathcal{O}_X(X)$. In particular, for any $\mathfrak{p} \in \text{Spec}(A)$, one has

$$\mathfrak{p} \in V(\mathcal{J}) \Leftrightarrow (\mathcal{J})_{\mathfrak{p}} \neq \mathcal{O}_{Y,\mathfrak{p}} \Leftrightarrow \ker(\tilde{f})_{\mathfrak{p}} \neq A_{\mathfrak{p}} \Leftrightarrow \ker(\tilde{f}) \subset \mathfrak{p} \Leftrightarrow \mathfrak{p} \in Z(\ker(\tilde{f})).$$

Therefore, $V(\mathcal{J}) = Z(\ker(\tilde{f}))$, where the second operator Z is the Zariski closed subset associated to $\ker(\tilde{f})$. Hence, $V(\mathcal{J}) \cong \text{Spec}(A / \ker(\tilde{f}))$ and $V(\mathcal{J})$ is a scheme. As one can

see, it is by gluing the closed subschemes of the form $\text{Spec}(A/\ker(\tilde{f}))$ that the scheme-theoretic image of f is obtained.

One says that f is **scheme-theoretic dominant** if $V(\ker(f^\#)) = Y$.

Remark 4.4: There is some application of the scheme-theoretic image (4.2) when we study the properties of fpqc descent [[34], Proposition 14.51], the dimension formula of the universally catenary base [[34], Proposition 14.107], the theorem of semi-continuity of fiber dimensions [[34], Theorem 14.110], and so on.

Proposition 4.1: Let $f : X \rightarrow Y$ be a morphism of finite type of schemes. Then the Zariski closure of the set-theoretic image $f(X)$ coincide with the reduced closed subscheme $V(\ker(f^\#))_{\text{red}}$ of $V(\ker(f^\#))$.

Proof: By Definition (4.2), it suffices to show it in the case that Y is affine. Let us write $Y = \text{Spec}(A)$. Take an affine open covering of X , the image of X equals to the union of the images of the covering. Thus, we can assume that $X = \text{Spec}(B)$. Then, we have a ring homomorphism $\tilde{f} : A \rightarrow B$. It remains to apply the commutative algebra.

$$\begin{aligned} \overline{f(X)} &= \overline{f(\text{Spec}(B))} = \overline{f(Z(\text{Nil}(B)))} = \overline{f(Z(\sqrt{(0)_B}))} = \overline{f(Z((0)_B))} = Z(\tilde{f}^{-1}(0_B)) \\ &= Z(\ker(\tilde{f})) = Z\left(\sqrt{\ker(\tilde{f})}\right) = V(\sqrt{\ker(f^\#)}) = V(\ker(f^\#))_{\text{red}} \end{aligned}$$

By the gluing process, $\overline{f(X)}$ coincides with the reduced subscheme of $V(\ker(f^\#))$. ■

Therefore, the Zariski closure of the image endowed with the reduced induced closed subscheme is actually the scheme-theoretic image in arithmetic geometry.

Corollary 4.3: Let $\varphi^* : X = \text{Spec}(B) \rightarrow Y = \text{Spec}(A)$ be a morphism of affine schemes associated with a ring homomorphism $\varphi : A \rightarrow B$. Then, the scheme-theoretic image of φ^* is exactly the Zariski closed subset $Z(\ker(\varphi)) = \{\mathfrak{p} \in \text{Spec}(A) \mid \mathfrak{p} \supseteq \ker(\varphi)\}$. In particular, if A is an integral domain with $K = \text{Frac}(A)$, the scheme-theoretic image of the generic fiber X_K of X in X under the morphism $i : X_K \rightarrow X$ is nothing but $\text{Spec}(B/\alpha)$, where $\alpha = \ker(B \rightarrow B \otimes_A K) = B_{\text{tor}}$. Moreover, if B is a torsion free A -module, the inclusion morphism $i : X_K \rightarrow X$ is scheme-theoretic dominant, i.e $V(\ker(i^\#)) = X$.

Proof: The first statement immediately follows from Corollary (4.1). When A is an integral domain, the generic fiber X_K of X is $\text{Spec}(B \otimes_A K)$ with an inclusion morphism i to $X = \text{Spec}(B)$. By Definition (4.2), the scheme-theoretic image of the generic fiber X_K under i is $Z(\ker(B \rightarrow B \otimes_A K)) \simeq \text{Spec}(B/\ker(B \rightarrow B \otimes_A K))$, where it is claimed that $\ker(B \rightarrow B \otimes_A K) = B_{\text{tor}}$. Indeed, for any $b \in B_{\text{tor}}$, there exists an $r \in A$ such that $rb = 0$, then $b \otimes_A 1 = b \otimes_A r(r^{-1}) = rb \otimes_A r^{-1} = 0$, which implies $B_{\text{tor}} \subseteq \ker(B \rightarrow B \otimes_A K)$.

Conversely, for any $b \in B$ with $b \otimes_A 1 = 0$, there exists an $s \in S = A \setminus \{0\}$ such that $sb = 0$ because $b \otimes_A 1 = 0$ in $B \otimes_A K$ if and only if $\frac{b}{1} = 0$ in $S^{-1}B$ thanks to

$$B \otimes_A K = S^{-1}A \otimes_A B \cong S^{-1}B.$$

Thus, $\ker(B \rightarrow B \otimes_A K) \subseteq B_{\text{tor}}$. ■

Proposition 4.2: Let $f : Z \rightarrow X$ be a morphism of schemes. Then, there is a unique closed subscheme Y of X satisfying that the morphism f factors through Y , and if Y' is any closed subscheme of X through which f factors, then $Y \rightarrow X$ factors through Y' also. Actually, Y is precisely the scheme-theoretic image $V(\ker(f^\#))$. If Z is reduced, then $Y = V(\ker(f^\#))$ coincides with the Zariski closure of the image $f(Z)$.

Proof: If Z is reduced, by Proposition (4.1), take $Y = \overline{f(Z)}$; while if Z is non-reduced, take $Y = \overline{f(Z_{\text{red}})}$, both of which are endowed with the reduced induced structure. The universal property comes from that of induced reduced closed subscheme structure. ■

The condition that a morphism is scheme-theoretic dominant is not just a condition on the underlying topological space.

Example 4.1: Consider the dual number scheme $\text{Spec}(K[\epsilon]/(\epsilon^2)) = \{(\epsilon)/(\epsilon^2)\}$ over a field K . Let $\varphi : \text{Spec}(K) \hookrightarrow \text{Spec}(K[\epsilon]/(\epsilon^2))$ be a surjection on the underlying topological space, by Proposition (4.2), of which the scheme-theoretic image $V(\ker(\varphi)) = \{(\epsilon)/(\epsilon^2)\}$ is the reduced closed subscheme of the non-reduced scheme $\text{Spec}(K[\epsilon]/(\epsilon^2))$.

Proposition 4.3: Let $f : X \rightarrow Y$ be a morphism of schemes. Then, f is scheme-theoretic dominant if and only if for any open subscheme U of Y , if $f|_{f^{-1}U} : f^{-1}U \rightarrow U$ factors as $f^{-1}U \rightarrow Z \hookrightarrow U$ for some closed subscheme Z of U , one must have $Z = U$. In particular, f is set-theoretic dominant, i.e. $\overline{f(X)} = Y$. Moreover, if Y is reduced, then f is scheme-theoretic dominant if and only if f is set-theoretic dominant.

Proof: Let U be any open subscheme of Y . If $f|_{f^{-1}U}$ factors through some closed subscheme Z of U , let $V = U \setminus Z$, which is also an open subscheme of U . It is claimed that $V = \emptyset$, hence $Z = U$. Indeed, the ring homomorphism $f^\#(V) : \mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V))$ is zero because f factors through Z and $f^{-1}(V) = \emptyset$. Thus, $\ker(f^\#(V)) = \mathcal{O}_Y(V)$. By the hypothesis that f is scheme-theoretic dominant, then $0 = \ker(f^\#(V)) = \mathcal{O}_Y(V)$. Conversely, for any open subset U of Y , take $Z = \overline{f(f^{-1}U)}$ endowed with the induced reduced closed subscheme structure. Then, $f|_{f^{-1}U}$ factors through Z , and by assumption, $Z = \overline{f(f^{-1}U)} = U$, which implies that $f|_{f^{-1}U}$ is injective, so is $f^\#$. If Y is reduced, the only closed subscheme of U with the same topological space as U is $U_{\text{red}} = U$. ■

Now, we will discuss the flatness of the scheme-theoretic image, which is very useful

when we study the integral points.

Theorem 4.1: Let S be an integral Dedekind scheme with its function field $K = \mathbf{k}(S)$. Let X be a scheme of finite type over S and denote $X_K = X \times_S K$ its generic fiber. If Y is a scheme over K and $Y \rightarrow X_K$ is a K -morphism of finite type, then the scheme-theoretic image of Y under the composition $f : Y \rightarrow X_K \hookrightarrow X$ is flat over S .

Proof: It is a local problem to show that $V(\ker(f^\#))$ is flat over S because the flatness is local on the target by [[34], Proposition 14.3]. Without loss of the generality, we can assume that $S = \operatorname{Spec}(R)$, where R is a Dedekind domain. If we cover X by an affine open covering $\{U_i = \operatorname{Spec}(A_i)\}$, where each A_i is a finitely generated R -algebra, let $V_i = f^{-1}(U_i)$ and the restriction of f on V_i by $f_i : V_i \rightarrow U_i$, by Definition (4.2), it suffices to show the flatness of the scheme-theoretic image of f_i , which is a local question.

It is obvious that Y is of finite type over K , hence Noetherian, because X_K is of finite type over K . Then, by [[43], Proposition II.5.8], the push-forward sheaf $(f_i)_* \mathcal{O}_{V_i}$ is a quasi-coherent sheaf on U_i . Thus, over any affine open subset $U_i = \operatorname{Spec}(A_i)$, the sheaf $(f_i)_* \mathcal{O}_{V_i}$ is isomorphic to the sheaf of modules $\widetilde{M_i}$ associated to some A_i -module M_i , which is also a finitely generated K -algebra as Y is of finite type over K . It is claimed that M_i is a torsion-free R -module. Indeed, for any $r \in R \setminus \{0\}$ and $m_i \in M_i$, if $rm_i = 0$, we immediately get $m_i = 0$ because r is invertible in K . However, the scheme-theoretic image of f_i in U_i is the closed subscheme $V(\ker(f_i^\#)) = \operatorname{Spec}(A_i/I_i)$ of U_i defined by the ideal $I_i = \ker(f_i^\#)$, where $f_i^\# : A_i \rightarrow M_i$ is the ring homomorphism corresponding to f_i and A_i/I_i is a torsion-free R -module because it is a sub R -module of M_i . Hence, A_i/I_i is flat over R by Lemma (4.1). It completes the proof. ■

Remark 4.5: The most common case of the theorem (4.1) is that $S = \operatorname{Spec}(R)$ for some Dedekind domain R with $K = \operatorname{Frac}(R)$, X is an algebraic variety over K and $\mathbf{X}$ is an integral model of X . Then, for any morphism $f : Y \rightarrow X$ of finite type, for example open immersions or closed immersions, the scheme-theoretic image of Y in $\mathbf{X}$ is flat over R .

Remark 4.6: In fact, Theorem (4.1) we prove is similar to [[34], Proposition 14.14], which says that the morphism $g : X \rightarrow S$ of schemes is flat over an integral Dedekind scheme S if and only if the scheme X is equal to the scheme-theoretic image (see its Remark 10.31) of its generic fiber X_K in X , and see Corollary (4.3) for the affine case.

Besides, flatness over a reduced base scheme is reasonably also easy to understand in geometric terms, and by [[34], Theorem 14.34], there is a valuative criterion for flatness. Let Y be a reduced locally Noetherian scheme, and $f : X \rightarrow Y$ be a morphism of schemes

that is locally of finite type. Then f is flat if every base change with respect to $\text{Spec } R \rightarrow Y$ is flat, where R is a DVR. It shows that understanding the flatness over DVRs is really the crucial case, at least as long as considering reduced base schemes, and see related results in [[34], Proposition 14.17], which says that the whole family equals the scheme-theoretic image of the generic fiber. Thus, it is easy to “flatten” a scheme X over Dedekind schemes by replacing X by the scheme-theoretic image of its generic fiber.

The following theorem tells us that the scheme-theoretic image of the closed subvariety Y of the generic fiber X in its integral model $\mathbf{X}$ is the smallest flat integral model of Y , i.e the smallest flat closed subscheme of $\mathbf{X}$ with the generic fiber Y .

Theorem 4.2: Let S be an integral Dedekind scheme with its generic point η . Let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Let $f : X \rightarrow S$ be a morphism of schemes and denote its generic fiber by $X_\eta = f^{-1}(\eta) = X \times_S K$. Let $i : X_\eta \hookrightarrow X$ be the inclusion morphism. For a closed subscheme Z^0 of X_η , denote $\overline{Z^0}$ the scheme-theoretic image of Z^0 in X . Then

(1) There is a bijection

$$\Psi : \left\{ \begin{array}{l} \text{closed subschemes } Z \subseteq X \\ \text{such that } Z \text{ is flat over } S \end{array} \right\} \leftrightarrow \{ \text{closed subschemes } Z^0 \subseteq X_\eta \}$$

$$Z \mapsto i^{-1}(Z) = Z_\eta = Z \times_S K$$

$$\overline{Z^0} \leftarrow Z^0.$$

(2) Let Y be an S -scheme also, and $Z^0 \subseteq X_\eta, W^0 \subseteq Y_\eta$ be two closed subschemes of generic fibers of X and Y respectively. Then

$$\overline{Z^0 \times_K W^0} = \overline{Z^0} \times_S \overline{W^0}.$$

Proof: (1) Suppose that Z is a closed subscheme of X which is flat over S , we have the following commutative diagram given by the fiber products

$$\begin{array}{ccc} & & \overline{Z_\eta} \\ & \nearrow \alpha & \downarrow \\ Z_\eta = i^{-1}(Z) & \hookrightarrow & Z \\ \downarrow \tilde{\varphi} & & \downarrow \varphi \\ X_\eta = X \times_S K & \xrightarrow{i} & X \\ \downarrow & & \downarrow f \\ \text{Spec}(K) & \hookrightarrow & S \end{array}$$

where φ is a closed immersion, α is scheme-theoretically dominant, and $\tilde{\varphi}$ is the base

change of φ , hence a closed immersion. Thus, $Z_\eta = i^{-1}(Z)$ is a closed subscheme of X_η . Moreover, since Z is flat over S , by remark (4.6), the scheme-theoretic image of the generic fiber Z_η in Z equals to Z , which implies Ψ is injective. Conversely, if Z^0 is a closed subscheme of X_η , then its scheme-theoretic image $\overline{Z^0}$ in X with respect to the morphism $Z^0 \hookrightarrow X_\eta \xrightarrow{i} X$ is a closed subscheme of X , which is flat over S by theorem (4.1). However, the generic fiber $i^{-1}(\overline{Z^0}) = \overline{Z^0} \times_S K = Z^0$, which implies Ψ is surjective.

(2) It is a local problem to show the fiber product equality. One can assume that $S = \text{Spec}(R)$, $X = \text{Spec}(A)$, and $Y = \text{Spec}(B)$. Then, we can write $Z^0 = \text{Spec}(A_K/J_A)$, where $A_K = K \otimes_R A$ and J_A is an ideal of A_K ; $W^0 = \text{Spec}(B_K/J_B)$, where $B_K = K \otimes_R B$ and J_B is an ideal of B_K . Under these notation, $\overline{Z^0} = \text{Spec}(A/I_A)$, where $I_A = \ker(A \rightarrow A_K/J_A)$, and $\overline{W^0} = \text{Spec}(B/I_B)$, where $I_B = \ker(B \rightarrow B_K/J_B)$. Thus, for the right handside, we have

$$\overline{Z^0} \times_S \overline{W^0} = \text{Spec}((A/I_A) \otimes_R (B/I_B)) = \text{Spec}((A \otimes_R B)/(I_A \otimes_R B + I_B \otimes_R A)).$$

For the left handside, we have

$$\overline{Z^0 \times_K W^0} = \text{Spec}((A \otimes_R B)/\ker(A \otimes_R B \rightarrow (A_K/J_A) \otimes_K (B_K/J_B))).$$

By a little commutative algebra, it is easy to show that the left handside equals to the right handside. Indeed, by [[27], Proposition V.8], one has the following commutative diagram

$$\begin{array}{ccc} Z^0 \times_K W^0 & \longrightarrow & \text{Spec}(K) \\ \downarrow & & \downarrow \\ \overline{Z^0} \times_S \overline{W^0} & \longrightarrow & S \end{array}$$

where $\overline{Z^0} \times_S \overline{W^0}$ is flat over S . By (1), one has $\overline{Z^0 \times_K W^0} = \overline{Z^0} \times_S \overline{W^0}$. ■

4.2 Main theorem II

Theorem 4.3: Let R be a Dedekind domain with $K = \text{Frac}(R)$. Let X_K be an algebraic variety over K . Then X_K admits a faithfully flat integral model $\mathbf{X}$ over R , which is also quasi-projective if X_K is affine. Moreover, if X_K is proper (resp. projective), X_K admits a faithfully flat proper (resp. projective) integral model $\mathbf{X}$ over R .

Proof: Firstly, if X_K is projective, the result is immediate by the similar argument of Theorem (3.4). One can embed X_K into a projective space $\mathbb{P}_K^n$ and take the scheme-

theoretic image $\mathbf{X}$ of X_K in $\mathbb{P}_R^n$, whose generic fiber is equal to X by Corollary (4.1). Since $\mathbf{X}$ is a closed subscheme of $\mathbb{P}_R^n$, $\mathbf{X}$ is projective over R . By Theorem (4.1), $\mathbf{X}$ is flat over R , hence faithfully flat over R due to the connectedness of $\text{Spec}(R)$ because the image of $\mathbf{X}$ is open and closed in $\text{Spec}(R)$.

Now, let us suppose that X_K is affine. It is notable that for any integral model $\mathbf{X}$ of X_K over R and take the largest closed subscheme $\mathbf{Y}$ of $\mathbf{X}$ which is flat over R , namely the scheme-theoretic image of the generic fiber X_K in $\mathbf{X}$ by Theorem (4.2), $\mathbf{Y}$ is not necessarily faithfully flat over R . Instead, we approach in another explicit way. By Lemma (4.4), we can choose $Y_K \subseteq \mathbb{P}_K^N$ be a projective variety containing X_K as an open dense subvariety. Consider the following relation

$$\begin{aligned} X_K &\hookrightarrow Y_K \hookrightarrow \mathbb{P}_K^N \hookrightarrow \mathbb{P}_R^N \\ Y_K &\longrightarrow \mathbf{Y} = \overline{Y_K} \\ Y_K \setminus X_K &\longrightarrow \mathbf{Z} = \overline{Y_K \setminus X_K}, \end{aligned}$$

where $\mathbf{Y}$ is the scheme-theoretic image of Y_K in $\mathbb{P}_R^N$ and $\mathbf{Z}$ is the scheme-theoretic image of $Y_K \setminus X_K$ in $\mathbb{P}_R^N$. Obviously, we have $\mathbf{Z} \subset \mathbf{Y}$. Let $\mathbf{X} = \mathbf{Y} \setminus \mathbf{Z}$. It is claimed that $\mathbf{X}$ is a faithfully flat integral model of X_K over R . Since $\mathbf{X}$ is an open subscheme of $\mathbf{Y}$ and $\mathbf{Y}$ is a closed subscheme of $\mathbb{P}_R^N$, then $\mathbf{X}$ is quasi-projective over R . Meanwhile, $\mathbf{X}$ is flat over R as $\mathbf{Y}$ is flat over R by Theorem (4.2). Furthermore, as $\mathbf{Y}$ and $\mathbf{Z}$ are both flat proper over R , hence faithfully flat over R by the same argument as the projective case above, then for all $v \in \text{Spec}(R)$, by [[53], Corollary 4.3.14], we have

$$\dim \mathbf{Z}_v = \dim \mathbf{Z}_K = \dim Y_K \setminus X_K < \dim Y_K = \dim \mathbf{Y}_K = \dim \mathbf{Y}_v,$$

where $\mathbf{Z}_v$ and $\mathbf{Y}_v$ stand for the fiber of $\mathbf{Z}$ and $\mathbf{Y}$ at v respectively, and the inequality follows from that X_K is open dense in Y_K . Therefore, $\mathbf{X}_v = \mathbf{Y}_v \setminus \mathbf{Z}_v \neq \emptyset$, and $\mathbf{X}$ is faithfully flat over R . By construction, the generic fiber of $\mathbf{X}$ is equal to X_K .

In the general case, since X_K is of finite type and separated over K , X_K can extend to a faithfully flat separated scheme $\mathbf{X}_1$ of finite type over some open subscheme $V \subseteq \text{Spec}(R)$ by spreading out, and see [[77], Appendix: Table 1]. Let U_K be a non-empty affine open subvariety of X_K . As shown in the affine case above, U_K possesses a faithfully flat integral model $\mathbf{U}$ over R . Shrinking V if necessary, by [[77], Theorem 3.2.1], we can extend the open immersion $U_K \hookrightarrow X_K$ to $\mathbf{U}_V = \mathbf{U} \times_R V \hookrightarrow (\mathbf{X}_1)_V = \mathbf{X}_1$, which is also an open immersion. Indeed, we have two commutative diagrams given by spreading out

$$\begin{array}{ccc}
 U_K = U \times_R K & \longrightarrow & \text{Spec}(K) \\
 \downarrow & & \downarrow \\
 U_V = U \times_R V & \longrightarrow & V \\
 \downarrow & & \downarrow \\
 U & \longrightarrow & \text{Spec}(R)
 \end{array}
 \qquad
 \begin{array}{ccc}
 X_K = X_1 \times_V K & \longrightarrow & \text{Spec}(K) \\
 \downarrow & & \downarrow \\
 X_1 & \longrightarrow & V \\
 & & \downarrow \\
 & & \text{Spec}(R).
 \end{array}$$

Thus, U_V is flat over R . Let X be the gluing of X_1 and U along U_V by Lemma (3.2). Then, X is a faithfully flat model of X_K over R , where the surjectivity follows from U and X_1 .

For the case that X_K is proper, the integral model X we obtained above is not proper over R in general. But by Nagata's embedding theorem^{[24] [58]}, it is known that X is an open subscheme of some scheme M , which is proper over R . Let M' be the scheme-theoretic image of X_K in M under the embedding $X_K \hookrightarrow X \hookrightarrow M$. It is clear that M' is proper over R since it is a closed subscheme of M . By theorem (4.1), M' is flat over R , hence faithfully flat. Thus, M' is a faithfully flat proper integral model of X_K over R . ■

Remark 4.7: By Remark (3.5), the model X is not always quasi-projective over R .

Corollary 4.4: Let S be an integral Dedekind scheme with its function field $K = k(S)$. Let X be an algebraic variety over K . Then there exists a faithfully flat separated scheme X of finite type over S such that $X \times_S K \simeq X$.

Proof: Apply Theorem (4.3), and glue the local pieces together by Lemma (3.2). ■

For convenience, we will give a functorial property of the scheme-theoretic image.

Lemma 4.6: Let S be a base scheme. Let X, Y, Z_1, Z_2 be S -schemes. Let $\pi_1 : Z_1 \rightarrow X$ and $\pi_2 : Z_2 \rightarrow Y$ be two S -morphisms. Denote the scheme-theoretic image of π_1 and π_2 by $\widetilde{Z}_1$ and $\widetilde{Z}_2$ respectively. If $f : X \rightarrow Y$ and $g : Z_1 \rightarrow Z_2$ are two S -morphisms such that $f \circ \pi_1 = \pi_2 \circ g$, there is an S -morphism $\tilde{g} : \widetilde{Z}_1 \rightarrow \widetilde{Z}_2$.

Proof: Let us consider the following commutative diagram

$$\begin{array}{ccccc}
 & & \widetilde{Z}_1 & & \\
 & \nearrow \beta_1 & \downarrow \tilde{g} & \nwarrow i_1 & \\
 Z_1 & \xrightarrow{\pi_1} & X & & \\
 \downarrow g & & \downarrow f & & \\
 Z_2 & \xrightarrow{\pi_2} & Y & & \\
 & \searrow & \downarrow & \swarrow & \\
 & & S & &
 \end{array}$$

where β_1 and β_2 are both scheme-theoretic dominant morphisms; i_1 and i_2 are both closed immersions. Let us define $\varphi = f \circ i_1 : \widetilde{Z}_1 \rightarrow Y$. It suffices to show that φ factors through $\widetilde{Z}_2$. By the commutative diagram, we have $i_2 \circ \beta_2 \circ g = \pi_2 \circ g = f \circ \pi_1 = f \circ i_1 \circ \beta_1 = \varphi \circ \beta_1$. Thus, φ factors through $\widetilde{Z}_2$ as β_1 is scheme-theoretic dominant. ■

Theorem 4.4: Let S be an integral Dedekind scheme with its function field K . Let Y be a scheme of finite type over S and $Y_K = Y \times_S K$ be its generic fiber. Let X be the scheme-theoretic image of the generic fiber Y_K in Y , by Theorem (4.2), which is the biggest closed subscheme such that flat over S . Then, for any flat S -scheme T , there exists a bijection

$$X(T) := \text{Hom}_S(T, X) \leftrightarrow \text{Hom}_S(T, Y) := Y(T).$$

In particular, it is harmless to take a flat integral model when studying integral points.

Proof: It is obvious that $X(T) \subseteq Y(T)$. It suffices to show that for any S -morphism $f : T \rightarrow Y$, f factors through X . Consider the following commutative diagram

$$\begin{array}{ccccc}
 & & \mathbf{T} & & \\
 & \nearrow & \downarrow \tilde{\varphi} & \searrow i_1 & \\
 T_K & & X & \xleftarrow{\tilde{f}} & T \\
 \downarrow \varphi & \nearrow & \downarrow i_2 & \searrow f & \\
 Y_K & & Y & & \\
 \downarrow & & \downarrow & & \\
 \text{Spec}(K) & \longrightarrow & S & &
 \end{array}$$

where $T_K = T \times_S K$ is the generic fiber of T , and $\mathbf{T}$ is the scheme-theoretic image of T_K in T that is equal to T by Remark (4.6), as T is flat over S . It is clear that φ is induced by f , which induces $\tilde{\varphi}$ by Lemma (4.6). Thus, one can take $\tilde{\varphi}$ as $\tilde{f}$. ■

Example 4.2: Let k be a global field, $S \subseteq \Omega_k$ be a finite subset containing ∞_k . Let X be an algebraic variety over k , and $\mathbf{Y}$ be an integral model of X over $\mathcal{O}_S$, not necessarily flat. Let $\mathbf{X}$ be the biggest closed subscheme of $\mathbf{Y}$ such that flat over $\mathcal{O}_S$, which is defined by the sheaf of ideals of $\mathcal{O}_S$ -torsion elements of $\mathcal{O}_{\mathbf{Y}}$, i.e the scheme-theoretic image of X in $\mathbf{Y}$ by Theorem (4.2). Then, by Theorem (4.4), for any flat $\mathcal{O}_S$ -scheme T , the canonical map $\mathbf{X}(T) \rightarrow \mathbf{Y}(T)$ is bijective. In particular, for any commutative integral $\mathcal{O}_S$ -algebra A , which is $\mathcal{O}_S$ -torsion free, hence flat over $\mathcal{O}_S$, the canonical map $\mathbf{Y}(A) \rightarrow \mathbf{X}(A)$ is bijective. Thus, we have $\mathbf{Y}(\mathcal{O}_S) = \mathbf{X}(\mathcal{O}_S)$ and $\mathbf{Y}(\mathcal{O}_v) = \mathbf{X}(\mathcal{O}_v)$ for $v \notin S$. Hence, when we study integral points on X , it is harmless to replace non-flat models by flat ones, which eliminates the bad fibers. Besides, it is also necessary to require models faithfully flat because of the existence of local points at $v \notin S$ when studying the integral Hasse principle (5.1).

CHAPTER 5 VERY STRONG APPROXIMATION FOR ALGEBRAIC TORI

In this chapter, without other statement, k is always a global field, Ω_k is the set of places of k and S is a non-empty finite subset of places of k containing ∞_k . Let $\mathcal{O}_S$ denote the ring of S -integers of k . Let X be an algebraic variety over k and $\mathbf{X}$ be an integral model of X over $\mathcal{O}_S$. Denote $P_S : X(\mathbf{A}_k) \rightarrow X(\mathbf{A}_k^S)$ the natural projecting, which is an open map.

5.1 Basic properties of SAP and VSAP

The integral points $\mathbf{X}(\mathcal{O}_S)$ can be identified with the intersection of rational points and S -adelic points $X(k) \cap \mathbf{X}(\mathbf{A}_{k,S})$.

Lemma 5.1: The set $\mathbf{X}(\mathcal{O}_S)$ of S -integral points on $\mathbf{X}$ can be identified with the set $X(k) \cap \mathbf{X}(\mathbf{A}_{k,S})$ of rational adelic points on X by the canonical injective maps:

$$\begin{array}{ccc} \mathbf{X}(\mathcal{O}_S) & \hookrightarrow & \mathbf{X}(\mathbf{A}_{k,S}) \\ \downarrow & & \downarrow \\ X(k) & \hookrightarrow & \prod_{v \in \Omega_k} X(k_v). \end{array}$$

In particular, if k is a number field, then the set $\mathbf{X}(\mathcal{O}_k)$ of integral points of $\mathbf{X}$ can be identified with $X(k) \cap \prod_{v \in \Omega} \mathbf{X}(\mathcal{O}_v)$ in the space $\prod_{v \in \Omega_k} X(k_v)$.

Proof: It is obvious that $\mathbf{X}(\mathcal{O}_S) \rightarrow X(k)$ and $\mathbf{X}(\mathcal{O}_S) \rightarrow \mathbf{X}(\mathbf{A}_{k,S})$ are injective since X and $\mathbf{X}$ are separated. Thus, $\mathbf{X}(\mathcal{O}_S) \subseteq \mathbf{X}(\mathbf{A}_{k,S}) \cap X(k)$ holds in $\prod_{v \in \Omega_k} X(k_v)$. Conversely, let $\sigma \in X(k)$ such that $\sigma \in \mathbf{X}(\mathcal{O}_v)$ in $X(k_v)$ for any $v \notin S$. Then $\sigma \in \mathbf{X}(\mathcal{O}_{k,v})$ for all $v \notin S$ since $k \cap \mathcal{O}_v = \mathcal{O}_{k,v}$, which is the localization of $\mathcal{O}_S$ at v . By spreading out [[53], Exercise 3.2.4], for any $v \notin S$, σ can extend to a morphism $U(v) \rightarrow \mathbf{X}$ for some open neighborhood $U(v)$ of $v \in \text{Spec}(\mathcal{O}_S)$. We can obtain $\sigma \in \mathbf{X}(\mathcal{O}_S)$ by gluing these morphisms. ■

Corollary 5.1: Let R be a Dedekind domain with $K = \text{Frac}(R)$ and $\mathbf{X}$ be a separated scheme of finite type over R with its generic fiber X_K . Let S be a finite subset consisting of finitely many non-zero prime ideals of R and denote $\mathcal{O}_S = \cap_{\mathfrak{p} \notin S} R_{\mathfrak{p}}$, which is also a Dedekind domain. Then, one can identify $\mathbf{X}(\mathcal{O}_S)$ with $X_K(K) \cap \left(\prod_{\mathfrak{p} \notin S} \mathbf{X}(R_{\mathfrak{p}}) \right)$. In particular, $\mathbf{X}(R) = X_K(K) \cap \left(\prod_{\mathfrak{p} \in \text{Spec}(R)} \mathbf{X}(R_{\mathfrak{p}}) \right)$.

Proof: It is similar to Lemma (5.1). ■

Definition 5.1: If the integral Brauer-Manin obstruction set $\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}$, which con-

tains the set $\mathbf{X}(\mathcal{O}_S)$ of S -integral points on $\mathbf{X}$, is sufficient to insure that $\mathbf{X}(\mathcal{O}_S) \neq \emptyset$, one says that **the Brauer-Manin obstruction is the only obstruction for the existence of integral points on $\mathbf{X}$** . The Brauer-Manin obstructions to integral points and their related approximation properties are always called **integral Brauer-Manin obstruction** roughly.

In fact, if $\text{Br}(X)$ is constant (10), then $\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)} = \mathbf{X}(\mathbf{A}_{k,S})$, in which case the definition (5.1) coincides with the classical integral Hasse principle.

Theorem 5.1: Let X be an algebraic variety over k . Then X satisfies SAP-BM off S if and only if the Brauer-Manin obstruction is the only obstruction for the existence of integral points for every integral model $\mathbf{X}$ of X over $\mathcal{O}_S$.

Proof: Let $\mathbf{X}$ be an integral model of X over $\mathcal{O}_S$ and suppose that set $\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)} \neq \emptyset$, which is an open subset in the Brauer-Manin obstruction set $X(\mathbf{A}_k)^{\text{Br}(X)}$ by an embedding $\mathbf{X}(\mathbf{A}_{k,S}) \rightarrow X(\mathbf{A}_k)$ with respect to the induced topology of the adelic topology (3.2.1). Since the natural projection map P_S is open, $P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}]$ is an open subset of $P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$. Thus, it follows from the SAP-BM off S that $\mathbf{X}(\mathcal{O}_S) = X(k) \cap P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}] \neq \emptyset$. Hence, the Brauer-Manin obstruction is the only obstruction for the existence of integral points on $\mathbf{X}$. One can also see the proof of Proposition (5.1).

Conversely, let U be a non-empty open subset of $P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$, which comes from a non-empty open subset $\tilde{U} \subseteq \mathbf{X}(\mathbf{A}_k^S)$ in the sense that $\tilde{U} \cap P_S[X(\mathbf{A}_k)^{\text{Br}(X)}] = U$. We claim that U meets the image of $X(k)$ in the S -adelic space $X(\mathbf{A}_k^S)$ under the diagonal map, in other words, each non-empty open subset of $P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ must contain some rational point of X , hence the set $X(k)$ of rational points is dense in $P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$. To prove our claim, let $z \in U$, and by the structure theorem of S -adelic space (3.1), there exists an integral model $\mathbf{X}$ of X over $\mathcal{O}_S$ such that $z \in \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \subseteq \tilde{U}$, which implies that $z \in \mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)} \neq \emptyset$ since $z \in P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ can be lift to an element of $X(\mathbf{A}_{k,S})^{\text{Br}(X)}$ with S -components lying in k_v . By the hypothesis, one obtains $\mathbf{X}(\mathcal{O}_S) \neq \emptyset$. Finally, note that in $X(\mathbf{A}_k^S)$ we have:

$$\text{Im}(X(k) \rightarrow X(\mathbf{A}_k^S)) \supseteq \text{Im}(\mathbf{X}(\mathcal{O}_S) \rightarrow X(\mathbf{A}_k^S)) \subseteq \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \subseteq \tilde{U}.$$

Thus, since $\mathbf{X}(\mathcal{O}_S) \neq \emptyset$, we have:

$$\begin{aligned} \tilde{U} \cap \text{Im}(X(k) \rightarrow X(\mathbf{A}_k^S)) &\supseteq \left(\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v) \right) \cap \text{Im}(\mathbf{X}(\mathcal{O}_S) \rightarrow X(\mathbf{A}_k^S)) \\ &= \text{Im}(\mathbf{X}(\mathcal{O}_S) \rightarrow X(\mathbf{A}_k^S)) \neq \emptyset. \end{aligned}$$

Therefore, $U \cap \text{Im}(X(k) \rightarrow X(\mathbf{A}_k^S)) \neq \emptyset$ and satisfies SAP-BM. ■

Theorem 5.2: Let X be an algebraic variety over k . Then X satisfies VSAP-BM off S if and only if the diagonal map $\mathbf{X}(\mathcal{O}_S) \rightarrow P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}]$ is bijective for every model $\mathbf{X}$ of X over $\mathcal{O}_S$.

Proof: The diagonal map $\mathbf{X}(\mathcal{O}_S) \rightarrow P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}]$ is injective naturally and it suffices to show the surjectivity. Let $y \in P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}]$, and there exists $x \in \mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)} \subseteq X(\mathbf{A}_k)^{\text{Br}(X)}$ such that $y = P_S(x)$. By assumption that X satisfies VSAP, there exists $\sigma \in X(k)$ such that $\sigma = P_S(x) = y$ under the diagonal map. Namely, for any $v \notin S$, under the diagonal embedding, $\sigma_v = x_v \in \mathbf{X}(\mathcal{O}_v) \cap X(k) = \mathbf{X}(\mathcal{O}_{k,v})$, thus $\sigma \in \mathbf{X}(\mathbf{A}_{k,S}) \cap X(k)$ which implies $\sigma \in \mathbf{X}(\mathcal{O}_S)$ by Lemma (5.1).

Conversely, it suffices to show that $P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ must come from some rational point. Let $x \in P_S[X(\mathbf{A}_k)^{\text{Br}(X)}] \subseteq X(\mathbf{A}_k^S)$ and $y \in X(\mathbf{A}_k)^{\text{Br}(X)}$ such that $x = P_S(y)$. By the structure theorem of S -adelic space (3.1), there exists an integral model $\mathbf{X}$ of X over $\mathcal{O}_S$ such that $x \in \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v)$, which implies $y \in \mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}$. Thus, $x = P_S(y) \in P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}]$. By the hypothesis, one can conclude that $x \in \mathbf{X}(\mathcal{O}_S) \subseteq X(k)$. ■

Remark 5.1: As shown in Theorem (5.2), the bijective diagonal map $\mathbf{X}(\mathcal{O}_S) \rightarrow P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}]$ holding for each model $\mathbf{X}$ implies that $X(k) = P_S[X(\mathbf{A}_k)^{\text{Br}(X)}] = \cup \mathbf{X}(\mathcal{O}_S)$, where the union runs through all integral models $\mathbf{X}$ of X over $\mathcal{O}_S$.

Inspired by the observations of Harari et al.^[41] and Liu et al.^[57], as well as discussions with my advisor Yong Hu, we propose the following proposition for VSAP.

Proposition 5.1: Let X be an algebraic variety. Suppose that X satisfies SAP-BM off S . If $\mathbf{X}(\mathcal{O}_S)$ is finite for any integral model $\mathbf{X}$, then the diagonal map $P_S : \mathbf{X}(\mathcal{O}_S) \rightarrow P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}]$ is bijective, which implies X satisfies VSAP-BM off S . In this case, SAP-BM off S is equivalent to VSAP-BM off S .

Proof: Let $x = (x_v)_{v \in \Omega_k} \in \mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)} \subseteq X(\mathbf{A}_k)^{\text{Br}(X)}$. Then $P_S(x) = (x_v)_{v \notin S} \in P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$. Since X satisfies SAP-BM, one has $\overline{X(k)} = P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$, then with respect to the open neighborhood $\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v)$ of $P_S(x)$, there must be a rational point $z \in X(k)$ such that $z \in \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v)$, which means $z \in \mathbf{X}(\mathcal{O}_v) \cap X(k) = \mathbf{X}(\mathcal{O}_{k,v})$, hence $z \in \mathbf{X}(\mathcal{O}_S)$. However, the image of z under the diagonal map is not exactly $P_S(x)$ in general, and the above argument just explains $\mathbf{X}(\mathcal{O}_S) \neq \emptyset$ again and one can see for Theorem (5.1). It remains to refine the open neighborhood of $P_S(x) = (x_v)_{v \notin S}$. For each $v \notin S$, set $T_v = \{z_v = z \in \mathbf{X}(\mathcal{O}_v) \mid z \in \mathbf{X}(\mathcal{O}_S), z_v \neq x_v\}$, which is a finite set because of the finiteness of $\mathbf{X}(\mathcal{O}_S)$. Let $U = \prod_{v \notin S} (\mathbf{X}(\mathcal{O}_v) \setminus T_v)$, which is also a non-empty open neighborhood of $P_S(x)$. By the SAP-BM of X , there is a $\tilde{z} \in X(k)$ such that $z \in U$. Hence,

$z \in \mathbf{X}(\mathcal{O}_S)$ and $P_S(z) = P_S(x)$, so the projection map $P_S : \mathbf{X}(\mathcal{O}_S) \rightarrow P_S[\mathbf{X}(\mathbf{A}_{k,S})^{\text{Br}(X)}]$ is bijective. By Theorem (5.2), VSAP-BM off S holds for X . ■

By Lemma(3.1), we can see that removing closed points from a variety might decrease the number of rational points on it. More precisely, removing a k -point reduces the set of k -rational points by exactly one. However, if the deleted point does not belong to the base field k , the set of k -rational points is completely unaffected

Example 5.1: Let X be a smooth hyperbolic curve of genus g over a number field k , which is a smooth projective curve of genus g minus a reduced effective divisor of degree d such that $2g - 2 + d > 0$. Then, VSAP-BM is equivalent to SAP-BM for X . More precisely, if $g \geq 2$, by Faltings^[29], the set $X(k)$ of rational points on X is finite, and hence there is no difference between SAP-BM off and VASP-BM for X . If $g = 1$, by Siegel's results in [[84], IX.3] for elliptic curves, $\mathbf{X}(\mathcal{O}_S)$ is finite for any integral model $\mathbf{X}$ of X over $\mathcal{O}_S$, hence by Proposition (5.1), SAP-BM implies VSAP-BM.

By the similar argument, we can obtain the analogous results for Conjecture (1.1).

Theorem 5.3: The Harari-Voloch conjecture (1.1) holds for any flat integral model $\mathbf{X}$ of the hyperbolic curve X over $\mathcal{O}_S$ if and only if the canonical map $X(k) \rightarrow X(\mathbf{A}_k^S)^{\text{Bs}(X)}$ is bijective.

Proof: Suppose that Conjecture (1.1) holds for all models of X and let $x = (x_v)_{v \notin S} \in X(\mathbf{A}_k^S)^{\text{Bs}(X)} \subseteq X(\mathbf{A}_k^S)$. By the structure theorem (3.1), there exists an integral model $\mathbf{X}$ of X over $\mathcal{O}_S$ such that $x \in \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v)$, arguably $x \in (\prod_{v \notin S} \mathbf{X}(\mathcal{O}_v))^{\text{Bs}(X)}$. Then, by the hypothesis, $x \in \mathbf{X}(\mathcal{O}_S) \subseteq X(k)$. Conversely, let $x = (x_v)_{v \notin S} \in \prod_{v \notin S} \mathbf{X}(\mathcal{O}_v)$ which is orthogonal to $\text{Bs}(X)$ and then $x \in X(\mathbf{A}_k^S)^{\text{Bs}(X)}$. By the assumption, $x \in X(k)$ and so $x_v \in \mathbf{X}(\mathcal{O}_{k,v})$, which implies $x \in \mathbf{X}(\mathcal{O}_S)$. ■

We can compare VSAP-BM off S with the integral Brauer-Manin obstruction set.

Theorem 5.4: If the Harari-Voloch Conjecture (1.1) holds for a rational hyperbolic curve X , then X satisfies the very strong approximation property with Brauer-Manin obstruction off S .

Proof: It follows from Theorem (5.2) and (5.3) directly and see also the Remark (1.5). ■

Remark 5.2: Conjecture (1.1) also implies^[41] a conjecture of Skolem.

Recall that an immersion morphism is a closed immersion followed by an open immersion [[35], I.4.2.1], which can be factored into an open immersion followed by a closed immersion if the immersion is quasi-compact by [[34], Remark 10.31].

Proposition 5.2: Let $f : X \rightarrow Y$ be an immersion of algebraic varieties over k . Suppose that VSAP-BM off S holds for Y , then the same holds for X . In particular, VSAP-BM off S holds for closed subvarieties and open subvarieties of Y .

Proof: Let us consider the following commutative diagram

$$\begin{array}{ccc}
 X(k) & \xhookrightarrow{f} & Y(k) \\
 \downarrow & & \downarrow \\
 X(\mathbf{A}_k)^{\text{Br}(X)} & \xhookrightarrow{f^*} & Y(\mathbf{A}_k)^{\text{Br}(Y)} \subseteq Y(\mathbf{A}_k) \\
 \downarrow P_S & & \downarrow P_S \\
 P_S[X(\mathbf{A}_k)^{\text{Br}(X)}] & \xhookrightarrow{f_S} & P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}] \subseteq Y(\mathbf{A}_k^S)
 \end{array}$$

where f , f^* and f_S are all injective maps since $f : X \rightarrow Y$ is an immersion, which is universally injective, then $X(\mathbf{A}_k) \rightarrow Y(\mathbf{A}_k)$ and $X(\mathbf{A}_k^S) \rightarrow Y(\mathbf{A}_k^S)$ are injective^[95]. Thus, when the sets $X(k), Y(k), X(\mathbf{A}_k^S)$ are viewed as subsets of $Y(\mathbf{A}_k^S)$ under the injective maps above and P_S , we can conclude that $Y(k) \cap X(\mathbf{A}_k^S) = X(k)$. More precisely, it is obvious that $X(k) \subseteq Y(k) \cap X(\mathbf{A}_k^S)$ and conversely let $y \in Y(k)$ with $(y)_v = x_v$ in $Y(k_v)$ for some $x_v \in X(k_v)$, for all $v \notin S$. For any fixed v_0 , by lemma (3.1), one has $(y)_{v_0} \in X(k_{v_0}) \cap Y(k) = X(k)$. Thus, $X(k) = Y(k) \cap X(\mathbf{A}_k^S)$. Then,

$$\begin{aligned}
 X(k) &= Y(k) \cap X(\mathbf{A}_k^S) = P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}] \cap X(\mathbf{A}_k^S) \\
 &= P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)} \cap X(\mathbf{A}_k)] \\
 &= P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]
 \end{aligned}$$

where P_S is surjective by (1.2) and Y satisfies VSAP. Hence, X satisfies VSAP. $\blacksquare$

Remark 5.3: Let $f : X \rightarrow Y$ be a universally injective morphism of algebraic varieties over k . Let K/k be arbitrary field extension. Then the induced map $f : X(K) \rightarrow Y(K)$ is injective. Indeed, in the proof of Proposition (5.2), we have shown $X(K) = Y(K) \cap X(\mathbf{A}_K)$, hence $X(K) = Y(K) \cap X(\mathbf{A}_K^S)$ for some finite subset S consisting of places of K . If f is an immersion, in a more algebraic setting, Proposition (5.2) showed that for any $y = (y)_v \in Y(K) \cap X(\mathbf{A}_K)$, the corresponding k -map $\tilde{y} : \mathcal{O}_{Y,y} \rightarrow K$ has a commutative diagram

$$\begin{array}{ccc}
 \mathcal{O}_{Y,y} & \xrightarrow{\tilde{y}} & K \\
 \downarrow & & \downarrow \\
 \mathcal{O}_{X,y_v} & \xrightarrow{\tilde{y}_v} & K_v
 \end{array}$$

where the left vertical arrow is surjective because of immersions. So, $\tilde{y}_v$ will land on K .

The proposition (5.2) might fail for SAP-BM off S even though for open immersions and see counter-examples for [Example 5.2, [10]]. Besides, there is an analogous result in [[41], Theorem 3] for Conjecture (1.1).

Example 5.2 (Harari-Voloch conjecture): Let X be a hyperbolic curve over k . Let $f : U \rightarrow X$ be a quasi-finite morphism between k -curves with that all fibers of f are of degree at most 4. Let S be a finite subset of Ω_k containing ∞_k . By the Main theorem II, f can be extended to a map between faithfully flat models $\mathcal{U}$ of U and $\mathcal{X}$ of X . If Conjecture (1.1) holds for X , then the integral Brauer-Manin obstruction (1-3) is the only obstruction. Moreover, if U is an open subscheme of X , Conjecture (1.1) holds for U .

Proposition 5.3: Let X, Y be algebraic varieties over k with rational points, namely $X(k) \neq \emptyset, Y(k) \neq \emptyset$. Then the product variety $X \times_k Y$ satisfies SAP-BM (resp. VSAP-BM) off S if and only if the same holds for both X and Y .

Proof: By functoriality [[20], Proposition 13.3.10] of the Brauer-Manin set, one can establish the following commutative diagram induced by the projections

$$\begin{array}{ccc} (X \times Y)(k) \subseteq (X \times Y)(\mathbf{A}_k)^{\text{Br}(X \times Y)} & \xrightarrow{f} & X(\mathbf{A}_k)^{\text{Br}(X)} \times Y(\mathbf{A}_k)^{\text{Br}(Y)} \supseteq X(k) \times Y(k) \\ \downarrow & & \downarrow \\ (X \times_k Y)(\mathbf{A}_k) & \xrightarrow{\simeq} & X(\mathbf{A}_k) \times Y(\mathbf{A}_k) \end{array}$$

where by the universal property of the fiber products, $(X \times_k Y)(\mathbf{A}_k)$ is canonically homeomorphic to the product $X(\mathbf{A}_k) \times Y(\mathbf{A}_k)$, and $(X \times_k Y)(k)$ is homeomorphic to $X(k) \times Y(k)$. Thus, f is an injection and any non-empty open subset of $P_S[(X \times_k Y)(\mathbf{A}_k)^{\text{Br}(X \times Y)}]$ is the trace of some $U \times V$ with U, V open subsets of $X(\mathbf{A}_k^S)$ and $Y(\mathbf{A}_k^S)$ respectively such that $U \cap P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ and $V \cap P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}]$ are both non-empty since

$$\begin{aligned} P_S[(X \times Y)(\mathbf{A}_k)^{\text{Br}(X \times Y)}] &\subseteq P_S[X(\mathbf{A}_k)^{\text{Br}(X)} \times Y(\mathbf{A}_k)^{\text{Br}(Y)}] \\ &= P_S[X(\mathbf{A}_k)^{\text{Br}(X)}] \times P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}] \subseteq X(\mathbf{A}_k^S) \times Y(\mathbf{A}_k^S). \end{aligned}$$

If X and Y satisfy VSAP-BM off S , then

$$\begin{aligned} (X \times_k Y)(k) &\subseteq P_S[(X \times_k Y)(\mathbf{A}_k)^{\text{Br}(X \times_k Y)}] \subseteq P_S[X(\mathbf{A}_k)^{\text{Br}(X)} \times Y(\mathbf{A}_k)^{\text{Br}(Y)}] \\ &= P_S[X(\mathbf{A}_k)^{\text{Br}(X)}] \times P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}] = X(k) \times Y(k) \end{aligned}$$

which implies $X \times_k Y$ satisfies VSAP-BM off S .

If X and Y satisfy SAP-BM off S , for any non-empty open subset of $P_S[(X \times Y)(\mathbf{A}_k)^{\text{Br}(X \times Y)}]$, which is the trace form of some $U \times V$ with U, V open subsets of $X(\mathbf{A}_k^S)$ and $Y(\mathbf{A}_k^S)$ respectively, there exist rational points $x \in X(k) \cap U$ and $y \in Y(k) \cap V$, therefore $(x, y) \in (X \times Y)(k) \cap (U \times V) \subseteq P_S[(X \times Y)(\mathbf{A}_k)^{\text{Br}(X \times Y)}] \cap (U \times V)$. Thus,

$X \times_k Y$ satisfies SAP-BM off S .

Conversely, note that a rational point $x_0 \in X(k)$ (resp. $y_0 \in Y(k)$) gives rise to a closed immersion $s_{x_0} : Y \rightarrow X \times Y$ (resp. $s_{y_0} : X \rightarrow X \times Y$) by base change mapping $y \in Y$ (resp. $x \in X$) to (x_0, y) (resp. (x, y_0))

$$\begin{array}{ccc} Y & \xrightarrow{s_{x_0}} & X \times Y \\ \downarrow & & \downarrow \\ \text{Spec}(k) & \xrightarrow{x_0} & X \end{array} \quad \begin{array}{ccc} X & \xrightarrow{s_{y_0}} & X \times Y \\ \downarrow & & \downarrow \\ \text{Spec}(k) & \xrightarrow{y_0} & Y \end{array} .$$

Thus, by Proposition (5.2), if $X \times_k Y$ satisfies VSAP-BM off S , so do X and Y .

If $X \times Y$ satisfies SAP-BM off S , let U be a non-empty open subset of $P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ and $U \times P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}]$ is a non-empty open subset of $P_S[X(\mathbf{A}_k)^{\text{Br}(X)}] \times P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}]$. Thus, $(U \times P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}]) \cap P_S[(X \times_k Y)(\mathbf{A}_k)^{\text{Br}(X \times_k Y)}]$ is a non-empty open subset of $P_S[(X \times_k Y)(\mathbf{A}_k)^{\text{Br}(X \times_k Y)}]$ because it contains $U \times Y(k)$. More precisely, $Y(k) \neq \emptyset$ and for any $(u, y) \in U \times Y(k)$, the closed immersion $s_y : X \rightarrow X \times Y$ induces a pull-back map $s_y^* : \text{Br}(X \times Y) \rightarrow \text{Br}(X)$ on Brauer groups. By the choice of u , $u = P_S(\tilde{u}) = (u_v)_{v \notin S}$ with $\tilde{u} = (u_v)_{v \in \Omega_k} \in X(\mathbf{A}_k)^{\text{Br}(X)}$. With respect to $(\tilde{u}, y) = ((u_v)_{v \in \Omega_k}, y) \in X(\mathbf{A}_k) \times Y(\mathbf{A}_k) \simeq (X \times Y)(\mathbf{A}_k)$, we have

$$\sum_{v \in \Omega_k} \text{inv}_v(A(u_v, y)) = \sum_{v \in \Omega_k} \text{inv}_v((s_y^* A)(u_v)) = 0$$

which means $(\tilde{u}, y) \in (X \times Y)(\mathbf{A}_k)^{\text{Br}(X \times Y)}$ and $(u, y) \in P_S[(X \times Y)(\mathbf{A}_k)^{\text{Br}(X \times Y)}]$. Thus, by the assumption, there exists a rational point $(x, y) \in (X \times_k Y)(k)$ contained in $U \times P_S[Y(\mathbf{A}_k)^{\text{Br}(Y)}]$, which means that $x \in X(k) \cap U$. So, X satisfies SAP-BM off S . Similarly, Y also satisfies SAP-BM off S . ■

Remark 5.4: It is validated that for any two smooth geometrically integral algebraic varieties X and Y over number fields k , we have $(X \times Y)(\mathbf{A}_k)^{\text{Br}(X \times Y)} = X(\mathbf{A}_k)^{\text{Br}(X)} \times Y(\mathbf{A}_k)^{\text{Br}(Y)}$, in which the projective case is obtained by Skorobogatov and Zarhin in [[87], Theorem 3] and the general case is due to Lv chang in [[59], Thm. 3.1].

5.2 Baby case of VSAP

Let $\Omega_{\mathbb{C}}$ be the set of complex places of k if k is a number field and $\Omega_{\mathbb{C}} = \emptyset$ otherwise. Write $\mathbf{A}_k^{\mathbb{C}}$ for the ring of prime-to- $\Omega_{\mathbb{C}}$ adèles. In particular, $\mathbf{A}_k^{\mathbb{C}} = \mathbf{A}_k$ if k is a totally real number field. Let X be an algebraic variety over k . Let $P_{\mathbb{C}} : X(\mathbf{A}_k) \rightarrow X(\mathbf{A}_k^{\mathbb{C}})$ denote the projection map and it is easy to see $P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}] = X(\mathbf{A}_k^{\mathbb{C}})^{\text{Br}(X)}$ since $\text{Br}(\mathbb{C}) = 0$, and

for any Brauer class $A \in \text{Br}(X)$, the evaluation map $\text{ev}_A : X(\mathbf{A}_k) \rightarrow \mathbb{Q}/\mathbb{Z}$ factors through the evaluation map $\text{ev}_A^{\mathbb{C}} : X(\mathbf{A}_k^{\mathbb{C}}) \rightarrow \mathbb{Q}/\mathbb{Z}$.

Lemma 5.2: Let $X = \text{Spec}(k) \sqcup \text{Spec}(k)$. Then

$$X(k) = X(\mathbf{A}_k^{\mathbb{C}})^{\text{Br}(X)} = P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}].$$

Proof: Note that X has exactly two rational points P_1 and P_2 , both of which are isomorphic to $\text{Spec}(k)$ when endowed the induced reduced closed subscheme structure, and the adelic space $X(\mathbf{A}_k)$ of X is the disjoint union of the two rational points P_1 and P_2 . Let $x = (x_v)_{v \in \Omega_k \setminus \Omega_{\mathbb{C}}} \in X(\mathbf{A}_k^{\mathbb{C}})^{\text{Br}(X)}$. Let $S_1 \subseteq \Omega_k \setminus \Omega_{\mathbb{C}}$ consist of the places v with $x_v = P_1$, and then the complement $S_2 = (\Omega_k \setminus \Omega_{\mathbb{C}}) \setminus S_1$ consist of the places v with $x_v = P_2$. It is claimed that $S_1 = \emptyset$ or $S_2 = \emptyset$. Otherwise, we can choose $v_1 \in S_1$ and $v_2 \in S_2$. Since neither $v_1 \in S_1$ nor $v_2 \in S_2$ is a complex place, we can find an element $\xi \in \text{Br}(k)$ with the image $\xi^* \in \text{Br}(X)$ under the canonical map $\text{Br}(k) \rightarrow \text{Br}(X)$ such that $\text{inv}_{v_1}(\xi) = \text{inv}_{v_2}(\xi) = \frac{1}{2}$ and $\text{inv}_v(\xi) = 0$ for $v \neq v_1, v_2$. Consider the element $\beta = (\xi, 0) \in \text{Br}(X) = \text{Br}(k) \oplus \text{Br}(k)$. Then by the choice of x , we have $0 = \sum_{v \in \Omega_k \setminus \Omega_{\mathbb{C}}} \text{inv}_v(\beta(x_v)) = \text{inv}_{v_1}(\xi) = \frac{1}{2}$, which contradicts. Thus, $x \in X(k)$. ■

Theorem 5.5: Let X be an algebraic variety over k . If X can be written as a disjoint union $X = X_1 \sqcup X_2$ of two open subschemes X_1 and X_2 , then

$$P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}] = P_{\mathbb{C}}[X_1(\mathbf{A}_k)^{\text{Br}(X_1)}] \bigsqcup P_{\mathbb{C}}[X_2(\mathbf{A}_k)^{\text{Br}(X_2)}].$$

In particular, if X is finite over k , or equivalently X is a zero-dimensional scheme over k , then X satisfies VSAP-BM off any finite subset places of k containing $\Omega_{\mathbb{C}}$.

Proof: It is obvious to see that $P_{\mathbb{C}}[X_1(\mathbf{A}_k)^{\text{Br}(X_1)}] \sqcup P_{\mathbb{C}}[X_2(\mathbf{A}_k)^{\text{Br}(X_2)}] \subseteq P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}]$ since $X_i(\mathbf{A}_k)^{\text{Br}(X_i)} \subseteq X(\mathbf{A}_k)^{\text{Br}(X)}$. Now, it suffices to show $P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}] \subseteq P_{\mathbb{C}}[X_1(\mathbf{A}_k)^{\text{Br}(X_1)}] \sqcup P_{\mathbb{C}}[X_2(\mathbf{A}_k)^{\text{Br}(X_2)}]$. Suppose otherwise, one can pick an element

$$(x_v)_{v \in \Omega_k \setminus \Omega_{\mathbb{C}}} \in P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}]$$

with $v_1, v_2 \in \Omega_k \setminus \Omega_{\mathbb{C}}$ such that $x_{v_1} \in X_1(k_{v_1})$ and $x_{v_2} \in X_2(k_{v_2})$. By the Brauer-Hasse-Noether exact sequence, one can choose $\xi_i \in \text{Br}(k)$ satisfying

$$\begin{cases} \text{inv}_v(\xi_1) \neq \text{inv}_v(\xi_2), \text{ if } v = v_1 \text{ or } v = v_2 \\ \text{inv}_v(\xi_1) = \text{inv}_v(\xi_2) = 0, \text{ otherwise} \\ \text{inv}_{v_1}(\xi_i) + \text{inv}_{v_2}(\xi_i) = 0, \text{ for } i = 1, 2. \end{cases}$$

Let ξ_i^* be the image of ξ_i under the canonical map $\text{Br}(k) \rightarrow \text{Br}(X_i)$ for $i = 1, 2$ and by [[20], Theorem 3.2.4], one has $(\xi_1^*, \xi_2^*) \in \text{Br}(X_1) \oplus \text{Br}(X_2) \simeq \text{Br}(X)$, which corresponds

an element $\xi_1^* + \xi_2^*$ in $\text{Br}(X)$. Thus, we have

$$\begin{aligned} 0 &= \sum_{v \in \Omega_k \setminus \Omega_{\mathbb{C}}} \text{inv}_v((\xi_1^*, \xi_2^*)(x_v)) = \sum_{v \in \Omega_k \setminus \Omega_{\mathbb{C}}} \text{inv}_v((\xi_1^* + \xi_2^*)(x_v)) \\ &= \text{inv}_{v_1}((\xi_1^* + \xi_2^*)(x_{v_1})) + \text{inv}_{v_2}((\xi_1^* + \xi_2^*)(x_{v_2})) \\ &= \text{inv}_{v_1}(\xi_1) + \text{inv}_{v_2}(\xi_2) = \text{inv}_{v_1}(\xi_1) - \text{inv}_{v_1}(\xi_2) \end{aligned}$$

where the last equality follows from the Brauer-Hasse-Noether exact sequence, and make a contradiction. Hence, $P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}] = P_{\mathbb{C}}[X_1(\mathbf{A}_k)^{\text{Br}(X_1)}] \sqcup P_{\mathbb{C}}[X_2(\mathbf{A}_k)^{\text{Br}(X_2)}]$.

If X is finite over k , then X is a zero-dimensional and affine algebraic variety. By [[20], Proposition 13.3.21], one may assume that X is reduced since $\text{Br}(X) = \text{Br}(X_{\text{red}})$ and $X_{\text{red}}(\mathbf{A}_k) = X(\mathbf{A}_k)$. Also, by above result, one may assume that X is local, hence we can write $X = \text{Spec}(K)$ for some finite field extension K over k . If the field extension K/k is non-trivial, by [[20], Theorem 13.1.5], essentially Chebotarev's theorem, there exists infinitely many places v of k such that the $K \otimes_k k_v$ has no direct summand isomorphic to k_v , and for such places v , we have $X(k_v) = \emptyset$, in particular $P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}] = \emptyset$. So, it is the trivial extension $K = k$ that provides rational points, in which case $X(k) = X(\mathbf{A}_k)$, hence $X(k) = P_{\mathbb{C}}[X(\mathbf{A}_k)^{\text{Br}(X)}]$ as similar to Lemma (5.2). ■

Corollary 5.2: Let $X = X_1 \sqcup X_2 \sqcup \dots \sqcup X_n$ be a disjoint union of algebraic varieties X_i over k . Then

$$X(\mathbf{A}_k^{\mathbb{C}})^{\text{Br}(X)} = X_1(\mathbf{A}_k^{\mathbb{C}})^{\text{Br}(X_1)} \sqcup X_2(\mathbf{A}_k^{\mathbb{C}})^{\text{Br}(X_2)} \sqcup \dots \sqcup X_n(\mathbf{A}_k^{\mathbb{C}})^{\text{Br}(X_n)}.$$

Proof: It follows from Theorem (5.5) directly. ■

Remark 5.5: The above statements are Brauer group versions of the results of [[88], Lemma 5.10, Prop 5.11, Prop 5.12].

The following example is a baby case of Theorem (5.11) investigated by Q. Liu and F. Xu with respect to finite closed subschemes of an algebraic torus, after the work of Stoll [[88], Thm 8.2] with respect to finite closed subschemes of a smooth projective curve of positive genus.

Example 5.3: Let us discuss the famous counter-example to the Hasse principle over $\mathbb{Q}$ which is defined by a reducible polynomial

$$(x^2 - 13)(x^2 - 17)(x^2 - 221) = 0. \quad (5-1)$$

Let X be the finite closed subscheme of $\mathbb{G}_{m, \mathbb{Q}}$ defined by Equation (5-1). It is well-known that X is everywhere locally solvable but not solvable over $\mathbb{Q}$. Indeed, it is obvious that X is solvable at archimedean places. For any odd prime number p , one can consider the

Legendre symbol

$$\left(\frac{13}{p}\right)\left(\frac{17}{p}\right)\left(\frac{221}{p}\right) = \left[\left(\frac{13}{p}\right)\left(\frac{17}{p}\right)\right]^2 = 1.$$

Thus, either $\left(\frac{13}{p}\right) = 1$ or $\left(\frac{17}{p}\right) = 1$, and by henselian lemma, X has $\mathbb{Q}_p$ -points. If $p = 2$, it follows from the fact that an element $a = 2^r u \in \mathbb{Q}_2$ with $r = v_2(a) \in \mathbb{Z}$ is a square if and only if r is even and $u \equiv 1 \pmod{8}$. In particular, 17 is a square in $\mathbb{Q}_2$. Hence, X is everywhere locally solvable. On the other hand, one considers the Brauer class $A = (x, 13) \in \text{Br}(\mathbb{G}_{m, \mathbb{Q}})$ given by a quaternion algebra $(x, 13)$. We want to show $X(\mathbf{A}_{\mathbb{Q}})^A = \emptyset$, hence $X(\mathbf{A}_{\mathbb{Q}})^{\text{Br}(X)} = \emptyset$ and $X(\mathbb{Q}) = \emptyset$. Take any $(x_v) \in X(\mathbf{A}_{\mathbb{Q}})$, where v is a prime p or the archimedean place. It is clear that $A(x_{\infty}) = 0$. If $p \neq 2, 13$, $A(x_p) = (x_p, 13) = 0$ since 13 and x_p are both in $\mathbb{Z}_p^*$ and then $(x_p, 13) \in \text{Br}(\mathbb{Z}_p) = 0$ by a simple computation with Legendre symbol in [[37], Example 9.11]. If $p = 13$, one can solve the equation (5-1) to obtain $x_{13} = \pm 2$ up to a square in $\mathbb{Q}_{13}^*$, and then $A(x_{13}) = (x_{13}, 13) = (\pm 2, 13) = \frac{1}{2}$ by [[37], Remark 14.12]. If $p = 2$, one can obtain $x_2 = \pm 1$ up to a square in $\mathbb{Q}_2^*$ and $A(x_2) = (x_2, 13) = (\pm 1, 13) = 0$. Therefore, one can conclude $\sum_{v \in \Omega_{\mathbb{Q}}} \text{inv}_v(A(x_v)) = \text{inv}_{13}(A(x_{13})) = \frac{1}{2} \neq 0$. Thus, $X(\mathbf{A}_{\mathbb{Q}})^{\text{Br}(X)} = \emptyset$. In particular, X satisfies VSAP-BM off some finite subset S of $\Omega_{\mathbb{Q}}$ containing $\infty_{\mathbb{Q}}$.

5.3 Main theorem III

For any topological group G , denote by $G^{\wedge}$ the profinite completion of G .

Let T be a torus over a global field k . **The group of characters of T** is defined as $\hat{T} = \text{Hom}_{k^S}(T_{k^S}, \mathbb{G}_{m, k^S})$, which is a continuous discrete $\text{Gal}(k^S/k)$ -module.

Recall that **the idèle group $\mathbf{I}_k$ of k** is defined as

$$\mathbf{I}_k = \prod'_{v \in \Omega_k} (k_v^*, \mathcal{O}_v^*),$$

which is a locally compact group equipped with its restricted product topology. It is known that $\mathbf{I}_k$ has a basis of open neighbourhoods of 1 consisting of subsets $\prod_{v \notin S} \mathcal{O}_v^* \times \prod_{v \in S} U_v$ with a finite subset $S \subseteq \Omega_k$ and an open subset U_v of k_v^* for each $v \in S$. Moreover, it is clear that $\mathbf{I}_k$ is furthermore a totally disconnected topological group when k is a function field and in the case of the number field, the group of finite idèles, consisting of idèles whose component at archimedean places is 1, is totally disconnected. Besides, it should be careful that the topology on $\mathbf{I}_k$ is not induced by that of $\prod_{v \in \Omega_k} k_v^*$ directly.

Definition 5.2: One defines **the idèle group $\mathbf{I}_{k^S}$ of k^S** as $\mathbf{I}_{k^S} = \varinjlim_K \mathbf{I}_K$, where K runs

all finite separable extensions of k . Warn that it is not the restricted product of $(k^s)^*$.

A point $x = (x_v)_{v \in \Omega_k} \in T(\mathbf{A}_k)$ induces an evaluation homomorphism as follows

$$\mathrm{ev}_x : \hat{T} \rightarrow \mathbf{I}_{k^s}$$

$$f : T_{k^s} \rightarrow \mathbb{G}_{m,k^s} \mapsto f(x) = (f(x_v))_{v \in \Omega},$$

which is well-defined because each local point $x_v \in T(k_v)$ corresponds to a pair $(\widetilde{x}_v, \tau_v : k(x_v) \rightarrow k_v)$, where $\widetilde{x}_v$ is a closed point of T and τ is a k -embedding, making $f(\widetilde{x}_v)$ a $k^s.k_v$ -point, hence a $(k_v)^s$ -point by [[37], Proposition 13.6]. Thus, the evaluation map ev_x gives rise to a group homomorphism

$$H^2(k, \hat{T}) \xrightarrow{x^*} H^2(k, \mathbf{I}_{k^s}) \simeq \bigoplus_{v \in \Omega_k} \mathrm{Br}(k_v) \xrightarrow{\sum_{v \in \Omega_k} \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z}, \quad (5-2)$$

where the isomorphism follows from the cohomology of idèle group $\mathbf{I}_{k^s}$ by [[37], Corollary 13.7]. Therefore, we obtain a pairing

$$\begin{aligned} T(\mathbf{A}_k) \times H^2(k, \hat{T}) &\rightarrow \mathbb{Q}/\mathbb{Z} \\ (x = (x)_{v \in \Omega_k}, \alpha) &\rightarrow \left(\sum_{v \in \Omega_k} \mathrm{inv}_v \right) (x^*(\alpha)), \end{aligned} \quad (5-3)$$

which induces a group homomorphism

$$\varphi : T(\mathbf{A}_k) \rightarrow \mathrm{Hom}_{\mathbb{Z}}(H^2(k, \hat{T}), \mathbb{Q}/\mathbb{Z}). \quad (5-4)$$

By[[37], Remark 10.8], $\mathrm{Hom}_{\mathbb{Z}}(H^2(k, \hat{T}), \mathbb{Q}/\mathbb{Z})$ coincides with Pontryagin dual $H^2(k, \hat{T})^\vee$.

Proposition 5.4: The pairing (5-3) is compatible with the Brauer-Manin pairing (1-2). Moreover, the Brauer-Manin obstruction set $T(\mathbf{A}_k)^{\mathrm{Br}(T)}$ is contained in the left kernel of the pairing (5-3). Namely, one has

$$T(\mathbf{A}_k)^{\mathrm{Br}(T)} \subseteq \ker(\varphi) = T(\mathbf{A}_k)^{H^2(k, \hat{T})}.$$

Proof: It is obvious that the geometric Picard group $\mathrm{Pic}(T_{k^s})$ of torus T is trivial because $T_{k^s} \simeq \mathbb{G}_{m,k^s}^n = \mathrm{Spec}(k^s[x_1, x_1^{-1}, \dots, x_n, x_n^{-1}])$ with $n = \dim(T)$ and the Picard group of an affine scheme is equal to the class group of its coordinate ring, which is trivial by [[43], II. Proposition 6.2]. Actually, the picard group of each quasi-trivial torus (e.g. split torus) is trivial. On the one hand, by Hochschild-Serre spectral sequence in [[68], III. Theorem 2.20], one has a spectral sequence

$$E_2^{p,q} = H^p(k, H_{\mathrm{\acute{e}t}}^q(T_{k^s}, \mathbb{G}_m)) \Rightarrow H_{\mathrm{\acute{e}t}}^{p+q}(T, \mathbb{G}_m), \quad (5-5)$$

which gives rise to an exact sequence in [[77], Proposition 6.7.1]

$$0 \rightarrow E_2^{1,0} \rightarrow H_{\text{ét}}^1 \rightarrow E_2^{0,1} \rightarrow E_2^{2,0} \rightarrow \ker[H_{\text{ét}}^2 \rightarrow E_2^{0,2}] \rightarrow E_2^{1,1} \rightarrow E_2^{3,0}.$$

By [[77], Proposition 6.6.1], it is clear that

$$\begin{aligned} E_2^{1,0} &= H^1(k, H_{\text{ét}}^0(T_{k^s}, \mathbb{G}_m)) = H^1(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times) \\ E_2^{0,1} &= H^0(k, H_{\text{ét}}^1(T_{k^s}, \mathbb{G}_m)) = H^0(k, \text{Pic}(T_{k^s})) = \text{Pic}(T_{k^s})^{\text{Gal}(k^s/k)} \\ E_2^{2,0} &= H^2(k, H_{\text{ét}}^0(T_{k^s}, \mathbb{G}_m)) = H^2(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times) \\ E_2^{0,2} &= H^0(k, H_{\text{ét}}^2(T_{k^s}, \mathbb{G}_m)) = H^0(k, \text{Br}(T_{k^s})) = \text{Br}(T_{k^s})^{\text{Gal}(k^s/k)} \\ E_2^{1,1} &= H^1(k, H_{\text{ét}}^1(T_{k^s}, \mathbb{G}_m)) = H^1(k, \text{Pic}(T_{k^s})) \\ E_2^{3,0} &= H^3(k, H_{\text{ét}}^0(T_{k^s}, \mathbb{G}_m)) = H^3(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times). \end{aligned}$$

Thus, one can obtain the following exact sequence (see also [[20], Proposition 5.4.2])

$$\begin{aligned} 0 \rightarrow H^1(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times) \rightarrow \text{Pic}(T) \rightarrow \text{Pic}(T_{k^s})^{\text{Gal}(k^s/k)} \rightarrow H^2(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times) \rightarrow \text{Br}_1(T) \\ \rightarrow H^1(k, \text{Pic}(T_{k^s})) \rightarrow H^3(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times), \end{aligned} \quad (5-6)$$

where $\text{Br}_1(T) = \ker[\text{Br}(T) \rightarrow \text{Br}(T_{k^s})]$ denotes the algebraic part of $\text{Br}(T)$. Since $\text{Pic}(T_{k^s}) = 0$, we have $H^1(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times) \simeq \text{Pic}(T)$ and $\text{Br}_1(T) \simeq H^2(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times)$.

On the other hand, the short split exact sequence of abelian groups

$$1 \rightarrow k^s \rightarrow \mathcal{O}_{T_{k^s}}(T_{k^s})^\times \rightarrow \hat{T} = \text{Hom}_{k^s}(T_{k^s}, \mathbb{G}_{m, k^s}) \rightarrow 1$$

induces an isomorphism (see also [[20], Proposition 9.1.2])

$$\text{Br}(k) \oplus H^2(k, \hat{T}) = H^2(k, \mathcal{O}_{T_{k^s}}(T_{k^s})^\times) \simeq \text{Br}_1(T) \hookrightarrow \text{Br}(T)$$

which commutes with the evaluation for any $x \in T(\mathbf{A}_k)$

$$\begin{array}{ccc} H^2(k, \hat{T}) & \hookrightarrow & \text{Br}(T) \\ \downarrow (\sum_{v \in \Omega_k} \text{inv}_v) \circ x^* & & \downarrow \sum_{v \in \Omega_k} \text{inv}_v \\ \bigoplus_{v \in \Omega_k} \text{Br}(k_v) & \xlongequal{\quad} & \bigoplus_{v \in \Omega_k} \text{Br}(k_v) \end{array}.$$

Thus, one has $T(\mathbf{A}_k)^{\text{Br}(T)} \subseteq \ker(\varphi) = T(\mathbf{A}_k)^{H^2(k, \hat{T})}$. ■

Here is a quick application of sequence (5-6), especially related to Conjecture (1.1).

Corollary 5.3: Let k be any field and X be an affine k -curve contained in $\mathbb{P}_k^1$ with $X(k) \neq \emptyset$. If k is perfect, or k is separably closed and X is proper over k , there is a split exact sequence

$$0 \rightarrow \text{Br}(k) \rightarrow \text{Br}(X) \rightarrow H^2(k, \mathcal{O}_{X_{k^s}}(X_{k^s})^\times / (k^s)^*) \rightarrow 0. \quad (5-7)$$

Moreover, if k is a global field and S is a finite subset of Ω_k containing ∞_k , then one has a split exact sequence

$$0 \rightarrow \text{Br}(k) \rightarrow \text{B}_S(X) \rightarrow \ker \left[H^2(k, \mathcal{O}(X_{k^S})^\times / k^{S*}) \xrightarrow{\text{Res}} \prod_{v \in S} H^2(k_v, \mathcal{O}(X_{k^S})^\times / k^{S*}) \right] \rightarrow 0.$$

Proof: On the one hand, since X is an affine open subscheme of $\mathbb{P}_k^1$, one has $\text{Pic}(X_{k^S}) = 0$ because of smoothness. On the other hand, by Tsen's theorem (see [[20], Theorem 1.2.14]) and [[20], Theorem 5.6.1], $\text{Br}(X_{k^S}) \subseteq \text{Br}(k^S(X_{k^S})) = 0$. Thus, by sequence (5-6), one can obtain $\text{Br}(X) \simeq H^2(k, \mathcal{O}(X_{k^S}))^\times$. The existence of k -rational point on X admits a split exact sequence of $\text{Gal}(k^S/k)$ -modules

$$1 \rightarrow k^{S*} \rightarrow \mathcal{O}(X_{k^S})^\times \rightarrow \mathcal{O}(X_{k^S})^\times / k^{S*} \rightarrow 1 \quad (5-8)$$

which induces a split exact sequence (5-7) above. When k is a global field, the sequence (5-7) holds over k_v for any $v \in S$ and the lemma follows easily. ■

Remark 5.6: The étale cohomology of split algebraic tori has been studied by many mathematicians, such as Magid^[60], Gille et al.^[32], Gille et al.^[33]. If k is a field of characteristic 0 and T is a torus over k of dimension n , there are non-canonical isomorphisms of abelian groups $H_{\text{ét}}^1(T_{k^S}, \mathbb{Q}/\mathbb{Z}) \simeq (\mathbb{Q}/\mathbb{Z})^n$, $\text{Br}(T_{k^S}) \simeq (\mathbb{Q}/\mathbb{Z})^{\frac{n(n-1)}{2}}$. If k is perfect field and $T \simeq \mathbb{G}_{m,k}^n$ is split over k , the natural map $\text{Br}(T) \rightarrow \text{Br}(T_{k^S})^{\text{Gal}(k^S/k)}$ is surjective. When k is algebraically closed, Harari et al.^[36] computed the Brauer group of a torsor.

Proposition 5.5: Let T be a torus over a global function field k . Then there exist canonical exact sequences:

$$1 \rightarrow T(k)^\wedge \rightarrow T(\mathbf{A}_k)^\wedge \rightarrow H^2(k, \widehat{T})^\vee \quad (5-9)$$

$$1 \rightarrow T(k) \rightarrow T(\mathbf{A}_k) \rightarrow H^2(k, \widehat{T})^\vee \quad (5-10)$$

where $\vee$ denotes the functor of Pontryagin dual $\text{Hom}_{\text{cont}}(_, \mathbb{Q}/\mathbb{Z})$.

Proof: See [[57], Proposition 3.4]. ■

Remark 5.7: The analogous result of Proposition (5.5) over number fields is obtained by Harari^[38], which is a special case of the generalized Cassels-Tate exact sequence of 1-motives (see [[67], theorem I 6.2.6] for Cassels-Tate exact sequence of abelian varieties) and can be used to measure the defect of strong approximation. Similarly, the defect of weak approximation of 1-motives can be measured by [[40], theorem 1.2].

Cao et al.^[10], and Wei^[97] obtained SAP-BM off ∞_k for smooth toric varieties over a number field k . Besides, after Harari^[38] obtained SAP-BM for algebraic tori over a

number field k off ∞_k , Cao et al.^[10] extended SAP-BM off ∞_k for algebraic tori to a relative version, and Demarche^[26] established SAP-BM off ∞_k for reductive groups.

Furthermore, we show that algebraic tori satisfies VSAP-BM.

Theorem 5.6: Let T_k be a torus over a global field k . Then T_k satisfies VSAP-BM off any finite subset S containing ∞_k , if one of the following conditions hold:

- (1) k is a global function field.
- (2) k is a number field and

$$\text{rank}_k(T_k) = \sum_{v \in \infty_k} \text{rank}_{k_v}(T_{k_v}),$$

where $\text{rank}_k(T_k)$ and $\text{rank}_{k_v}(T_{k_v})$ are respectively the ranks of the free abelian group of rational characters $\text{Hom}_k(T_k, \mathbb{G}_{m,k})$ and $\text{Hom}_{k_v}(T_{k_v}, \mathbb{G}_{m,k_v})$.

In case (2) above, $T_k(k)$ is discrete, hence closed in $T_k(\mathbf{A}_k^S)$.

Proof: By Proposition (5.4), it suffices to show $\ker(\varphi) = T(\mathbf{A}_k)^{H^2(k, \hat{T})} = T(k)$. When k is a global function field, it follows from Proposition (5.5) directly. When k is a number field, under our assumption, by [[83], Main theorem], a generalization of Dirichlet's unit theorem, the S -unit group of T over k

$$T_k(S) = T_k(k) \cap \left(\prod_{v \in S} T_k(k_v) \times \prod_{v \notin S} T_k(k_v)^c \right)$$

is isomorphic to the finite group $T_k(k) \cap \prod_{v \in \Omega_k} T_k(k_v)^c$, where $T_k(k_v)^c$ is the unique maximal compact subgroup of $T_k(k_v)$ and also open in $T_k(k_v)$. In particular, the intersection of $T_k(k)$ and an open subgroup of $T(\mathbf{A}_k)$ is finite, which implies that $T_k(k)$ is discrete, hence closed in $T_k(\mathbf{A}_k)$ because any discrete group in a Hausdorff topological group is closed. It remains to follow Théorème 2 and also its Remarques of Harari^[38],

$$0 \rightarrow \overline{H^0(k, M)} \rightarrow \mathbf{P}_0(M) \rightarrow H^1(k, M^*)^\vee \rightarrow \text{III}^1(M) \rightarrow 0, \quad (5-11)$$

where $\text{III}^1(M)$ is the Tate-Shafarevich group of M and we take the 1-motive $M = [0 \rightarrow T_k]$, then $\mathbf{P}_0(M) = T(\mathbf{A}_k)$, $H^1(k, M^*)^\vee = H^2(k, T_k)^\vee$, $\overline{H^0(k, M)} = \overline{T_k(k)} = T_k(k)$. ■

Example 5.4: If k is a number field, the following tori satisfy the rank condition (2):

- (1) T_k is split torus over k and, $k = \mathbb{Q}$ or k is a quadratic imaginary field.
- (2) k is a totally real field, and

$$T_k = \prod_{i=1}^d \text{Res}_{L_i/k}^1(\mathbb{G}_{m,L_i}) \quad \text{or} \quad T_k = \prod_{i=1}^d \text{Res}_{L_i/k}(\mathbb{G}_{m,L_i})$$

where L_i are totally imaginary quadratic extensions of k (i.e L_i are CM-fields over k)

and $\text{Res}_{L_i/k}^1(\mathbb{G}_{m,L_i})$ denotes the norm one torus, namely the kernel of the norm map $\text{Res}_{L_i/k}(\mathbb{G}_{m,L_i}) \rightarrow \mathbb{G}_{m,k}$. Actually, for $v \in \infty_k$,

Torus	dim	$\text{rank}_k / \text{rank}_{k_v}$	$\text{rank}_{k^s} / \text{rank}_{k_v^s}$
$\text{Res}_{L_i/k}(\mathbb{G}_{m,L_i})$	2	1	2
$\text{Res}_{L_i/k}^1(\mathbb{G}_{m,L_i})$	1	0	1
$\text{Res}_{L_i/k}(\mathbb{G}_{m,L_i})_{k_v}$	2	1	2
$\text{Res}_{L_i/k}^1(\mathbb{G}_{m,L_i})_{k_v}$	1	0	1

More precisely, let us denote the k -th projection $\epsilon_k : \text{Res}_{L_i/k}(\mathbb{G}_{m,L_i}) \times_k k^s \rightarrow \mathbb{G}_{m,k^s}^2$. Then, $\widehat{\text{Res}_{L_i/k}(\mathbb{G}_{m,L_i})} \simeq \mathbb{Z}\epsilon_1 \oplus \mathbb{Z}\epsilon_2$ and

$$\text{Hom}_k(\text{Res}_{L_i/k}(\mathbb{G}_{m,L_i}), \mathbb{G}_{m,k}) = \text{Hom}_k(\text{Res}_{L_i/k}(\mathbb{G}_{m,L_i}), \mathbb{G}_{m,k})^{\text{Gal}(k^s/k)} \simeq (\epsilon_1 + \epsilon_2)\mathbb{Z}.$$

Thus, $\text{Hom}_k(\text{Res}_{L_i/k}^1(\mathbb{G}_{m,L_i}), \mathbb{G}_{m,k}) \simeq \widehat{\text{Res}_{L_i/k}(\mathbb{G}_{m,L_i})} / \mathbb{Z}(\epsilon_1 + \epsilon_2) \simeq \mathbb{Z}\theta$ with $\theta \equiv \epsilon_1 \pmod{\epsilon_1 + \epsilon_2}$, whose $\text{Gal}(k^s/k)$ -invariant is trivial.

(3) T_k is isogeneous to any finite product of the above tori over k .

Theorem 5.7: Let X be a subvariety of a k -torus T as in Theorem (5.6), which is a closed subvariety of an open subvariety of T . Then for any finite subset S of places of k containing ∞_k , VSAP-BM off S holds for X . In particular, when $S = \infty_k$, we obtain

$$X(k) = P_{\infty_k}[X(\mathbf{A}_k)^{\text{Br}(X)}].$$

Proof: This follows from Proposition (5.1) and Theorem (5.6) directly. ■

Example 5.5: For the following algebraic varieties X over $\mathbb{Q}$, VSAP-BM off $\infty_{\mathbb{Q}}$ holds.

(1) Let X be an algebraic variety over $\mathbb{Q}$ defined by the diagonal equations

$$\sum_{i=1}^d a_i x_i^{n_i} = a_0 \quad \text{with} \quad \prod_{i=1}^d x_i \neq 0 \quad (5-12)$$

where $a_0, a_i \in \mathbb{Q}$, $n_i \in \mathbb{N} \setminus \{0\}$ for $1 \leq i \leq d$ and $\prod_{i=1}^d x_i \neq 0$ means each variable x_i is invertible in the coordinate ring, defining $\mathbb{G}_{m,\mathbb{Q}}^d$. Thus, X is a closed subvariety of $\mathbb{G}_{m,\mathbb{Q}}^d$.

(2) Let E be an elliptic curve over $\mathbb{Q}$ with identity $\mathcal{O}$ and $x_1, x_2, x_3 \in E(\mathbb{Q})[3]$ be three distinct 3-torsion rational points on E . Take $D = x_1 + x_2 + x_3$ and then $X = E \setminus \{x_1, x_2, x_3\} \subseteq \mathbb{P}_{\mathbb{Q}}^2 \setminus \{L_1, L_2, L_3\} \simeq \{[x : y : z] \mid xyz \neq 0\} \simeq \mathbb{G}_{m,\mathbb{Q}}^2$ via D , where L_i is the tangent line of E at x_i . More precisely, let $D_i = D \cap L_i$ be the intersection divisors that are exactly hyperplane sections, hence linear equivalent. By Bézout's theorem, there are 9 intersection points with multiplicity. In resulting $\mathbb{P}_{\mathbb{Q}}^2$ given by D , by construction x_1, x_2, x_3 are colinear, hence $3x_1 \sim 3x_2 \sim 3x_3 \sim D \sim \mathcal{O}$. Thus, $L_i.D = 3$ and $D_i = \{x_i\}$ with

multiplicity 3. Moreover, the condition of 3-torsion makes use of symmetric intersection configurations, which is not necessary. For example, one can take an elliptic curve E defined by the equation

$$Y^2Z = X^3 - 3X^2Z + 4XZ^2,$$

and take $x_1 = [0 : 1 : 0]$, $x_2 = [0 : 0 : 1] \in E(\mathbb{Q})[2]$ and $x_3 = [2 : 2 : 1] \in E(\mathbb{Q})[4]$.

(3) Let $X = \mathbb{P}_{\mathbb{Q}}^1 \setminus \{x\}$ with an imaginary quadratic point x (a closed point of $\mathbb{P}_{\mathbb{Q}}^1$ with imaginary quadratic residue field), which is a norm one torus. More precisely, one can write the residue field of x as $K = \mathbb{Q}(\sqrt{-d})$, where d is a square-free integer and $d \neq 0, 1$, and x corresponds to a morphism $\text{Spec}(K) \rightarrow \mathbb{P}_{\mathbb{Q}}^1$ which base changing to K gives two K -points, the identity and the conjugate embedding of K into itself, denoted by P_1 and P_2 . When base changing to K , $X = \mathbb{P}_K^1 \setminus \{x_K\} = \mathbb{P}_K^1 \setminus \{P_1, P_2\} \simeq \mathbb{G}_{m,K}$. Thus, X is a $\mathbb{Q}$ -torus split over K , hence a norm torus since $\dim(X) = 1$ (see [[77], Example 5.5.5]).

(4) By (2) of Example (5.4), with the same argument as (3), one can take $X = \prod_i^d (\mathbb{P}_{\mathbb{Q}}^1 \setminus \{x_i\})$, where x_i are imaginary quadratic points of $\mathbb{P}_k^1$.

Corollary 5.4: Let E be an elliptic curve over k with $E(k)$ and $\text{III}(E/k)$ finite, then for any affine open subscheme U of E and any finite subset $S \subseteq \Omega_k$ containing ∞_k , the Harari-Voloch Conjecture (1.1) holds.

Proof: By Example (5.2), it suffices to show the Harari-Voloch conjecture (1.1) for the elliptic curve E . Since $E(k)$ is a finite and E is proper over k , by an S -part analogy of the sequence (5-11) (see also [[38], Proposition 3]), there is a sequence of Cassels-Tate like exact sequence

$$0 \rightarrow E(k) \rightarrow \prod_{v \notin S} E(k_v) \rightarrow B_S(E)^{\vee}.$$

Thus, Conjecture (1.1) is true for E . ■

Corollary 5.5: Let X be an algebraic variety over k , where $k = \mathbb{Q}$, or k is a CM-field, or k is a global function field. Then for any finite subset $S \subseteq \Omega_k$ containing ∞_k , there exists an open dense subscheme U of X such that U satisfies VSAP-BM off S .

Proof: By Theorem (5.7), Corollary (5.2) and Example (5.4), it suffices to show that X contains an open dense subscheme U consisting of a finite disjoint union of subvarieties of split tori over k . When restricted to each irreducible component, we can assume that X is irreducible and even integral since $X_{\text{red}}(\mathbf{A}_k) = X(\mathbf{A}_k)$. Shrinking further if necessary, by the definition of adelic space, we can assume that X is an affine variety. Then, we can embed X as a closed subvariety into some affine space $\mathbb{A}_k^n$ with minimal n . Similarly to (2)

of Example (5.5), let T be a split torus obtained as the complement of n general position hyperplanes in $\mathbb{A}_k^n$, which is an open subvariety of $\mathbb{A}_k^n$. Then $U = X \cap T$ is non-empty by the minimality of n , a closed subvariety of T , and also an open subvariety of X . ■

Remark 5.8: For an open subscheme U of a smooth geometrically integral algebraic variety X over k , if $\text{Br}(U)/\text{Br}(k)$ is finite, Colliot-Thélène and Xu obtained [[22], Proposition 2.6] that the SAP-BM of U implies the SAP-BM of X . It is natural ask if the finiteness condition for $\text{Br}(U)/\text{Br}(k)$ is necessary and Corollary (5.5) gives a negative answer.

In the end of this section, we give a counter-example in which VSAP-BM fails.

Example 5.6: Let k be a number field and X be an algebraic variety over k admitting a non-constant k -morphism $f : \mathbb{A}_k^1 \rightarrow X$. Then for any $\infty_k \subseteq S \subseteq \Omega_k$, X fails VSAP-BM off S . It suffices to show that $X(k) \rightarrow P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ is not surjective. Note that the image $f(\mathbb{A}_k^1)$ is of dimension 1 and an irreducible constructible set by Chevalley's theorem (see [[43], Exercise II. 3.19]), hence locally closed in X , which is an integral subvariety of X when endowed with reduced structure and actually nothing but the reduced component of the scheme-theoretic image of f from Proposition (4.1). Thus, by Proposition (5.2), it is allowed to replace X by $f(\mathbb{A}_k^1)$ and suppose that f is surjective. By Corollary (3.1), we can extend f to a morphism $\pi : \mathbb{P}_k^1 \rightarrow \bar{X}$, where $\bar{X}$ is a compactification of X . It is clear that π is a quasi-finite proper morphism, hence finite and surjective by Zariski's main theorem. Then, $\pi^{-1}(X) = \mathbb{A}_k^1$ and $f = \pi|_{\mathbb{A}_k^1}$ is a finite morphism. By Lüroth's theorem (see [[53], Corollary 7.4.23]), any curve that admits a non-constant rational map from $\mathbb{P}_k^1$ is itself a k -rational curve (i.e birational to $\mathbb{P}_k^1$), and thus the normalization of $\bar{X}$ (the unique smooth projective curve birational to $\bar{X}$) must be the projective line $\mathbb{P}_k^1$. Thus, we can replace f by the normalization map of X and furthermore suppose that f is birational. Let V be the smooth locus of X and then $U = f^{-1}(V) \simeq V$ because the normalization map is an isomorphism when restricted to the smooth locus of target. Hence, the injective map

$$P_S[U(\mathbf{A}_k)] \cong P_S[V(\mathbf{A}_k)] \rightarrow P_S[X(\mathbf{A}_k)]$$

factors through $P_S[U(\mathbf{A}_k)] \subseteq P_S[\mathbb{A}_k^1(\mathbf{A}_k)] = P_S[\mathbb{A}_k^1(\mathbf{A}_k)^{\text{Br}(\mathbb{A}_k^1)}] \rightarrow P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ because $\text{Br}(k) \rightarrow \text{Br}(\mathbb{A}_k^1)$ is an isomorphism when k is a perfect field by [[20], Theorem 5.6.1]. Therefore, $P_S[U(\mathbf{A}_k)] \subseteq P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$. If the map $X(k) \rightarrow P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ is surjective, as $X(k)$ is countable, $P_S[X(\mathbf{A}_k)^{\text{Br}(X)}]$ is countable, which means $P_S[U(\mathbf{A}_k)]$ is countable. This is impossible because $U(\mathbf{A}_k)$ is uncountable.

Remark 5.9: In Example (5.6), it is validated that any curve admitting a non-constant morphism from $\mathbb{A}_k^1$ is a k -rational curve whose normalization is exactly $\mathbb{A}_k^1$. By Riemann-

Roch theorem, we can prove that any integral smooth algebraic k -curve that is not proper must be affine. Moreover, one can show that any separated one-dimensional scheme of finite type over k , none of whose irreducible components is proper, is affine.

5.4 The Harari-Voloch conjecture over global function fields

In this section, we will give some evidence and list some results of the Harari-Voloch conjecture obtained by Q.Liu and F.Xu in [57]. Furthermore, we will demonstrate a new conjecture (1.2) proposed by them, which is related to Adelic Mordell-Lang conjecture.

Theorem 5.8: Let $X = \mathbb{P}_{\mathbb{Q}}^1 \setminus D$ be a hyperbolic rational curve (1) over a global field k , where D is an effective separable reduced divisor of degree at least 3. Let T be a torus over k such that the group of characters $\widehat{T}$ is isomorphic to $\text{Div}_{D_{k^S}}^0(\mathbb{P}_{k^S}^1)$ as $\text{Gal}(k^S/k)$ -module, which denotes the group of divisors of degree 0 on $\mathbb{P}_{k^S}^1$ supported in D_{k^S} . Fix a non-empty finite set $\infty_k \subseteq S \subseteq \Omega_k$. Then

(1) There exists a closed immersion $j : X \rightarrow T$ such that the induced group homomorphism $\mathcal{O}(T_{k^S})^\times / (k^S)^\times \rightarrow \mathcal{O}(X_{k^S})^\times / (k^S)^\times$ is an isomorphism.

(2) Let $B_{1,S}(T) = \ker[H^2(k, \widehat{T}) \rightarrow \prod_{v \in S} H^2(k_v, \widehat{T})]$. Then

$$\overline{T(k)}^S = T(\mathbf{A}_k^S)^{B_{1,S}(T)}$$

where $\overline{T(k)}^S$ is the topological closure of $T(k)$ inside $T(\mathbf{A}_k^S)$.

(3) The Harari-Voloch conjecture is true for all flat models $\mathbf{X}$ of X over $\mathcal{O}_S$ if and only if $\overline{T(k)}^S \cap X(\mathbf{A}_k^S) = X(k)$.

Proof: This follows from Lemma 4.4 and Proposition 4.6 in [57]. ■

Theorem 5.9: The Harari-Voloch conjecture is true if k is a global function field and X is not defined over a finite field.

Proof: By Theorem (5.8), one can embed X into a torus T corresponding to $\text{Gal}(k^S/k)$ -module $\text{Div}_{D_{k^S}}^0(\mathbb{P}_{k^S}^1)$ as a closed subscheme and it suffices to show that $\overline{T(k)}^S \cap X(\mathbf{A}_k^S) \subseteq X(k)$. For any $x \in \overline{T(k)}^S \cap X(\mathbf{A}_k^S)$, by the definition of adelic topology (3.2.1), there exists a finite set $\tilde{S} \supseteq S$ such that T can extend to a group scheme $\mathbf{T}$ of finite type over $\mathcal{O}_{\tilde{S}}$ with

$$x \in \prod_{\tilde{S} \setminus S} T(k_v) \times \prod_{v \notin \tilde{S}} \mathbf{T}(\mathcal{O}_v),$$

which implies $x \in \overline{\mathbf{T}(\mathcal{O}_{\tilde{S}})}$, where $\overline{\mathbf{T}(\mathcal{O}_{\tilde{S}})}$ is the topological closure of $\mathbf{T}(\mathcal{O}_{\tilde{S}})$ in $\prod_{v \notin \tilde{S}} T(k_v)$, because $\prod_{v \notin \tilde{S}} \mathbf{T}(\mathcal{O}_v)$ is an open subset of $\mathbf{T}(\mathbf{A}_k^{\tilde{S}})$ and intersects with $T(k)$ is equal to $\mathbf{T}(\mathcal{O}_{\tilde{S}})$. Since $\mathbf{T}(\mathcal{O}_{\tilde{S}})$ is a finitely generated abelian group (Dirichlet's unit

theorem), thus

$$x \in \overline{T(\mathcal{O}_{\tilde{S}})} \cap X(\mathbf{A}_k^{\tilde{S}}) \subseteq \overline{T(\mathcal{O}_{\tilde{S}})} \cap \prod_{v \notin \tilde{S}} X(k_v) = T(\mathcal{O}_{\tilde{S}}) \cap X(k) \subseteq X(k),$$

where the equality follows from [[89], theorem 1]. The proof is finished. $\blacksquare$

Theorem 5.10 ([57], Corollary 5.11): Let D be a reduced effective divisor of $\mathbb{P}_{\mathbb{Q}}^1$ supported in at least two points. Let $X = \mathbb{P}_{\mathbb{Q}}^1 \setminus D$ be the complement of D and $S = \infty_{\mathbb{Q}}$. Then $X(\mathbb{Q}) = X(\mathbf{A}_{\mathbb{Q}}^S)^{\text{Bs}(X)}$. In particular, the Harari-Voloch conjecture holds for X and S if and only if $\deg(D) \geq 3$, equivalently if X is a rational hyperbolic curve.

Example 5.7: Let X be a non-empty open subset of $Y = \mathbb{P}_{\mathbb{Q}}^1 \setminus D$, where D consists of either three rational points, or two imaginary quadratic points, or one point of each type (a rational point with an imaginary quadratic point). Then Harari-Voloch conjecture (1.1) holds for X with $S = \infty_{\mathbb{Q}}$. Indeed, the torus that the curve Y is embedded into following Theorem (5.8) satisfies the condition (2) of Theorem (5.6). In particular, $\overline{T(\mathbb{Q})}^S = T(\mathbb{Q})$. By Theorem (5.8), $\overline{T(\mathbb{Q})}^S \cap Y(\mathbf{A}_k^S) = T(\mathbb{Q}) \cap Y(\mathbf{A}_k^S) = Y(\mathbb{Q})$ and then Harari-Voloch conjecture holds for Y . By Example (5.2), Harari-Voloch conjecture holds for X .

As for the condition 3 in Theorem (5.8), Liu and Xu obtained a general equality.

Theorem 5.11 ([57], Theorem 5.7): Let T be a torus over k . If Z is a finite closed subscheme of T , then

$$Z(k) = \overline{T(k)}^S \cap Z(\mathbf{A}_k^S)$$

where $\overline{T(k)}^S$ is the topological closure of $T(k)$ inside $T(\mathbf{A}_k^S)$.

Recently, an analogous result of Theorem (5.11) has been obtained by Creutz^[7].

Theorem 5.12 ([7], Theorem 1.1/1.2): Let A be an abelian variety over a global function field k . Assume that $A_{\bar{k}}$ has no non-zero isotrivial quotient. Then for any closed subvariety X of A , $\overline{X(k)} = X(\mathbf{A}_k) \cap \overline{A(k)}$ holds, where $\overline{X(k)}$ (resp. $\overline{A(k)}$) is the topological closure of $X(k)$ (resp. $A(k)$) in the adelic space $X(\mathbf{A}_k)$ (resp. $A(\mathbf{A}_k)$). Moreover, $\overline{X(k)} = X(\mathbf{A}_k)^{\text{Br}(X)}$ if and only if the maximal divisible subgroup of $\text{III}(A)$ is trivial, if and only if $\overline{A(k)} = A(\mathbf{A}_k)^{\text{Br}(A)}$.

Remark 5.10: Theorem (5.12) is a conjecture of Poonen and Voloch^[78].

As an application of Theorem (5.11), the result of Example (5.2) can be generalized to any quasi-finite morphism of hyperbolic rational curves.

Theorem 5.13: Let $f : X \rightarrow Y$ be a quasi-finite morphism of curves over a global field k , where X is a rational hyperbolic curve. Let S be a finite subset of Ω_k containing ∞_k . If $Y(\mathbf{A}_k^S)^{\text{Bs}(Y)} = Y(k)$, then $X(\mathbf{A}_k^S)^{\text{Bs}(X)} = X(k)$. In particular, the Harari-Voloch

conjecture holds for X .

Proof: Let $x \in X(\mathbf{A}_k^S)^{\text{Bs}(X)}$. Then there exists $y \in Y(\mathbf{A}_k^S)^{\text{Bs}(Y)} = Y(k)$ such that $y = f(x)$ by the functoriality of the Brauer-Manin pairing. Since f is quasi-finite and y corresponds to a closed point of Y , $Z = f^{-1}(y)$ is a finite closed subscheme over k and $x \in Z(\mathbf{A}_k^S)$. Let $j : X \rightarrow T$ be the closed immersion obtained in Theorem (5.8) and $\overline{T(k)}^S$ is the topological closure of $T(k)$ in $T(\mathbf{A}_k^S)$. Then the induced group homomorphism $\mathcal{O}(T_{k^S})^\times / (k^S)^\times \rightarrow \mathcal{O}(X_{k^S})^\times / (k^S)^\times$ is an isomorphism. Since $\widehat{T} \simeq \mathcal{O}(T_{k^S})^\times / (k^S)^\times$ (5-8), by corollary (5.3), there is a split exact sequence induced by j

$$0 \rightarrow \text{Br}(k) \rightarrow \text{B}_S(X) \xrightarrow{j^*} \text{B}_{1,S}(T) \rightarrow 0$$

which gives an injective map $X(\mathbf{A}_k^S)^{\text{Bs}(X)} \rightarrow T(\mathbf{A}_k^S)^{\text{B}_{1,S}(T)}$. By theorem (5.8) (2), one has

$$\overline{T(k)}^S = T(\mathbf{A}_k^S)^{\text{B}_{1,S}(T)}.$$

Thus, $x \in X(\mathbf{A}_k^S) \cap Z(\mathbf{A}_k^S) \subseteq \overline{T(k)}^S \cap Z(\mathbf{A}_k^S) = Z(k) \subseteq X(k)$ by Theorem (5.11). $\blacksquare$

Conversely, inspired by the adelic Mordell-Lang conjecture, Liu and Xu proposed a “S-adelic Mordell-Lang conjecture” for algebraic tori.

Conjecture 1.2 (Liu and Xu): Let k be a global field. Let X be a hyperbolic rational curve over k and $j : X \rightarrow T$ be the closed immersion into a k -torus T from Theorem (5.8). If G is a finitely generated subgroup of $T(k)$ and $\bar{G}^S$ is the topological closure of G inside $T(\mathbf{A}_k^S)$, then there is a finite subscheme $Z \subseteq X$ such that $X(\mathbf{A}_k^S) \cap \bar{G}^S \subseteq Z(\mathbf{A}_k^S)$.

Remark 5.11: Let A be an abelian variety over a global field k and X be a closed subvariety of A . The **Mordell-Lang conjecture** describes $X(k)$ as the set of rational points on a finite union of special subvarieties, which was proved by Faltings^[29] in the number field case and by Hrushovski^[45] in the function field case. More precisely, if A has no isotrivial isogeny factors, there is a finite union of cosets $Y \subseteq X$ such that $X(k) = Y(k)$, where a **coset** in A is a subvariety of the form $C = a + A'$ with $a \in A(\bar{k})$ and an abelian subvariety $A' \subseteq A$, namely a k -torsor under A' .

The **adelic Mordell-Lang conjecture** says that there is a finite union of cosets $Y \subseteq X$ such that $\overline{A(k)} \cap X(\mathbf{A}_k) \subseteq Y(\mathbf{A}_k)$, where $\overline{A(k)}$ is the topological closure of $A(k)$ in $\prod_{v \in \Omega_k} A(k_v)$ and $\overline{A(k)} \simeq A(k) \times \widehat{\mathbb{Z}}$ due to Serre and Milne in 1972. Equivalently, the **adelic Mordell-Lang conjecture** says that each sequence of k -rational points on abelian variety A approaching X must approach Y , which was proved by Stoll^[88] over number fields and by Poonen et al.^[78] over function fields both when $X_{\bar{k}}$ is coset-free, i.e X does not contain any coset of positive dimension. Induced from a new proof of the Mordell-

Lang conjecture given by Wisson^[96] recently, the adelic Mordell-Lang conjecture is true for any X over global function fields if A admits no non-zero isotrivial quotient.

The Mordell-Weil Sieve conjecture says that for any closed subvariety X of A , the equality $\overline{X(k)} = X(\mathbf{A}_k) \cap \overline{A(k)}$ holds, which has recently been validated to be equivalent to the adelic Mordell-Lang Conjecture by Creutz^[7]. The Mordell-Weil Sieve conjecture has been asked in various forms by Scharaschkin^[80], Skorobogatov^[85], Stoll^[88], Poonen et al.^[78], which has been established by Creutz^[7] and see Theorem (5.12).

Until now, it is less well understood the “S-adelic Mordell-Lang Conjecture” for algebraic tori proposed by Liu-Xu above, which is still widely open and worth considering.

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ACKNOWLEDGEMENTS

I would like to sincerely thank Professor Yong Hu, Professor Zhan Li, and Professor Hui Gao from Southern University of Science and Technology for their guidance and support in both my academic pursuits and daily life.

I am deeply grateful to Professor Yong Hu, who provided sufficient funding for my participation in various academic activities, allowing me to engage in academic exchanges and learning without any financial concerns. Besides, I am particularly grateful to Professor Yong Hu again, for numerous insightful discussions throughout this research. From our regular conversations, I have acquired many essential techniques and gained a deeper understanding of the frontiers of arithmetic geometry, especially related topics about rational points on algebraic varieties.

I also wish to extend my gratitude to Teacher Zhongwei Yang from Southwest Jiaotong University for his encouragement and guidance during my undergraduate studies.

I would like to thank my senior fellow student, Jing Liu, for his valuable suggestions and discussions on the proof of Theorem (4.4). Also, I am deeply grateful to Professor Qing Liu for his helpful explanations on Main Theorem I at the Conference on Arithmetic Groups held at SUSTech in June 2025. Moreover, I wish to express my sincere gratitude to my friends Chuning Du, Binhe Huang, Jincheng Hu, Yulei Liu, Yixuan Zhou for their companionship and constant encouragement throughout my research.

Finally, I would like to express my deepest gratitude to my family for their constant companionship and unwavering support along mathematical journey.

再次重逢的世界

从湘西的大山，到巴蜀的盆地，再到纸醉金迷的湾区，也不过一千多公里的路，我吃了很多的苦，走了二十多年，才走到今天这里。

泥潭里长大的人，生来就不会顺从天意。这两年间，无数次有人问我，是否后悔来到这里。问征夫以前路，恨晨光之熹微。尽管没有人托举，更没有人铺路，我却仍然感到庆幸，至少我拥有选择的机会，可以大胆地走自己的路。我没有成为别人，我成为了我自己。只要我还在，这个世界就在。

在求学路上，我最要感谢的人是我的导师胡勇教授，在他悉心的指导下，我顺利地完成了这篇硕士论文。大家都说他招收学生极为严格，然而当年连二次互反律都没学明白的我，却成为了他的学生。出于在学术方面对热点的追求，我为他制造了许多麻烦，因此很感谢他对我无比的包容。记得第一次正式见面，我便直言不讳，表示不想做二次型的问题，他不但没有生气，还很尊重我的想法。这两年里，我也时常为自己没有提前预约，便闯入他的办公室而感到惭愧。即便如此，他还是愿意停下手里的事情，不厌其烦地跟我讨论数学，回答我各式各样的问题。我很感谢胡勇教授在学术上对我的鼓励，是他一次次给我信心，让我在这条路上继续走下去。他时常安慰我，告诉我不用着急，还有时间；他总能将心比心，说自己在我这个年纪，一样也没学明白。他不要求我做出来的成果有多么宏大，而是希望我在学术上能踏实地走好每一步，要求我写进论文一定都是我自己明白的。

在资金方面，我很感谢胡勇教授资助我参加厦门大学 Abel 簇，西湖大学 Diophantine 日，武汉大学 Topos 的短期课程和学术会议；他经费短缺，跟高辉老师筹措经费，仍资助我本科来南方科技大学访问两个月；我想要买书，他便把自己仅剩下来一千多块的教学改革经费拿来给我们复印书籍。我很感谢胡勇教授在我硕士期间开设了数论和学术英语写作这两门课程，他的课堂无微不至，有对我们这些后辈们的期望和勉励，或许还有些许羡慕。他希望作为学生的我们能够超过他。他比我还要勤奋，几乎每天早上九点钟，就来到办公室；每天晚上九点，他办公室的灯大都还亮着。他学术上的刚毅和生活上的乐观，深深地打动了我。他就像一束火炬，像一座灯塔，一直指引着我的方向。

在南方科技大学攻读硕士的两年里，学业上为我指导，帮我解惑的老师还有很多。我有幸遇到高辉教授开设抽象代数和交换代数这两门基础课程，他课堂的证明组织得非常流畅，最重要的是他让我真切地感受到了代数的美妙。在恰逢毕

业的季节，我总算能腾出时间去学习本科没有学明白的代数拓扑。朱一飞教授的代数拓扑课堂，让我体会到了复杂代数工具背后的几何直观，这正是我最缺乏的，也是最需要的。我很感谢李展教授当年的推荐，很感谢他为我争取了一个面试的机会，让我得以展现自己。我也很感谢邬龙挺教授开设 Toric 簇的课程，我很珍惜这样的学习机会，以后难以有人再这样教我。

其次，我很想感谢法国波尔多大学刘青教授在学业上为我提供的建议和帮助。三年前，我从成都赴厦门参加了刘青教授在厦门大学开设的 Mordell—Weil 定理短课，厦门高崎机场成为了我人生第一次降落的机场，也种下了我心中那颗算术几何的种子。第一次在厦门，刘青老师问我学过什么知识，我只看了一些 Zaraski 拓扑，刘老师告诉我说那是最基本的内容；研究生一年级的時候，我去厦门参加了刘青老师的 Abel 簇短课，刘青老师问了我的学业，我说在南方科技大学跟胡勇念硕士，学 Brauer-Manin 障碍和有理点的问题。后来，我的硕士课题主要是围绕着刘青教授和徐飞教授合作的一篇文章开展，在深圳的算术几何会议上，他给我解释了论文当中的一部分细节，让我对读懂他们的工作更有信心。在毕业之际，我很感谢刘青教授能够与我通过电话探讨博士阶段的课题，并将我的硕士论文及困惑一并邮件抄送给首都师范大学的徐飞教授，为我未来指明了更多可能的方向。

我很感谢在南方科技大学遇到的每一位朋友，跟你们相伴的每一幕汇聚成了我的硕士生涯。在学术启蒙的道路上，我很感谢我的大师兄刘靖博士和计算数学领域的戴咏诚博士为我提供了许多建议和帮助。我的室友伍俊然，他在的日子里，总能把我照顾的很好；数学中心的黄斌和，他总是很热情，只要在数学中心讨论问题，他就会帮我拉一杯咖啡，他的拉花技术已经如火纯青。在科研之余，我和黄斌和，胡锦涛，杜楚宁，彭欣常在一起高谈阔论，这些时光欢乐却短暂。在周末闲暇，我和陈俊廷，杨都，我们一起走遍了深圳的很多角落，吃到了很多美食。我很感谢金鹏荣博士，他经常与我分享数学，为我弥补了很多几何上的直观。我很感谢我的同门刘昱磊，陪我一起上了多门专业课和讨论班，度过了难熬的时光。我很感谢李航师兄和吴晓明学长，他们经常约我打羽毛球锻炼身体，在球场上也很照顾我。

在硕士期间，我两次自费前往北京，参加代数与数论暑期学校。我很感谢中国科学技术大学的汪立言师兄，他在第一次暑校专门预订了一间双床房，为我提供了在北京的住宿。在第二次暑校，我特别感谢哈尔滨工业大学的田乙胜教授，他同意我参加他在中科院主持的研讨班，带领我学习了约化代数群的知识并开展相关前沿研究。这是我这两年里数学上成长最快的一次。

从拿到教师资格证，到真正地站上讲台，我很感谢钟华教授，他为我提供了线性代数的助教岗位，并常与我探讨线性代数的教学要点和难点。我担任了三次线

性代数和一次初等数论的助教，遇见了一批批温柔又炽热的学生，见证了他们的成长和青春。跟你们的相遇是我在硕士两年里最幸福的一件事。记得我人生中的第一堂课，下课时，教室里响起了雷鸣般的掌声，这一幕幕都曾深深地触动我。

在这两年的生活上，我特别感谢我的父母，他们做着最平凡的工作，却用一双手托举着我，让我走到今天这里。我似乎跟她们之间隔着无法跨越的距离，靠近了，就失去了狼的野性；远离了，会放不下心里的那份牵挂。靠近你们，就靠近了痛苦，这种痛苦是细碎的，是无孔不入的。它是一边读着论文做着研究，一边照顾老人的饮食起居；它是我每次回来，都能清晰得看见这片土地上的慢，和这片土地上被时代甩在后面的人；它是父母日益佝偻的背，是每天求神拜佛、可生活依然未见保佑的阿大，是那个患有双向情感障碍的嬢嬢，是那个要落叶归根，死活不肯去医院的阿公，是那只陪我长大，却不知葬在何处的老狗豆丁。我爱他们，可我也怕他们，怕自己跟他们同化，怕自己再也迈不出那一步。而远离你们，就远离了幸福。这种幸福是具体的，是泪流满面的。它是一家人吃团圆唠家常的欢聚；是听着奶奶念叨的歪门邪道；是家里零食塞满了我的行李箱；是临走前掐着时间做出的那顿饭菜。可我知道，就像格林注定属于草原，我的一部分也永远留在了这篇土地上。无论我跑得多快，跑得多远，只要一回头，就能看见那些山，那些水，和那些站在家门口送别的人。——只愿岁岁平安。

日月虽有被遮蔽之时，只要本体未失，终能重放光明；江河即使遭遇险阻，只要源头不竭，终能贯通入海。我也感谢这两年亲眼目睹了人性的复杂，那些肮脏阴暗的手段，尖酸刻薄的嘴脸，教会了我学会沉默，让我懂得不用对每个人都保持热情。我骄傲的是，即使在阴暗的人和事上吃过亏，我依然不屑成为那样的人。

最后，要感谢自己的坚韧不屈，感谢自己见到了这个世界的难堪后，依然选择勇敢地走下去。后来，我发现，这个世界没有岸，只有越来越会游泳的自己。尽管有时候的确很难，有的难处没有人能真正帮得了，惟有自己迈过去。没人会知道，曾经从湘西的山沟沟，通过高考，误打误撞考上数学专业的那个人，他一个人走了多少路，才见到了，那些别人口中，所谓好的数学。

夕阳惹红了操场，黄昏撒了糖。青春却在塘朗山这来来回回的坡里散去了，在智华楼外蝉鸣倒数的钟声里渐行渐远。我们一遍遍奋力向上奔走，气喘吁吁，跨过崎岖；一路上怀揣热忱，奔赴前路。不知不觉间，少年意气淹没在了沿途风尘。那些年的夏至未至，旧时光里最初的记忆，振华路口闪烁的交通灯，像离别又像是重逢。这么多年后，当我飞奔向你们时，橘生淮南的暗恋，耿耿余淮的瞬间，像释怀也像是遗憾。那些追光的日子，藏着暗格里的秘密，和那个——最好的我们。

这个人也许永远不会回来了，也许“明天”回来。

RESUME AND ACADEMIC ACHIEVEMENTS

Resume

Born on 12/21/2001 in Xiangxi Tujia and Miao Autonomous Prefecture, Hunan, China, the applicant, Wei Jie, studied and obtained a bachelor's degree in Mathematics at Southwest Jiaotong University (SWJTU), from September 2020 to June 2024, supervised by Prof. Yang Zhongwei. After that, the applicant was admitted to Southern University of Science and Technology (SUSTech) in September 2024 to pursue a Master's degree in pure Mathematics. During the graduate study, the applicant was supervised by Prof. Hu Yong, and devoted into the problems in arithmetic geometry, especially integral points/rational points on curves and surfaces, the Brauer-Manin obstruction and related topics.

Awards:

- (1) Excellent Graduate of SUSTech, 2026.
- (2) Outstanding Teaching Assistant, Linear Algebra, SUSTech, 2026.
- (3) Special-grade Research Assistantship Scholarship, SUSTech, 2025.
- (4) Excellent Graduate of Sichuan Province, SWJTU, 2024.

Work experience:

- (1) Fall 2025; basic number theory; teaching assistant.
- (2) Fall 2024; linear algebra; teaching assistant.
- (3) Fall 2023; CIB FINTECH, Chengdu Branch, the Research and Development Center of Enterprise Finance and Investment Bank; C Language engineer, intern.